

(19) **United States**
 (12) **Patent Application Publication** (10) **Pub. No.: US 2025/0283121 A1**
Vugt-Van Lutz et al. (43) **Pub. Date: Sep. 11, 2025**

(54) **RECOMBINANT YEAST CELL**

Publication Classification

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(51) **Int. Cl.**
C12P 7/06 (2006.01)
C07K 14/39 (2006.01)
C12N 9/12 (2006.01)
C12N 9/34 (2006.01)
C12N 9/88 (2006.01)
 (52) **U.S. Cl.**
 CPC *C12P 7/06* (2013.01); *C07K 14/39* (2013.01); *C12N 9/1205* (2013.01); *C12N 9/2428* (2013.01); *C12N 9/88* (2013.01); *C12Y 207/01019* (2013.01); *C12Y 302/01003* (2013.01); *C12Y 401/01039* (2013.01)

(21) Appl. No.: **18/578,590**
 (22) PCT Filed: **Jul. 7, 2022**
 (86) PCT No.: **PCT/EP2022/068983**
 § 371 (c)(1),
 (2) Date: **Jan. 11, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/220,905, filed on Jul. 12, 2021.

Foreign Application Priority Data

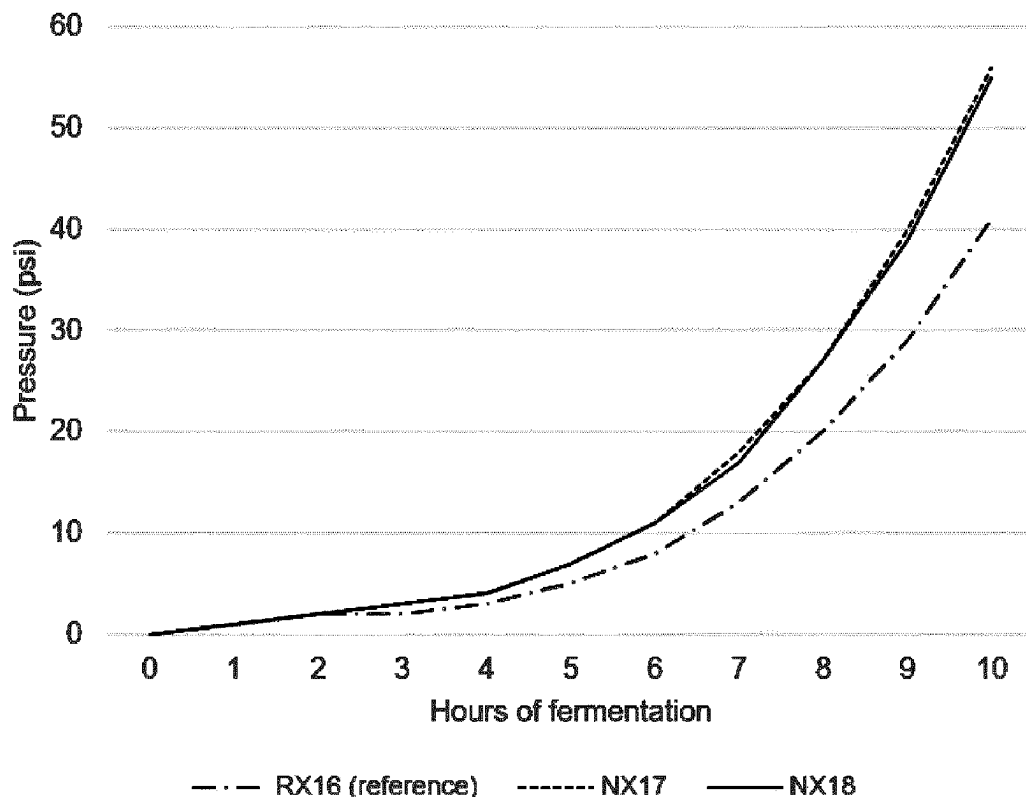
Jul. 12, 2021 (EP) 21185147.2

(57) **ABSTRACT**

A recombinant yeast cell that functionally expresses: —a nucleic acid sequence encoding a protein having glycerol dehydrogenase activity; —a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity; and —a nucleic acid sequence encoding a protein having glycerol transporter activity, wherein the expression of the nucleic acid sequence encoding the protein having glycerol transporter activity is under control of a promoter (the “GT promoter”), which GT promoter has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more, and a process for the production of ethanol using such recombinant yeast cell.

Specification includes a Sequence Listing.

Ethanol and CO₂ gas production



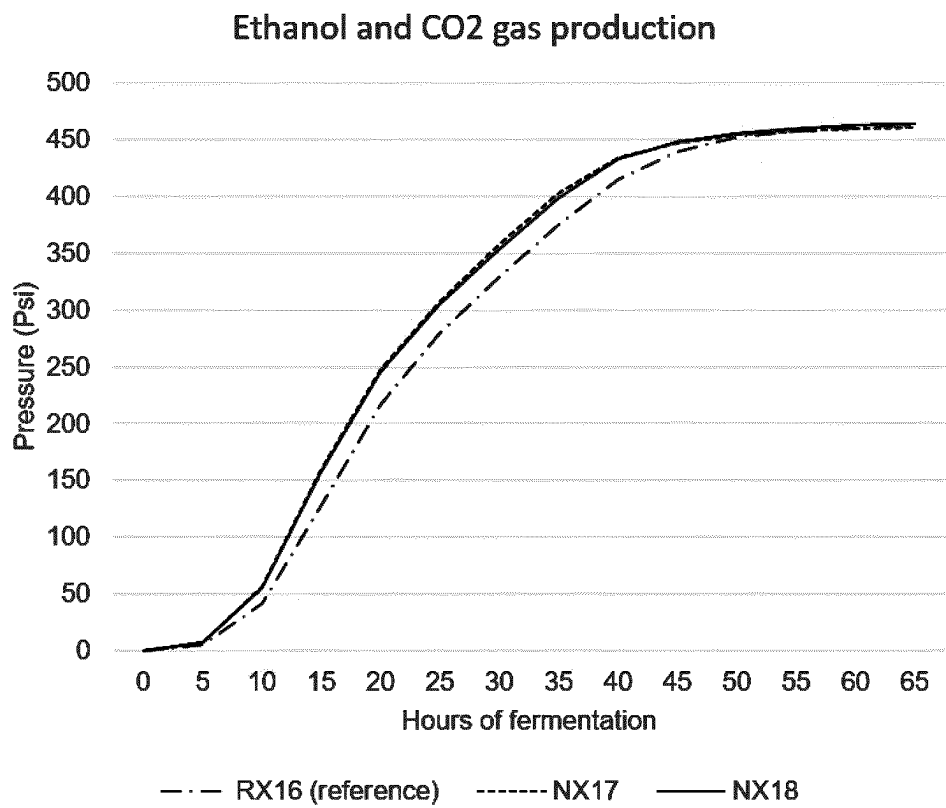


Fig. 1

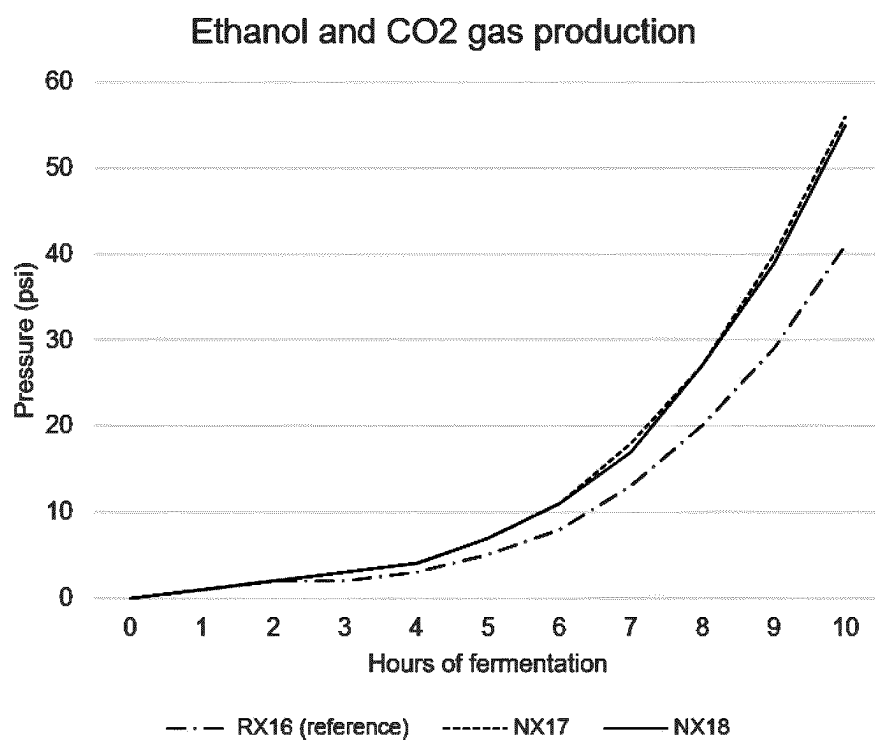


Fig. 2

RECOMBINANT YEAST CELL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a U.S. National Stage Application under 35 U.S.C. § 371 of International Application No. PCT/EP2022/068983, filed Jul. 7, 2022, which claims priority to U.S. Application No. 63/220,905, filed Jul. 12, 2021, and European Application No. 21185147.2, filed Jul. 12, 2021, all of which are hereby incorporated by reference in their entireties.

INCORPORATION BY REFERENCE OF THE SEQUENCE LISTING

[0002] The present application is being filed with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled IFF34060-US-PCT_SEQ_Listing.xml, created on Jul. 9, 2024, which is 161,064 in size. The information in the electronic format of the Sequence Listing is incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0003] The invention relates to a recombinant yeast cell and to a process for the production of ethanol wherein said recombinant yeast cell is used.

BACKGROUND OF THE INVENTION

[0004] Microbial fermentation processes are applied to industrial production of a broad and rapidly expanding range of chemical compounds from renewable carbohydrate feedstocks. Especially in anaerobic fermentation processes, redox balancing of the cofactor couple NADH/NAD⁺ can cause important constraints on product yields. This challenge is exemplified by the formation of glycerol as major by-product in the industrial production of—for instance—fuel ethanol by *Saccharomyces cerevisiae*, a direct consequence of the need to re-oxidize NADH formed in biosynthetic reactions.

[0005] Ethanol production by *Saccharomyces cerevisiae* is currently, by volume, the single largest fermentation process in industrial biotechnology. Various approaches have been proposed to improve the fermentative properties of organisms used in industrial biotechnology by genetic modification. A major challenge relating to the stoichiometry of yeast-based production of ethanol, is that substantial amounts of NADH-dependent side-products such as glycerol are generally formed as a by-product, especially under anaerobic and oxygen-limited conditions or under conditions where respiration is otherwise constrained or absent. It has been estimated that, in typical industrial ethanol processes, up to about 4 wt. % of the sugar feedstock is converted into glycerol (Nissen et al., “Anaerobic and aerobic batch cultivations of *Saccharomyces cerevisiae* mutants impaired in glycerol Synthesis”, (2000), Yeast, vol. 16, pages 463-474). Under conditions that are ideal for anaerobic growth, the conversion into glycerol may even be higher, up to about 10%.

[0006] Glycerol production under anaerobic conditions is primarily linked to redox metabolism. During anaerobic growth of *S. cerevisiae*, sugar dissimilation occurs via alcoholic fermentation. In this process, the NADH formed in the glycolytic glyceraldehyde-3-phosphate dehydrogenase reaction is reoxidized by converting acetaldehyde, formed

by decarboxylation of pyruvate to ethanol via NAD⁺-dependent alcohol dehydrogenase. The fixed stoichiometry of this redox-neutral dissimilatory pathway causes problems when a net reduction of NAD⁺ to NADH occurs elsewhere in metabolism. Under anaerobic conditions, NADH reoxidation in *S. cerevisiae* is strictly dependent on reduction of sugar to glycerol. Glycerol formation is initiated by reduction of the glycolytic intermediate dihydroxyacetone phosphate (DHAP) to glycerol 3-phosphate (glycerol-3P), a reaction catalyzed by NAD⁺-dependent glycerol 3-phosphate dehydrogenase. Subsequently, the glycerol 3-phosphate formed in this reaction is hydrolysed by glycerol-3-phosphatase to yield glycerol and inorganic phosphate. Consequently, glycerol is a major by-product during anaerobic production of ethanol by *S. cerevisiae*, which is undesired as it reduces overall conversion of sugar to ethanol. Further, the presence of glycerol in effluents of ethanol production plants may impose costs for waste-water treatment.

[0007] In the literature, several different approaches have been reported that could help to reduce the amount of the byproduct glycerol and divert carbon to ethanol resulting in an increased yield of ethanol per gram of fermented carbohydrate.

[0008] WO2015/028583 describes a yeast cell that is genetically modified comprising: a) one or more nucleic acid sequence encoding a glycerol dehydrogenase (E.C. 1.1.1.6); b) one or more nucleic acid sequence encoding a dihydroxyacetone kinase (E.C. 2.7.1.28 or E.C. 2.7.1.29) and c) one or more nucleic acid sequence encoding a glycerol transporter. In addition, the cell may comprise one or more nucleic acid sequences encoding a NAD⁺-dependent acetylating acetaldehyde dehydrogenase. WO2015/028583 further describes a process comprising the preparation of a fermentation product from acetate and from a fermentable carbohydrate—in particular a carbohydrate selected from the group of glucose, fructose, sucrose, maltose, xylose, arabinose, galactose and mannose—which preparation is carried out under anaerobic conditions using the above yeast cell.

[0009] WO2015/028583 explains that as acetic acid is often considered to be the most toxic compound present in hydrolysates, there is a desire to further decrease the acetate (acetic acid) concentration in hydrolysates. It is mentioned that one way of increasing the anaerobic acetate conversion potential of the yeast is by introducing a glycerol conversion pathway that for example converts externally added glycerol forcing the yeast cell to convert more acetic acid in order to maintain the redox balance.

[0010] In the examples and Table 11, WO2015/028583 illustrates that especially transformant T5, including a glycerol transporter originating from *Zygosaccharomyces rouxii*, resulted in the conversion of more glycerol, relative to the reference strain. Also more acetic acid was consumed. The ethanol titer, however, was not the highest in case of this T5, because not all sugars were consumed. Hence, although good results are obtained with the yeast cell and process described in WO2015/028583, there is still room for further improvement.

[0011] It would be an advancement in the art to provide a yeast and process for the production of ethanol wherein the yeast comprises a glycerol conversion pathway and/or a glycerol transporter, similar to the yeast in WO2015/028583, but wherein the speed of the sugar conversion and/or the total amount of sugar consumed is improved.

SUMMARY OF THE INVENTION

[0012] The inventors have now surprising found that the process and yeast cell of WO2015/028583 can be even further improved by promoting the glycerol transporter with a specific promoter.

[0013] Accordingly the invention provides a recombinant yeast cell functionally expressing:

[0014] a nucleic acid sequence encoding a protein having glycerol dehydrogenase activity (preferably within enzyme class E.C. 1.1.1.6);

[0015] a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (preferably within enzyme class E.C. 2.7.1.28 or E.C. 2.7.1.29); and

[0016] a nucleic acid sequence encoding a protein having glycerol transporter activity,

wherein the expression of the nucleic acid sequence encoding the protein having glycerol transporter activity is under control of a promoter (the "GT promoter"), which GT promoter has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more.

[0017] In addition, the invention provides a process for the production of ethanol, comprising converting a carbon source, such as a carbohydrate or another organic carbon source, using the above recombinant yeast cell, suitably thereby forming ethanol.

[0018] Advantageously, use of the above recombinant yeast cell and/or the above process results in an improved speed of the sugar conversion and/or a higher total amount of sugar consumed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] The invention is illustrated by the following figures:

[0020] FIG. 1: Ethanol and CO₂ gas production during the full 66 hours of the fermentation of corn mash with respectively reference strain RX16, new strain NX17 and new strain NX18 as described in Example 5 and illustrated in Table 18.

[0021] FIG. 2: Ethanol and CO₂ gas production during the first 10 hours of the fermentation of corn mash with respectively reference strain RX16, new strain NX17 and new strain NX18 as described in Example 5 and illustrated in Table 19.

BRIEF DESCRIPTION OF THE SEQUENCE LISTING

[0022] This application contains a Sequence Listing in computer readable form, which is incorporated herein by reference. An overview is provided by Table 1 below.

TABLE 1

Overview of sequence listings:			
SEQ ID NO:	Enzyme Name	Origin	Type
SEQ ID NO: 1	Glycerol transporter	<i>S. pombe</i>	protein
SEQ ID NO: 2	Glycerol transporter	<i>Plasmodium falciparum</i> ;	protein
SEQ ID NO: 3	Glycerol transporter (T3) (aquaglyceroporin AQP9)	<i>Danio rerio</i>	protein
SEQ ID NO: 4	Glycerol transporter	<i>Xenopus tropicalis</i> ;	protein
SEQ ID NO: 5	Glycerol transporter (T5) (ZYRO0E0121)	<i>Z. rouxii</i>	protein
SEQ ID NO: 6	Glycerol transporter (T3) (aquaglyceroporin AQP9)	<i>Danio rerio</i>	DNA
SEQ ID NO: 7	Glycerol transporter (T5) (ZYRO0E0121)	<i>Z. rouxii</i>	DNA
SEQ ID NO: 8	Promoter Motif	Artificial sequence	DNA
SEQ ID NO: 9	Promoter Motif	Artificial sequence	DNA
SEQ ID NO: 10	Promoter ANB1	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 11	Promoter DAN1	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 12	Promoter HEM13	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 13	Glycerol dehydrogenase (Kpne_gldA)	<i>Klebsiella pneumoniae</i>	protein
SEQ ID NO: 14	Glycerol dehydrogenase (Eaer_gldA)	<i>Enterococcus aerogenes</i>	protein
SEQ ID NO: 15	Glycerol dehydrogenase (Ya_gldA)	<i>Yersinia aldovae</i>	protein
SEQ ID NO: 16	Glycerol dehydrogenase (Ec_gldA)	<i>Escherichia coli</i>	protein
SEQ ID NO: 17	Glycerol dehydrogenase (Ec_gldA)	Artificial sequence; based on gldA gene from <i>Escherichia coli</i> - codon optimized	DNA
SEQ ID NO: 18	Dihydroxyacetone kinase (DAK1)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 19	Dihydroxyacetone kinase (DAK2)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 20	Dihydroxyacetone kinase (Kpne_dhaK)	<i>Klebsiella pneumoniae</i>	protein
SEQ ID NO: 21	Dihydroxyacetone kinase (Ylip_DAK1)	<i>Yarrowia lipolytica</i>	protein
SEQ ID NO: 22	Dihydroxyacetone kinase (Spom_DAK1)	<i>Schizosaccharomyces pombe</i>	protein
SEQ ID NO: 23	Dihydroxyacetone kinase (DAK1)	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 24	Dihydroxyacetone kinase (DAK2)	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 25	Ribulose-1,5-biphosphate carboxylase oxygenase (RuBisCO) as previously described in WO2014/129898	<i>Thiobacillus denitrificans</i>	protein

TABLE 1-continued

Overview of sequence listings:			
SEQ ID NO:	Enzyme Name	Origin	Type
SEQ ID NO: 26	Ribulose-1,5-biphosphate carboxylase oxygenase (RuBisCO) as previously described in WO2014/129898	Artificial sequence; based on cbbM gene from <i>Thiobacillus denitrificans</i> - codon optimized	DNA
SEQ ID NO: 27	Phosphoribulokinase (PRK) as previously described in WO2014/129898	<i>Spinacea oleracea</i>	protein
SEQ ID NO: 28	Phosphoribulokinase (PRK) as previously described in WO2014/129898	Artificial sequence; based on prk gene from <i>Spinacea oleracea</i> , codon optimized	DNA
SEQ ID NO: 29	(long) chaperone groEL as previously described in WO2014/129898	Artificial sequence; Translated protein sequence based on GroEL of <i>Escherichia coli</i>	protein
SEQ ID NO: 30	(long) chaperone groEL as previously described in WO2014/129898	Artificial sequence; based on GroEL gene from <i>Escherichia coli</i> , codon optimized	DNA
SEQ ID NO: 31	(short) chaperone groES as previously described in WO2014/129898	Artificial sequence; Translated protein sequence based on GroES of <i>Escherichia coli</i>	protein
SEQ ID NO: 32	(short) chaperone groES as previously described in WO2014/129898	Artificial sequence; based on GroES gene from <i>Escherichia coli</i> , codon optimized	DNA
SEQ ID NO: 33	Phosphoketolase	<i>Bifidobacterium animalis</i>	protein
SEQ ID NO: 34	Phosphoketolase	<i>Bifidobacterium adolescentis</i>	protein
SEQ ID NO: 35	Phosphoketolase	<i>Bifidobacterium lactis</i>	protein
SEQ ID NO: 36	Phosphoketolase	<i>Leuconostoc mesenteroides</i>	protein
SEQ ID NO: 37	Phosphotransacetylase	<i>Bacillus subtilis</i>	protein
SEQ ID NO: 38	Phosphotransacetylase	<i>Lactobacillus plantarum</i>	protein
SEQ ID NO: 39	Phosphotransacetylase	<i>Bifidobacterium adoloscentis</i>	protein
SEQ ID NO: 40	Phosphotransacetylase	<i>Methosarcina thermophila</i>	protein
SEQ ID NO: 41	Acetate kinase	<i>Bifidobacterium adolescentis</i>	protein
SEQ ID NO: 42	Acetate kinase	<i>Aspergillus nidulans</i>	protein
SEQ ID NO: 43	Bifunctional NAD ⁺ dependent acetylating acetaldehyde/alcohol dehydrogenase (adhE)	<i>Escherichia coli</i>	protein
SEQ ID NO: 44	Ethanolamine utilizing protein (eutE)	<i>Escherichia coli</i>	protein
SEQ ID NO: 45	acetaldehyde dehydrogenase (acdH)	<i>Lactobacillus plantarum</i>	protein
SEQ ID NO: 46	acetaldehyde dehydrogenase (acdH)	<i>Listeria innocua</i>	protein
SEQ ID NO: 47	acetaldehyde/alcohol dehydrogenase (adhE)	<i>Staphylococcus aureus</i>	protein
SEQ ID NO: 48	acetylating acetaldehyde dehydrogenase (mhpF)	<i>Escherichia coli</i>	protein
SEQ ID NO: 49	Acetyl-coenzyme A synthetase 1 (ACS1)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 50	Acetyl-coenzyme A synthetase 2 (ACS2)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 51	Alcohol dehydrogenase 1 (ADH1)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 52	Alcohol dehydrogenase 2 (ADH2)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 53	Alcohol dehydrogenase 3 (ADH3)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 54	Alcohol dehydrogenase 4 (ADH4)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 55	Alcohol dehydrogenase 5 (ADH5)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 56	Glucoamylase (GA)	<i>Punctularia strigosozonata</i>	protein
SEQ ID NO: 57	Glucoamylase (GA)	<i>Punctularia strigosozonata</i>	protein
SEQ ID NO: 58	Glucoamylase (GA)	Artificial sequence, based on <i>Punctularia strigosozonata</i>	protein
SEQ ID NO: 59	Fragment A	Artificial sequence	DNA
SEQ ID NO: 60	Fragment B	Artificial sequence	DNA
SEQ ID NO: 61	Fragment C	Artificial sequence	DNA
SEQ ID NO: 62	Fragment D	Artificial sequence	DNA
SEQ ID NO: 63	Fragment E	Artificial sequence	DNA
SEQ ID NO: 64	Fragment F	Artificial sequence	DNA
SEQ ID NO: 65	Fragment G	Artificial sequence	DNA
SEQ ID NO: 66	Connector 25	Artificial sequence	DNA
SEQ ID NO: 67	Connector 2A	Artificial sequence	DNA
SEQ ID NO: 68	Connector 2B	Artificial sequence	DNA
SEQ ID NO: 69	Connector 2C	Artificial sequence	DNA
SEQ ID NO: 70	Sc_INT7.03_FLANK5	Artificial sequence	DNA
SEQ ID NO: 71	Sc_INT7.03_FLANK3	Artificial sequence	DNA
SEQ ID NO: 72	INT28_FLANK5	Artificial sequence	DNA
SEQ ID NO: 73	INT28_FLANK3	Artificial sequence	DNA
SEQ ID NO: 74	NR1 nitrate reductase [Uniprot accession no: Q9XGW5]	<i>Zea mays</i>	Protein

TABLE 1-continued

Overview of sequence listings:			
SEQ ID NO:	Enzyme Name	Origin	Type
SEQ ID NO: 75	NIA Nitrate reductase [Uniprot accession no: P23312]	<i>Spinacia oleracea</i>	Protein
SEQ ID NO: 76	nirD Nitrite reductase (small subunit)	<i>Escherichia coli</i>	Protein
SEQ ID NO: 77	nirB Nitrite reductase (large subunit)	<i>Escherichia coli</i>	Protein
SEQ ID NO: 78	niiA Nitrite reductase	<i>Emmericella nidulans</i> (<i>Aspergillus nidulans</i>)	Protein
SEQ ID NO: 79	YNT1 Nitrate/Nitrite transporter	<i>Pichia angusta</i> (<i>Hansenula polymorpha</i>)	Protein

[0023] In the context of this patent application, each of the above protein/amino acid sequences is preferably encoded by a DNA/nucleic acid sequence that is codon-pair optimized for expression in a yeast, more preferably for expression in a *Saccharomyces cerevisiae* yeast.

DETAILED DESCRIPTION OF THE INVENTION

Definitions

[0024] Unless defined otherwise or clearly indicated by context, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art.

[0025] Throughout the present specification and the accompanying claims, the words “comprise” and “include” and variations such as “comprises”, “comprising”, “includes” and “including” are to be interpreted inclusively. That is, these words are intended to convey the possible inclusion of other elements or integers not specifically recited, where the context allows.

[0026] The articles “a” and “an” are used herein to refer to one or to more than one (i.e. to one or at least one) of the grammatical object of the article. By way of example, “an element” may mean one element or more than one element. When referring to a noun (e.g. a compound, an additive, etc.) in the singular, the plural is meant to be included. Thus, when referring to a specific moiety, e.g. “gene”, this means “at least one” of that gene, e.g. “at least one gene”, unless specified otherwise.

[0027] When referring to a compound of which several isomers exist (e.g. a D and an L enantiomer), the compound in principle includes all enantiomers, diastereomers and cis/trans isomers of that compound that may be used in the particular aspect of the invention; in particular when referring to such as compound, it includes the natural isomer(s).

[0028] Unless explicitly indicated otherwise, the various embodiments of the invention described herein can be cross-combined.

[0029] The term “carbon source” refers to a source of carbon, preferably a compound or molecule comprising carbon. Preferably the carbon source is a carbohydrate. A carbohydrate is understood herein to be an organic compound made of carbon, oxygen and hydrogen. Suitably the carbon source may be selected from the group consisting of mono-, di- and/or polysaccharides, acids and acid salts. More preferably the carbon source is a compound selected from the group consisting of glucose, arabinose, xylose, galactose, mannose, rhamnose, fructose, glycerol, and acetic acid or a salt thereof.

[0030] The terms “dry matter” and “dry solids”, abbreviated respectively as “DM” and “DS”, are used interchangeably herein and refer to material remaining after removal of water. Dry matter content can be determined by any method known to the person skilled in the art therefore.

[0031] The term “ferment”, and variations thereof such as “fermenting”, “fermentation” and/or “fermentative”, is used herein in a classical sense, i.e. to indicate that a process is or has been carried out under anaerobic conditions. An anaerobic fermentation is herein defined to be a fermentation carried out under anaerobic conditions. Anaerobic conditions are herein defined as conditions without any oxygen or in which essentially no oxygen is consumed by the yeast cell. Conditions in which essentially no oxygen is consumed suitably corresponds to an oxygen consumption of less than 5 mmol/l·h⁻¹, in particular to an oxygen consumption of less than 2.5 mmol/l·h⁻¹, or less than 1 mmol/l·h⁻¹. More preferably 0 mmol/L/h is consumed (i.e. oxygen consumption is not detectable). This suitably corresponds to a dissolved oxygen concentration in a culture broth of less than 5% of air saturation, more suitably to a dissolved oxygen concentration of less than 1% of air saturation, or less than 0.2% of air saturation.

[0032] The term “fermentation process” refers to a process for the preparation or production of a fermentation product.

[0033] The term “cell” refers to a eukaryotic or prokaryotic organism, preferably occurring as a single cell. In the present invention the cell is a recombinant yeast cell. That is, the recombinant cell is selected from the group of genera consisting of yeast.

[0034] The terms “yeast” and “yeast cell” are used herein interchangeably and refer to a phylogenetically diverse group of single-celled fungi, most of which are in the division of Ascomycota and Basidiomycota. The budding yeasts (“true yeasts”) are classified in the order Saccharomycetales. The yeast cell according to the invention is preferably a yeast cell derived from the genus of *Saccharomyces*. More preferably the yeast cell is a yeast cell of the species *Saccharomyces cerevisiae*.

[0035] The term “recombinant”, for example referring to a “recombinant yeast”, a “recombinant cell”, “recombinant micro-organism” and/or “recombinant strain” as used herein, refers to a yeast, cell, micro-organism or strain, respectively, containing nucleic acid which is the result of one or more genetic modifications. Simply put the yeast, cell, micro-organism or strain contains a different combination of nucleic acid from (either of) its parent(s). To construe a recombinant yeast, cell, micro-organism or strain, recombinant DNA technique(s) and/or another mutagenic technique(s) can be used. For example a recombinant yeast

and/or a recombinant yeast cell may comprise nucleic acid not present in the corresponding wild-type yeast and/or cell, which nucleic acid has been introduced into that yeast and/or yeast cell using recombinant DNA techniques (i.e. a transgenic yeast and/or cell), or which nucleic acid not present in said wild-type yeast and/or cell is the result of one or more mutations—for example using recombinant DNA techniques or another mutagenesis technique such as UV-irradiation—in a nucleic acid sequence present in said wild-type yeast and/or yeast cell (such as a gene encoding a wild-type polypeptide) or wherein the nucleic acid sequence of a gene has been modified to target the polypeptide product (encoding it) towards another cellular compartment. Further, the term “recombinant” may suitably relate to a yeast, cell, micro-organism or strain from which nucleic acid sequences have been removed, for example using recombinant DNA techniques.

[0036] By a recombinant yeast comprising or having a certain activity is herein understood that the recombinant yeast may comprise one or more nucleic acid sequences encoding for a protein having such activity. Hence allowing the recombinant yeast to functionally express such a protein or enzyme.

[0037] The term “functionally expressing” means that there is a functioning transcription of the relevant nucleic acid sequence, allowing the nucleic acid sequence to actually be transcribed, for example resulting in the synthesis of a protein.

[0038] The term “transgenic” as used herein, for example referring to a “transgenic yeast” and/or a “transgenic cell”, refers to a yeast and/or cell, respectively, containing nucleic acid not naturally occurring in that yeast and/or cell and which has been introduced into that yeast and/or cell using for example recombinant DNA techniques, such as a recombinant yeast and/or cell.

[0039] The term “mutated” as used herein regarding proteins or polypeptides means that, as compared to the wild-type or naturally occurring protein or polypeptide sequence, at least one amino acid has been replaced with a different amino acid, inserted into, or deleted from the amino acid sequence. The replacement, insertion or deletion of the amino acid can for example be achieved via mutagenesis of nucleic acids encoding these amino acids. Mutagenesis is a well-known method in the art, and includes, for example, site-directed mutagenesis by means of PCR or via oligonucleotide-mediated mutagenesis as described in Sambrook et al., *Molecular Cloning—A Laboratory Manual*, 2nd ed., Vol. 1-3 (1989), published by Cold Spring Harbor Publishing).

[0040] The term “mutated” as used herein regarding genes means that, as compared to the wild-type or naturally occurring nucleic acid sequence, at least one nucleotide in the nucleic acid sequence of a gene or a regulatory sequence thereof, has been replaced with a different nucleotide, inserted into, or deleted from the nucleic acid sequence. The replacement, insertion or deletion of the amino acid can for example be achieved via mutagenesis, resulting for example in the transcription of a protein sequence with a qualitatively or quantitatively altered function or the knock-out of that gene. In the context of this invention an “altered gene” has the same meaning as a mutated gene.

[0041] The term “gen” or “gene”, as used herein, refers to a nucleic acid sequence that can be transcribed into mRNAs that are then translated into protein. A gene encoding for a

certain protein refers to the one or more nucleic acid sequence(s) encoding for such a protein.

[0042] The term “nucleic acid” or “nucleotide” as used herein, refers to a monomer unit in a deoxyribonucleotide or ribonucleotide polymer, i.e. a polynucleotide, in either single or double-stranded form, and unless otherwise limited, encompasses known analogues having the essential nature of natural nucleotides in that they hybridize to single-stranded nucleic acids in a manner similar to naturally occurring nucleotides (e.g., peptide nucleic acids). For example, a certain enzyme that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to the reference nucleotide sequence encoding the enzyme. A polynucleotide can be full-length or a subsequence of a native or heterologous structural or regulatory gene. Unless otherwise indicated, the term includes reference to the specified sequence as well as the complementary sequence thereof. Thus, DNAs or RNAs with backbones modified for stability or for other reasons are “polynucleotides” as that term is intended herein. Moreover, DNAs or RNAs comprising unusual bases, such as inosine, or modified bases, such as tritylated bases, to name just two examples, are polynucleotides as the term is used herein. It will be appreciated that a great variety of modifications have been made to DNA and RNA that serve many useful purposes known to those of skill in the art. The term polynucleotide as it is employed herein embraces such chemically, enzymatically or metabolically modified forms of polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including among other things, simple and complex cells.

[0043] The terms “nucleotide sequence” and “nucleic acid sequence” are used interchangeably herein. An example of a nucleic acid sequence is a DNA sequence.

[0044] The terms “polypeptide”, “peptide” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues, for example illustrated by an amino acid sequence. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers. The essential nature of such analogues of naturally occurring amino acids is that, when incorporated into a protein, that protein is specifically reactive to antibodies elicited to the same protein but consisting entirely of naturally occurring amino acids. The terms “polypeptide”, “peptide” and “protein” are also inclusive of modifications including, but not limited to, glycosylation, lipid attachment, sulphation, gamma-carboxylation of glutamic acid residues, hydroxylation and ADP-ribosylation.

[0045] The term “enzyme” refers herein to a protein having a catalytic function. Where a protein catalyzes a certain biological reaction, the terms “protein” and “enzyme” may be used interchangeably herein. When an enzyme is mentioned with reference to an enzyme class (EC), the enzyme class is a class wherein the enzyme is classified or may be classified, on the basis of the Enzyme Nomenclature provided by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (NC-IUBMB), which nomenclature may be found at <http://www.chem.qmul.ac.uk/iubmb/enzyme/>. Other suitable enzymes that have not (yet) been classified in a specified class but may be classified as such, are meant to be included.

[0046] If referred herein to a protein or a nucleic acid sequence, such as a gene, by reference to an accession number, this number in particular is used to refer to a protein or nucleic acid sequence (gene) having a sequence as can be found via www.ncbi.nlm.nih.gov/, (as available on 1 Oct. 2020) unless specified otherwise.

[0047] Every nucleic acid sequence herein that encodes a polypeptide also includes any conservatively modified variants thereof. This includes that, by reference to the genetic code, it describes every possible silent variation of the nucleic acid. The term “conservatively modified variants” applies to both amino acid and nucleic acid sequences. With respect to particular nucleic acid sequences, conservatively modified variants refers to those nucleic acids which encode identical or conservatively modified variants of the amino acid sequences due to the degeneracy of the genetic code. The term “degeneracy of the genetic code” refers to the fact that a large number of functionally identical nucleic acids encode any given protein. For instance, the codons GCA, GCC, GCG and GCU all encode the amino acid alanine. Thus, at every position where an alanine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded polypeptide. Such nucleic acid variations are “silent variations” and represent one species of conservatively modified variation.

[0048] The term “functional homologue” (or in short “homologue”) of a polypeptide and/or amino acid sequence having a specific sequence (e.g. “SEQ ID NO: X”), as used herein, refers to a polypeptide and/or amino acid sequence comprising said specific sequence with the proviso that one or more amino acids are mutated, substituted, deleted, added, and/or inserted, and which polypeptide has (qualitatively) the same enzymatic functionality for substrate conversion.

[0049] The term “functional homologue” (or in short “homologue”) of a polynucleotide and/or nucleic acid sequence having a specific sequence (e.g. “SEQ ID NO: X”), as used herein, refers to a polynucleotide and/or nucleic acid sequence comprising said specific sequence with the proviso that one or more nucleic acids are mutated, substituted, deleted, added, and/or inserted, and which polynucleotide encodes for a polypeptide sequence that has (qualitatively) the same enzymatic functionality for substrate conversion. With respect to nucleic acid sequences, the term functional homologue is meant to include nucleic acid sequences which differ from another nucleic acid sequence due to the degeneracy of the genetic code and encode the same polypeptide sequence.

[0050] Sequence identity is herein defined as a relationship between two or more amino acid (polypeptide or protein) sequences or two or more nucleic acid (polynucleotide) sequences, as determined by comparing the sequences. Usually, sequence identities or similarities are compared over the whole length of the sequences compared. In the art, “identity” also means the degree of sequence relatedness between amino acid or nucleic acid sequences, as the case may be, as determined by the match between strings of such sequences.

[0051] Amino acid or nucleotide sequences are said to be homologous when exhibiting a certain level of similarity. Two sequences being homologous indicate a common evolutionary origin. Whether two homologous sequences are closely related or more distantly related is indicated by “percent identity” or “percent similarity”, which is high or

low respectively. Although disputed, to indicate “percent identity” or “percent similarity”, “level of homology” or “percent homology” are frequently used interchangeably. A comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. The skilled person will be aware of the fact that several different computer programs are available to align two sequences and determine the homology between two sequences (Kruskal et al., “*An overview of sequence comparison: Time warps, string edits, and macromolecules*”, (1983), Society for Industrial and Applied Mathematics (SIAM), Vol 25, No. 2, pages 201-237 and D. and the handbook edited by Sankoff and J. B. Kruskal, (ed.), “*Time warps, string edits and macromolecules: the theory and practice of sequence comparison*”, (1983), pp. 1-44, published by Addison-Wesley Publishing Company, Massachusetts USA).

[0052] The percent identity between two amino acid sequences can be determined using the Needleman and Wunsch algorithm for the alignment of two sequences. (Needleman et al “*A General Method Applicable to the Search for Similarities in the Amino Acid Sequence of Two Proteins*” (1970) J. Mol. Biol. Vol. 48, pages 443-453). The algorithm aligns amino acid sequences as well as nucleotide sequences. The Needleman-Wunsch algorithm has been implemented in the computer program NEEDLE. For the purpose of this invention the NEEDLE program from the EMBOS package is used (version 2.8.0 or higher, see Rice et al, “*EMBOSS: The European Molecular Biology Open Software Suite*” (2000), Trends in Genetics vol. 16, (6) pages 276-277, <http://emboss.bioinformatics.nl/>). For protein sequences, EBLSUM62 is used for the substitution matrix. For nucleotide sequences, EDNAFULL is used. Other matrices can be specified. The optional parameters used for alignment of amino acid sequences are a gap-open penalty of 10 and a gap extension penalty of 0.5. The skilled person will appreciate that all these different parameters will yield slightly different results but that the overall percentage identity of two sequences is not significantly altered when using different algorithms.

[0053] The homology or identity is the percentage of identical matches between the two full sequences over the total aligned region including any gaps or extensions. The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment including the gaps. The identity defined as herein can be obtained from NEEDLE and is labelled in the output of the program as “IDENTITY”.

[0054] The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment after subtraction of the total number of gaps in the alignment. The identity defined as herein can be obtained from NEEDLE by using the NOBRIEF option and is labelled in the output of the program as “longest-identity”.

[0055] A variant of a nucleotide or amino acid sequence disclosed herein may also be defined as a nucleotide or amino acid sequence having one or more substitutions, insertions and/or deletions as compared to the nucleotide or amino acid sequence specifically disclosed herein (e.g. in the sequence listing).

[0056] Optionally, in determining the degree of amino acid similarity, the skilled person may also take into account so-called “conservative” amino acid substitutions, as will be clear to the skilled person. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is asparagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine, and tryptophan; a group of amino acids having basic side chains is lysine, arginine, and histidine; and a group of amino acids having sulphur-containing side chains is cysteine and methionine. In an embodiment, conservative amino acid substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, and asparagine-glutamine. Substitutional variants of the amino acid sequence disclosed herein are those in which at least one residue in the disclosed sequences has been removed and a different residue inserted in its place. Preferably, the amino acid change is conservative. In an embodiment, conservative substitutions for each of the naturally occurring amino acids are as follows: Ala to Ser; Arg to Lys; Asn to Gln or His; Asp to Glu; Cys to Ser or Ala; Gln to Asn; Glu to Asp; Gly to Pro; His to Asn or Gln; Ile to Leu or Val; Leu to Ile or Val; Lys to Arg; Gln or Glu; Met to Leu or Ile; Phe to Met, Leu or Tyr; Ser to Thr; Thr to Ser; Trp to Tyr; Tyr to Trp or Phe; and, Val to Ile or Leu.

[0057] Nucleotide sequences of the invention may also be defined by their capability to hybridise with parts of specific nucleotide sequences disclosed herein, respectively, under moderate, or preferably under stringent hybridisation conditions. Stringent hybridisation conditions are herein defined as conditions that allow a nucleic acid sequence of at least about 25, preferably about 50 nucleotides, 75 or 100 and most preferably of about 200 or more nucleotides, to hybridise at a temperature of about 65° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at 65° C. in a solution comprising about 0.1 M salt, or less, preferably 0.2×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having about 90% or more sequence identity. Moderate conditions are herein defined as conditions that allow a nucleic acid sequences of at least 50 nucleotides, preferably of about 200 or more nucleotides, to hybridise at a temperature of about 45° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at room temperature in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours, and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having up to 50% sequence identity. The person skilled in the art

will be able to modify these hybridisation conditions in order to specifically identify sequences varying in identity between 50% and 90%.

[0058] “Expression” refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

[0059] “Overexpression” refers to expression of a gene, respectively a nucleic acid sequence, by a recombinant cell in excess to its expression in a corresponding wild-type cell. Such overexpression can for example be arranged for by: increasing the frequency of transcription of one or more nucleic acid sequences, for example by operational linking of the nucleic acid sequence to a promoter functional within the recombinant cell; and/or by increasing the number of copies of a certain nucleic acid sequence.

[0060] The terms “upregulate”, “upregulated” and “upregulation” refer to a process by which a cell increases the quantity of a cellular component, such as RNA or protein. Such an upregulation may be in response to or caused by a genetic modification.

[0061] By the term “pathway” or “metabolic pathway” is herein understood a series of chemical reactions in a cell that build and breakdown molecules.

[0062] Nucleic acid sequences (i.e. polynucleotides) or proteins (i.e. polypeptides) may be native or heterologous to the genome of the host cell.

[0063] “Native”, “homologous” or “endogenous” with respect to a host cell, means that the nucleic acid sequence does naturally occur in the genome of the host cell or that the protein is naturally produced by that cell. The terms “native”, “homologous” and “endogenous” are used interchangeably herein.

[0064] As used herein, “heterologous” may refer to a nucleic acid sequence or a protein. For example, “heterologous”, with respect to the host cell, may refer to a polynucleotide that does not naturally occur in that way in the genome of the host cell or that a polypeptide or protein is not naturally produced in that manner by that cell. A heterologous nucleic acid sequence is a nucleic acid that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. For example, a promoter operably linked to a native structural gene is from a species different from that from which the structural gene is derived, or, if from the same species, one or both are substantially modified from their original form. A heterologous protein may originate from a foreign species or, if from the same species, is substantially modified from its original form by deliberate human intervention. That is, heterologous protein expression involves expression of a protein that is not naturally expressed in that way in the host cell. The term “heterologous expression” refers to the expression of heterologous nucleic acids in a host cell. The expression of heterologous proteins in eukaryotic host cell systems such as yeast are well known to those of skill in the art. A polynucleotide comprising a nucleic acid sequence of a gene encoding a certain protein or enzyme with a specific activity can be expressed in such a eukaryotic system. In some embodiments, transformed/transfected cells may be employed as expression systems for the expression of the enzymes. Expression of heterologous proteins in yeast is well known. Sherman, F., et al., *Methods in Yeast Genetics*, (1986), published by Cold Spring Harbor Laboratory, is a well-recognized work describing the various methods avail-

able to express proteins in yeast. Two widely utilized yeasts are *Saccharomyces cerevisiae* and *Pichia pastoris*. Vectors, strains, and protocols for expression in *Saccharomyces* and *Pichia* are known in the art and available from commercial suppliers (e.g., Invitrogen). Suitable vectors usually have expression control sequences, such as promoters, including 3-phosphoglycerate kinase or alcohol oxidase, and an origin of replication, termination sequences and the like as desired.

[0065] As used herein “promoter” is a DNA sequence that directs the transcription of a (structural) gene or other (part of) nucleic acid sequence. Suitably, a promoter is located in the 5'-region of a gene, proximal to the transcriptional start site of a (structural) gene. Promoter sequences may be constitutive, inducible or repressible. In an embodiment there is no (external) inducer needed.

[0066] The term “vector” as used herein, includes reference to an autosomal expression vector and to an integration vector used for integration into the chromosome.

[0067] The term “expression vector” refers to a DNA molecule, linear or circular, that comprises a segment encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and may optionally include one or more origins of replication, one or more selectable markers, an enhancer, a polyadenylation signal, and the like. Expression vectors are generally derived from plasmid or viral DNA, or may contain elements of both. In particular an expression vector comprises a nucleic acid sequence that comprises in the 5' to 3' direction and operably linked: (a) a yeast-recognized transcription and translation initiation region, (b) a coding sequence for a polypeptide of interest, and (c) a yeast-recognized transcription and translation termination region.

[0068] “Plasmid” refers to autonomously replicating extrachromosomal DNA which is not integrated into a microorganism's genome and is usually circular in nature.

[0069] An “integration vector” refers to a DNA molecule, linear or circular, that can be incorporated in a microorganism's genome and provides for stable inheritance of a gene encoding a polypeptide of interest. The integration vector generally comprises one or more segments comprising a gene sequence encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and one or more segments that drive the incorporation of the gene of interest into the genome of the target cell, usually by the process of homologous recombination. Typically, the integration vector will be one which can be transferred into the target cell, but which has a replicon which is nonfunctional in that organism. Integration of the segment comprising the gene of interest may be selected if an appropriate marker is included within that segment.

[0070] By “host cell” is herein understood a cell, such as a yeast cell, that is to be transformed with one or more nucleic acid sequences encoding for one or more heterologous proteins, to construe a transformed cell, also referred to as a recombinant cell. For example, the transformed cell may contain a vector and may support the replication and/or expression of the vector.

[0071] “Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide into a host cell, irrespective of the method used for the

insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome. “Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide (i.e. an exogenous nucleic acid sequence) into a host cell, irrespective of the method used for the insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome.

[0072] By “constitutive expression” and “constitutively expressing” is herein understood that there is a continuous transcription of a nucleic acid sequence. That is, the nucleic acid sequence is transcribed in an ongoing manner. Constitutively expressed genes are always “on”.

[0073] By “anaerobic constitutive expression” is herein understood that nucleic acid sequence is constitutively expressed in an organism under anaerobic conditions. That is, under anaerobic conditions the nucleic acid sequence is transcribed in an ongoing manner, i.e. under such anaerobic conditions the genes are always “on”.

[0074] By “disruption” is herein understood any disruption of activity, including, but not limited to, deletion, mutation and reduction of the affinity of the disrupted gene and expression of RNA complementary to such disrupted gene. It includes all nucleic acid modifications such as nucleotide deletions or substitutions, gene knock-outs, and other actions which affect the translation or transcription of the corresponding polypeptide and/or which affect the enzymatic (specific) activity, its substrate specificity, and/or or stability. It also includes modifications that may be targeted on the coding sequence or on the promoter of the gene. A gene disruptant is a cell that has one or more disruptions of the respective gene. Native to yeast herein is understood as that the gene is present in the yeast cell before the disruption.

[0075] The term “encoding” has the same meaning as “coding for”. Thus, by way of example, “one or more genes encoding a protein having activity X” has the same meaning as “one or more genes coding for a protein having activity X”.

[0076] As far as genes or nucleic acid sequences encoding a protein or an enzyme are concerned, the phrase “a nucleic acid sequence encoding a X”, respectively “one or more nucleic acid sequences encoding a X”, wherein X denotes a certain protein or (enzymatic) activity, has the same meaning as “a nucleic acid sequence encoding a protein having X activity”, respectively “one or more nucleic acid sequences encoding a protein having X activity”. Thus, by way of example, “one or more nucleic acid sequences encoding a transketolase” has the same meaning as “one or more nucleic acid sequences encoding a protein having transketolase activity”. As indicated above, the article “a” refers to “one or more”.

[0077] By a “redox sink” is herein understood a metabolic pathway that, overall, consumes or oxidizes NADH into NAD⁺ and/or prevents or reduces the consumption or reduction of NAD⁺ into NADH. A non-native metabolic pathway is a metabolic pathway that does not occur in the corresponding wild-type cell. Hence, a non-native metabolic pathway forming a redox sink is preferably a non-native metabolic pathway that, as compared to a corresponding wild-type yeast cell, increases NADH consumption and/or

decreases NAD⁺ consumption. By increasing NADH consumption and/or decreasing NAD⁺ consumption advantageously an (additional) non-native redox sink can be created within the cell.

[0078] The abbreviation “NADH” refers to reduced, hydrogenated form of nicotinamide adenine dinucleotide. The abbreviation “NAD⁺” refers to the oxidized form of nicotinamide adenine dinucleotide. Nicotinamide adenine dinucleotide may act as a so-called cofactor, assisting in biochemical reactions and/or transformations in a cell. [074] “NADH dependent” or “NAD⁺ dependent” is herein equivalent to NADH specific and “NADH dependency” or “NAD⁺ dependency” is herein equivalent to NADH specificity.

[0079] By a “NADH dependent” or “NAD⁺ dependent” enzyme is herein understood an enzyme that is exclusively depended on NADH/NAD⁺ as a co-factor or that is predominantly dependent on NADH/NAD⁺ as a cofactor, i.e. as contrasted to other types of co-factor. By an “exclusive NADH/NAD⁺ dependent” enzyme is herein understood an enzyme that has an absolute requirement for NADH/NAD⁺ over NADPH/NADP⁺. That is, it is only active when NADH/NAD⁺ is applied as cofactor. By a “predominantly NADH/NDA⁺-dependent” enzyme is herein understood an enzyme that has a higher specificity and/or a higher catalytic efficiency for NADH/NAD⁺ as a cofactor than for NADPH/NADP⁺ as a cofactor.

[0080] The enzyme’s specificity characteristics can be described by the formula:

$$1 < K_m \text{NADP}^+ / K_m \text{NAD}^+ < \infty \text{ (infinity)}$$

wherein K_m is the so-called Michaelis constant.

[0081] For a predominantly NADH-dependent enzyme, preferably $K_m \text{NADP}^+ / K_m \text{NAD}^+$ is between 1 and 1000, between 1 and 500, between 1 and 200, between 1 and 100, between 1 and 50, between 1 and 10, between 5 and 100, between 5 and 50, between 5 and 20 or between 5 and 10.

[0082] The K_m ’s for the enzymes herein can be determined as enzyme specific, for NAD⁺ and NADP⁺ respectively, using know analysis techniques, calculations and protocols. These are described for instance in Lodish et al., Molecular Cell Biology 6th Edition, Ed. Freeman, pages 80 and 81, e.g. FIG. 3-22. For an predominantly NADH-dependent enzyme, preferably the ratio of the catalytic efficiency for NADPH/NADP⁺ as a cofactor $(k_{cat}/K_m)^{\text{NADP}^+}$ to NADH/NAD⁺ as cofactor $(k_{cat}/K_m)^{\text{NAD}^+}$, i.e. the catalytic efficiency ratio $(k_{cat}/K_m)^{\text{NADP}^+} : (k_{cat}/K_m)^{\text{NAD}^+}$, is more than 1:1, more preferably equal to or more than 2:1, still more preferably equal to or more than 5:1, even more preferably equal to or more than 10:1, yet even more preferably equal to or more than 20:1, even still more preferably equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the predominantly NADH-dependent enzyme may have a catalytic efficiency ratio $(k_{cat}/K_m)^{\text{NADP}^+} : (k_{cat}/K_m)^{\text{NAD}^+}$ of equal to or less than 1,000,000,000:1 (i.e. $1 \cdot 10^9:1$).

The Yeast Cell

[0083] The recombinant yeast cell is preferably a yeast cell, or derived from a yeast cell, from the genus of Saccharomycetaceae or the genus of Schizosaccharomycetaceae. That is, preferably the host cell from which the recombinant yeast cell is derived is a yeast cell from the genus of Saccharomycetaceae or the genus of Schizosaccharomycetaceae.

[0084] Examples of suitable yeast cells include *Saccharomyces*, such as *Saccharomyces cerevisiae*, *Saccharomyces eubayanus*, *Saccharomyces jurei*, *Saccharomyces pastorianus*, *Saccharomyces beticus*, *Saccharomyces fermentati*, *Saccharomyces paradoxus*, *Saccharomyces uvarum* and *Saccharomyces bayanus*.

[0085] Examples of suitable yeast cells further include *Schizosaccharomyces*, such as *Schizosaccharomyces pombe*, *Schizosaccharomyces japonicus*, *Schizosaccharomyces octosporus* and *Schizosaccharomyces cryophilus*.

[0086] Other exemplary yeasts include *Torulaspora* such as *Torulaspora delbrueckii*; *Kluyveromyces* such as *Kluyveromyces marxianus*; *Pichia* such as *Pichia stipitis*, *Pichia pastoris* or *Pichia angusta*; *Zygosaccharomyces* such as *Zygosaccharomyces bailii*; *Brettanomyces* such as *Brettanomyces inter medius*; *Brettanomyces bruxellensis*, *Brettanomyces anomalus*, *Brettanomyces custersianus*, *Brettanomyces naardenensis*, *Brettanomyces nanus*, *Dekkera bruxellensis* and *Dekkera anomala*; *Metschnikowia*, *Issatchenkia*, such as *Issatchenkia orientalis*, *Kloeckera* such as *Kloeckera apiculata*; and *Aureobasidium* such as *Aureobasidium pullulans*.

[0087] The yeast cell is preferably a yeast cell of the genus *Schizosaccharomyces*, herein also referred to as a *Schizosaccharomyces* yeast cell, or a yeast cell of the genus *Saccharomyces*, herein also referred to as a *Saccharomyces* yeast cell. More preferably the yeast cell is a yeast cell derived from a yeast cell of the species *Saccharomyces cerevisiae*, herein also referred to as a *Saccharomyces cerevisiae* yeast cell. That is, preferably the host cell from which the recombinant yeast cell is derived is a yeast cell from the species *Saccharomyces cerevisiae*.

[0088] Preferably the yeast cell is an industrial yeast cell. The living environments of yeast cells in industrial processes are significantly different from that in the laboratory. Industrial yeast cells must be able to perform well under multiple environmental conditions which may vary during the process. Such variations include changes in nutrient sources, pH, ethanol concentration, temperature, oxygen concentration, etc., which together have potential impact on the cellular growth and ethanol production of the yeast cell. An industrial yeast cell can be understood to refer to a yeast cell that, when compared to a laboratory counterpart, has a more robust performance. That is, when compared to a laboratory counterpart, the industrial yeast cell shows less variation in performance when one or more environmental conditions selected from the group of nutrient sources, pH, ethanol concentration, temperature, oxygen concentration, are varied during fermentation. Preferably, the yeast cell is constructed on the basis of an industrial yeast cell as a host, wherein the construction is conducted as described herein-after. Examples of industrial yeast cells are Ethanol Red® (Fermentis) Fermiol® (DSM) and Thermosacc® (Lallemand).

[0089] The recombinant yeast cell described herein may be derived from any host cell capable of producing a fermentation product. Preferably the host cell is a yeast cell, more preferably an industrial yeast cell as described herein above. Preferably the yeast cell described herein is derived from a host cell having the ability to produce ethanol.

[0090] The yeast cell described herein may be derived from the host cell through any technique known by one skilled in the art to be suitable therefore. Such techniques may include any one or more of mutagenesis, recombinant DNA technology (including, but not limited to, CRISPR-CAS techniques), selective and/or adaptive evolution, mating, cell fusion, and/or cytoduction between yeast strains. Suitably the one or more desired genes are incorporated in the yeast cell by a combination of one or more of the above techniques.

[0091] The recombinant yeast cells according to the invention are preferably inhibitor tolerant, i.e. they can withstand common inhibitors at the level that they typically have with common pretreatment and hydrolysis conditions, so that the recombinant yeast cells can find broad application, i.e. it has high applicability for different feedstock, different pretreatment methods and different hydrolysis conditions. In an embodiment the recombinant yeast cell is inhibitor tolerant. Inhibitor tolerance is resistance to inhibiting compounds. The presence and level of inhibitory compounds in lignocellulose may vary widely with variation of feedstock, pretreatment method hydrolysis process. Examples of categories of inhibitors are carboxylic acids, furans and/or phenolic compounds. Examples of carboxylic acids are lactic acid, acetic acid or formic acid. Examples of furans are furfural and hydroxy-methylfurfural. Examples of phenolic compounds are vanillin, syringic acid, ferulic acid and coumaric acid. The typical amounts of inhibitors are for carboxylic acids: several grams per liter, up to 20 grams per liter or more, depending on the feedstock, the pretreatment and the hydrolysis conditions. For furans: several hundreds of milligrams per liter up to several grams per liter, depending on the feedstock, the pretreatment and the hydrolysis conditions. For phenolics: several tens of milligrams per liter, up to a gram per liter, depending on the feedstock, the pretreatment and the hydrolysis conditions.

[0092] In an embodiment, the recombinant yeast cell is a cell that is naturally capable of alcoholic fermentation, preferably, anaerobic alcoholic fermentation. A recombinant yeast cell preferably has a high tolerance to ethanol, a high tolerance to low pH (i.e. capable of growth at a pH lower than about 5, about 4, about 3, or about 2.5) and towards organic and/or a high tolerance to elevated temperatures.

Glycerol Transporter

[0093] The recombinant yeast comprises a nucleic acid sequence encoding a protein having glycerol transporter activity. By glycerol transporter activity is herein understood the activity of transporting glycerol across the membrane of the recombinant yeast cell.

[0094] The glycerol transporter can suitably allow the recombinant yeast cell to transport glycerol, that is externally available in the medium (e.g. from the backset in corn

mash) or secreted after internal cellular synthesis, into the cell. Subsequently the recombinant yeast cell can convert the glycerol to ethanol with help of for example a suitable glycerol dehydrogenase and/or a suitable dihydroxyacetone kinase.

[0095] The protein having glycerol transporter activity is herein also referred to as “glycerol transporter enzyme”, “glycerol transporter protein” or simply “glycerol transporter”. The protein having glycerol transporter activity is also abbreviated herein as “GT”. Preferences for the glycerol transporter protein and the nucleic sequences encoding for such are as described in WO2015/028583, incorporated herein by reference.

[0096] As explained in detail below, preferably the recombinant yeast cell comprises glycerol-proton symporter activity. That is, preferably the protein having glycerol transporter activity is a protein having glycerol-proton symporter activity and preferably the nucleic acid sequence encoding a protein having glycerol transporter activity is a nucleic acid sequence encoding a protein having glycerol-proton symporter activity. Preferably the recombinant yeast cell functionally expresses such nucleic acid sequence encoding for a protein having glycerol-proton symporter activity. Still more preferably the recombinant yeast cell comprises a heterologous glucose-tolerant gene encoding a protein with glycerol-proton symporter activity, suitably allowing the recombinant yeast cell to functionally express such a protein.

[0097] Nowadays many glycerol transporters (such as channels, facilitators and symporters) have been identified, characterized biochemically and the corresponding genes have been cloned (Neves et al., “*Yeast orthologues associated with glycerol transport and metabolism*”, (2004), FEMS Yeast Res. Vol. 5, pages 51-62 and Neves et al “*New insights on glycerol transport in *Saccharomyces cerevisiae**”, (2004), FEBS Letters 565 (2004) 160-162), both incorporated herein by reference).

[0098] As explained in WO2015/028583, in case of *S. cerevisiae*, four different genes have been implicated with glycerol transport (see Table 4 of WO2015/028583, incorporated herein by reference): FPS1, GUP1, GUP2 and STL1.

[0099] In WO2015/028583, five alternative proteins were selected, heterologous to *S. cerevisiae*. These glycerol transporters, either being a facilitator, a channel, a uniporter or a symporter, were shown, upon overexpression in strains having anaerobic glycerol and acetic acid conversion pathways, to result in improved glycerol uptake activity in yeast cells.

[0100] Preferably the recombinant yeast cell in the present invention functionally expresses one or more nucleic acid sequence(s) and/or corresponding proteins as listed in Table 2 below, or a functional homologue of any of these having a nucleic acid sequence, respectively amino acid sequence, with at least 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 98 or 99% nucleic acid sequence identity, respectively amino acid sequence identity, therewith. Specific examples of suitable protein(s) having glycerol transporter activity and their sequence identity with the protein first listed are summarized in Table 3(a) to 3(e).

TABLE 2

Preferred glycerol transporter proteins and genes encoding for such:			
Species	Gene Name	Protein Sequence	DNA Sequence
<i>Schizosaccharomyces pombe</i>	SPAC977.17	SEQ ID NO: 1	—
<i>Plasmodium falciparum</i>	CAC88373	SEQ ID NO: 2	—
<i>Danio rerio</i>	AQP9	SEQ ID NO: 3	—
	NP_001171215		
<i>Xenopus tropicalis</i>	NP_001087946	SEQ ID NO: 4	SEQ ID NO: 6
<i>Zygosaccharomyces rouxii</i>	ZYRO0E01210p	SEQ ID NO: 5	SEQ ID NO: 7

TABLE 3

Description	Identity (%)	Accession number
a) SPAC977.17 from <i>Schizosaccharomyces pombe</i> and proteins with a similar amino acid sequence identity.		
MIP water channel (predicted) [<i>Schizosaccharomyces pombe</i> 972h-] > sp Q9P7T9.1 Y17H	100	NP_592788.1
hypothetical protein TDEL_0C00200 [<i>Torulaspora delbrueckii</i>] > emb CCE90909.1	54	XP_003680120.1
KLTH0B00440p [<i>Lachancea thermotolerans</i>] > emb CAR21378.1	55	XP_002551817.1
YFL054C-like protein [<i>Saccharomyces cerevisiae</i> × <i>Saccharomyces kudriavzevii</i> VIN7]	55	EHN02649.1
BN860_19284g1_1 [<i>Zygosaccharomyces bailii</i> CLIB 213]	53	CDF87997.1
aquaglyceroporin, putative [<i>Aspergillus fumigatus</i> A1163]	38	EDP47128.1
b) CAC88373 from <i>Plasmodium falciparum</i> and proteins with a similar amino acid sequence identity.		
aquaglyceroporin [<i>Plasmodium falciparum</i> 3D7] > pdb 3C02	100	XP_001348009.1
putative aquaglyceroporin [<i>Plasmodium vivax</i>]	66	ADD39010.1
glycerol uptake facilitator protein [<i>Pseudomonas viridiflava</i>] > gb EKN47076.1	38	WP_004881860.1
glycerol transporter [<i>Pseudomonas syringae</i> group genomosp. 3] > gb EGH58018.1	38	WP_007248745.1
glycerol uptake facilitator GlpF [<i>Caloramator australicus</i>] > emb CCC57748.1	33	WP_008907471.1
PREDICTED: aquaporin-10 [<i>Gorilla gorilla gorilla</i>]	33	XP_004026862.1
c) AQP9 (NP_001171215) from <i>Danio rerio</i> and proteins with a similar amino acid sequence identity.		
aquaporin-9 [<i>Danio rerio</i>] > gb ACB10576.1 aquaporin-9b [<i>Danio rerio</i>]	100	NP_001171215.1
aquaglyceroporin [<i>Osmerus mordax</i>]	73	ABG24574.1
PREDICTED: aquaporin-9 [<i>Gorilla gorilla gorilla</i>]	51	XP_004056310.1
aquaporin 3 (Gill blood group) [<i>Xenopus laevis</i>] > emb CAA10517.1	52	NP_001081876.1
aquaporin-3 [<i>Xenopus laevis</i>]		
d) NP_001087946 from <i>Xenopus tropicalis</i> and proteins with a similar amino acid sequence identity.		
aquaporin 3 (Gill blood group) [<i>Xenopus laevis</i>] > gb AAH77577.1	100	NP_001087946.1
aquaporin 3 (Gill blood group) [<i>Xenopus (Silurana) tropicalis</i>] > emb CAJ82459.1	94	NP_001016845.1
PREDICTED: aquaporin-3 [<i>Gorilla gorilla gorilla</i>]	78	XP_004047976.1
PREDICTED: aquaporin-3-like [<i>Takifugu rubripes</i>]	65	XP_003975282.1
aquaporin [<i>Sparus aurata</i>]	50	AAR13054.1
e) ZYRO0E01210p from <i>Zygosaccharomyces rouxii</i> and proteins with a similar amino acid sequence identity.		
ZYRO0E01210p [<i>Zygosaccharomyces rouxii</i>] > emb CAR30744.1	100	XP_002498999.1
ZYRO0E01210p [<i>Zygosaccharomyces rouxii</i>]		
BN860_18536g1_1 [<i>Zygosaccharomyces bailii</i> CLIB 213]	82	CDF87965.1
hypothetical protein TDEL_0B07220 [<i>Torulaspora delbrueckii</i>] > emb CCE90851.1 hypothetical protein TDEL_0B07220 [<i>Torulaspora delbrueckii</i>]	66	XP_003680062.1
Stl1p [<i>Saccharomyces cerevisiae</i> S288c] > sp P39932.2	66	NP_010825.3
sugar transporter STL1 [<i>Candida albicans</i> WO-1]	64	EEQ46634.1
monosaccharide transporter [<i>Cryptococcus gattii</i> WM276] > gb ADV21423.1	45	XP_003193210.1

[0101] As indicated above, the recombinant yeast preferably comprises glycerol-proton symporter activity. That is, the recombinant yeast preferably comprises one or more nucleic acid sequences encoding for a heterologous protein having glycerol-proton symporter activity. A preferred example of such glycerol-proton symporter proteins are STL1 proteins. STL1 proteins belong to the category of “Sugar Transporter-Like proteins” and can be subject to glucose-induced inactivation. STL1 proteins are glycerol proton symporters of the plasma membrane, they can be strongly but transiently induced when cells are subjected to osmotic shock. Preferably the glycerol transporter protein is a STL1 protein and preferably the nucleic acid sequence encoding for the protein having glycerol transporter activity is a nucleic acid sequence encoding for a STL1 protein. Preferably the recombinant yeast cell comprises a nucleic acid sequence encoding a protein having glycerol transporter activity, wherein the protein having glycerol transporter activity is a STL1 protein, most preferably a STL1 protein derived from *Zygosaccharomyces rouxii*.

[0102] More preferably the recombinant yeast comprises one or more glucose-tolerant nucleic acid sequence(s) encoding one or more heterologous protein(s) with glycerol-proton symporter activity.

[0103] Preferably the protein having glycerol transporter activity comprises or consists of:

[0104] an amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5; or

[0105] a functional homologue of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5; or

[0106] a functional homologue of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5.

[0107] More preferably the recombinant yeast comprises, respectively functionally expresses, a nucleic acid sequence encoding for a protein comprising an amino acid sequence represented by SEQ ID NO: 1, 2, 3, 4 or 5, most preferably represented by SEQ ID NO: 5.

[0108] The proteins having an amino acid sequence of SEQ ID NO: 3 or SEQ ID NO: 5 and functional homologues thereof are most preferred.

[0109] Preferable the nucleic acid sequence encoding the protein having glycerol transporter activity comprises or consists of:

[0110] a nucleic acid sequence of SEQ ID NO: 6 or SEQ ID NO: 7; or

[0111] a functional homologue of SEQ ID NO: 6 or SEQ ID NO: 7, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 6 or SEQ ID NO: 7; or

[0112] a functional homologue of SEQ ID NO: 6 or SEQ ID NO: 7, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 6 or SEQ ID NO: 7, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 6 or SEQ ID NO: 7.

[0113] More preferably the recombinant yeast comprises a glucose-tolerant STL gene, most preferably a STL1 protein derived from *Zygosaccharomyces rouxii*.

[0114] The nucleic acid sequence (e.g. the gene) encoding for the glycerol transporter protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2015/028583, herein incorporated by reference.

GT Promoter

[0115] The recombinant yeast cell functionally expresses, a nucleic acid sequence encoding a protein having glycerol transporter activity, wherein the expression of the nucleic acid sequence encoding the protein having glycerol transporter activity is under control of a promoter (the “GT promoter”), which GT promoter has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more. Herewith is suitably meant that the expression of the glycerol transporter (“GT”) is at least a factor 2 higher under anaerobic conditions than under aerobic conditions. The above can alternatively be phrased as the recombinant yeast cell functionally expressing one or more nucleic acid sequences encoding for a glycerol transporter, wherein the glycerol transporter is under control of a promoter (the “GT promoter”) which has a GT expression ratio anaerobic/aerobic of 2 or more.

[0116] The GT promoter can suitably be operably linked to the nucleic acid sequence encoding the protein having glycerol transporter activity. Preferably, the GT promoter is located in the 5'-region of a glycerol transporter gene, more preferably it is located proximal to the transcriptional start site of a glycerol transporter gene. As indicated above, the glycerol transporter gene is preferably a glycerol-proton symporter gene and more preferably an STL1 gene.

[0117] Preferably the GT promoter is ROX1 repressed. ROX1 is herein Heme-dependent repressor of hypoxic gene(s); that mediates aerobic transcriptional repression of hypoxia induced genes such as COX5b and CYC7; the repressor function is regulated through decreased promoter occupancy in response to oxidative stress; and contains an HMG domain that is responsible for DNA bending activity; involved in the hyperosmotic stress resistance. ROX1 is regulated by oxygen.

[0118] Without wishing to be limited by any kind of theory it is believed that the regulation of ROX1 may function as follows: According to Kwast et al., “*Genomic Analysis of Anaerobically induced genes in Saccharomyces cerevisiae: Functional roles of ROX1 and other factors in mediating the anoxic response*”, (2002), Journal of bacteriology vol 184, no 1 pages 250-265, herein incorporated by reference: “Although Rox1 functions in an O₂-independent manner, its expression is oxygen (heme) dependent, activated by the heme-dependent transcription factor Hap1 [19]. Thus, as oxygen levels fall to those that limit heme biosynthesis [20], ROX1 is no longer transcribed [21], its protein levels fall [22], and the genes it regulates are de-repressed”. Further details and suitable motifs are provided by Keng, T. (1992), “HAP1 and ROX1 form a regulatory pathway in the repression of HEM13 transcription in *Saccharomyces cerevisiae*”, Mol. Cell. Biol. 12: pages 2616-2623, and Ter Kinde and de Steensma, “*A microarray-assisted screen for potential Hap1 and Rox1 target genes in Saccharomyces cerevisiae*”, (2002), Yeast 19: pages 825-840, incorporated herein by reference.

[0119] Preferably, the GT promoter comprises a ROX1 binding motif. The GT promoter may suitably comprise one or more ROX1 binding motif(s).

[0120] Preferably the GT promoter can comprise in its nucleic acid sequence a copy, or one or more copies, of the motif NNNATTGTTNNN (illustrated by SEQ ID NO: 8), wherein “N” represents a nucleic acid chosen from the group consisting of Adenine (A), Guanine (G), Cytosine (C) and Thymine (T).

[0121] More preferably, the GT promoter comprises or consists of a nucleic acid sequence that is identical to the nucleic acid sequence of the, preferably native, promoter of a gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C and SML1, or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. The reference to a native promoter is herein to the promoter that is native to the host cell.

[0122] Most preferably the recombinant yeast cell according to the invention is a recombinant yeast cell, wherein the GT promoter is the, preferably native, promoter of a gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5 and HEM13, or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. Preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell and preferably the GT promoter is a native promoter of a *Saccharomyces cerevisiae* gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C and SML1.

[0123] In addition or in the alternative, the GT promoter preferably comprises in its nucleic acid sequence one or more copies of the motifs: TCGTTYAG and/or AAAAAT-TGTTGA (illustrated by SEQ ID NO: 9), wherein “Y” represents C or T.

[0124] The GT promoter can also comprise or consist of a nucleic acid sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a DAN, TIR or PAU gene.

[0125] Preferably, the GT promoter comprises or consists of a nucleic acid sequence that is the same as that of the, preferably native, promoter of a gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4 or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. The reference to a native promoter is herein to the promoter that is native to the host cell.

[0126] Preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell and preferably the GT promoter is a native promoter of a *Saccharomyces cerevisiae* gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4.

[0127] More suitably, the GT promoter can comprise or consist of a sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, and YLL025W or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith.

[0128] More preferably the recombinant yeast cell according to the invention is a recombinant yeast cell, wherein the GT promoter is the, preferably native, promoter of a, preferably native, gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5 and HEM13, or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith.

[0129] Most preferably the recombinant yeast cell is a recombinant yeast cell, wherein the GT promoter is the native promoter of ANB1, DAN1 or HEM13 of *Saccharomyces cerevisiae*.

[0130] The promoter is herein also simply abbreviated respectively as ANB1 promoter, DAN1 promoter and HEM13 promoter.

[0131] The nucleic acid sequence of the *S. cerevisiae* ANB1 promoter is illustrated in SEQ ID NO: 10. The nucleic acid sequence of the *S. cerevisiae* DAN1 promoter is illustrated in SEQ ID NO: 11. The nucleic acid sequence of the *S. cerevisiae* HEM13 promoter is illustrated in SEQ ID NO: 12.

[0132] Preferable the GT promoter comprises or consists of:

[0133] a nucleic acid sequence of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12; or

[0134] a functional homologue of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12; or

[0135] a functional homologue of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12.

[0136] The GT promoter can also be a synthetic oligonucleotide. That is, the GT promoter may be a product of artificial oligonucleotide synthesis. Artificial oligonucleotide synthesis is a method in synthetic biology that is used to create artificial oligonucleotides, such as genes, in the laboratory. Commercial gene synthesis services are now available from numerous companies worldwide, some of which have built their business model around this task. Current gene synthesis approaches are most often based on a combination of organic chemistry and molecular biological techniques and entire genes may be synthesized “de novo”, without the need for precursor template DNA.

[0137] The GT promoter preferably has a GT expression ratio anaerobic/aerobic of 2 or more, preferably of 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. That is, the GT promoter preferably has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more, preferably of 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. Preferably the expression of the glycerol transporter enzyme (“GT”) is thus at least a factor 2, at least a factor 3, at least a factor 4, at least a factor 5, at least a factor 6, at least a factor 7, at least a factor 8, at least a factor 9, at least a factor 10, at least a factor 20 or at least a factor 50, higher under anaerobic conditions than under aerobic conditions.

[0138] There is no upper limit, and the GT promoter can be a GT promoter that allows the promoted glycerol transporter gene to be expressed only at anaerobic conditions and not at aerobic conditions. That is, preferably the recombinant yeast cell is a recombinant yeast cell, wherein the GT promoter enables expression only during anaerobic conditions.

[0139] For practical reasons a GT expression ratio anaerobic/aerobic in the range from equal to or more than 2 to equal to or less than 10×10 (i.e. 10^{10}) or to or less than 10×4 (i.e. 10^4) can be considered.

[0140] As indicated above, “Expression” herein refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

[0141] The GT expression ratio can for example be determined by measuring the amount of glycerol transporter (GT) protein of cells grown under aerobic and anaerobic conditions. The amount of GT protein can be determined by proteomics.

[0142] It is also possible to determine the level or glycerol transporter (GT) expression ratio by measuring the glycerol transporter (GT) activity of cells grown under aerobic and anaerobic conditions, e.g. in a cell-free extract.

[0143] In addition or in the alternative to the above, the level or GT expression ratio can be determined by measuring the transcription level (e.g. as amount of mRNA) of the Glycerol transporter gene of cells grown under aerobic and anaerobic conditions. The skilled person knows how to determine translation levels using methods commonly known in the art, e.g. Q-PCR, real-time PCR, northern blot, RNA-seq.

[0144] The GT promoter advantageously enables higher expression of the glycerol transporter during anaerobic conditions than under aerobic conditions. In the process according to the invention, the recombinant yeast cell preferably expresses the glycerol transporter, where the amount of the glycerol transporter expressed under anaerobic conditions is a multiplication factor higher than the amount of glycerol transporter expressed under aerobic conditions and wherein this multiplication factor is preferably 2 or more, more preferably 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more.

Glycerol Dehydrogenase

[0145] The recombinant yeast cell also functionally expresses a nucleic acid sequence encoding a protein having glycerol dehydrogenase activity.

[0146] The recombinant yeast cell may comprise a NAD⁺ dependent glycerol dehydrogenase (EC 1.1.1.6) and/or a NADP⁺ dependent glycerol dehydrogenase (EC 1.1.1.72). That is, the recombinant yeast cell may comprise a nucleic acid sequence encoding a protein having NAD⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.6) and/or a nucleic acid sequence encoding a protein having NADP⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.72).

[0147] Preferably the protein having glycerol dehydrogenase activity is a protein having NAD⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.6) and preferably the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a protein having NAD⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.6). Such protein may be from bacterial origin or for instance from fungal origin. An example is gldA from *E. coli*.

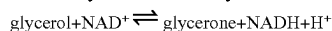
[0148] In an alternative or additional embodiment, a NADP⁺ dependent glycerol dehydrogenase can be present (EC 1.1.1.72).

[0149] A protein having glycerol dehydrogenase activity is herein also referred to as “glycerol dehydrogenase protein”, “glycerol dehydrogenase enzyme” or simply as “glyc-

erol dehydrogenase". In analogy thereto a protein having NAD⁺ dependent glycerol dehydrogenase activity is herein also referred to as "NAD⁺ dependent glycerol dehydrogenase protein", "NAD⁺ dependent glycerol dehydrogenase enzyme" or simply as "NAD⁺ dependent glycerol dehydrogenase". The glycerol dehydrogenase is abbreviated as GLD.

[0150] Preferences for a glycerol dehydrogenase and the nucleic sequences encoding for such are as described in WO2015028582, incorporated herein by reference.

[0151] NAD⁺ dependent glycerol dehydrogenase (EC 1.1.1.6) is an enzyme that catalyzes the chemical reaction:



[0152] Thus, the two substrates of this enzyme are glycerol and NAD⁺, whereas its three products are glycerone, NADH, and H⁺. Glyceron and dihydroxyacetone are herein synonyms.

[0153] This glycerol dehydrogenase enzyme belongs to the family of oxidoreductases, specifically those acting on the CH—OH group of donor with NAD⁺ or NADP⁺ as acceptor. The systematic name of this enzyme class is glycerol:NAD⁺ 2-oxidoreductase. Other names in common use include glycerin dehydrogenase, and NAD⁺-dependent glycerol dehydrogenase. This enzyme participates in glycerolipid metabolism. A glycerol dehydrogenase protein may be further defined by its amino acid sequence. Likewise a glycerol dehydrogenase protein may be further defined by a nucleotide sequence encoding the glycerol dehydrogenase protein. As explained in detail above under definitions, a certain glycerol dehydrogenase protein that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the glycerol dehydrogenase protein.

[0154] Preferably the nucleic acid sequence encoding the protein having glycerol dehydrogenase activity is a heterologous nucleic acid sequence. Preferably the protein having glycerol dehydrogenase activity is a heterologous protein having NAD⁺ dependent glycerol dehydrogenase activity.

[0155] If the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase, the recombinant yeast cell preferably further comprises suitable co-factors to enhance the activity of the glycerol dehydrogenase. For example, the recombinant yeast cell may comprise zinc, zinc ions or zinc salts and/or one or more pathways to include such in the cell.

[0156] Suitable examples of heterologous proteins having glycerol dehydrogenase activity include the glycerol dehydrogenase proteins of respectively *Klebsiella pneumoniae*, *Enterococcus aerogenes*, *Yersinia aldovae*, and *Escherichia coli*. Their amino acid sequences of such proteins have been illustrated respectively by SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 and SEQ ID NO: 16.

[0157] A preferred glycerol dehydrogenase protein is the glycerol dehydrogenase protein encoded by the *gldA* gene from *E. coli*. SEQ ID NO: 16 shows the amino acid sequence of this preferred NAD⁺ dependent glycerol dehydrogenase protein, encoded by the *gldA* gene from *E. coli*. The nucleic acid sequence of the *gldA* gene of *E. coli* is illustrated by SEQ ID NO: 17.

[0158] The recombinant yeast cell therefore most preferably comprises a heterologous nucleotide sequence encoding a protein having NAD⁺ dependent glycerol dehydroge-

nase activity (E.C. 1.1.1.6) derived from *E. coli*, optionally codon-optimized for the host cell, as exemplified by the nucleic acid sequence shown in SEQ ID NO:17.

[0159] Preferably the protein having glycerol dehydrogenase activity thus comprises or consists of:

[0160] an amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16; or

[0161] a functional homologue of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16; or

[0162] a functional homologue of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16.

The protein having an amino acid sequence of SEQ ID NO: 16 and functional homologues thereof are most preferred.

[0163] Preferable the nucleic acid sequence encoding the protein having glycerol dehydrogenase activity comprises or consists of:

[0164] a nucleic acid sequence of SEQ ID NO: 17; or

[0165] a functional homologue of SEQ ID NO: 17, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 17; or

[0166] a functional homologue of SEQ ID NO: 17, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 17, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 17.

[0167] The nucleic acid sequence (e.g. the gene) encoding for the glycerol dehydrogenase protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2015/028583, herein incorporated by reference.

[0168] Further examples of suitable glycerol dehydrogenases are listed in table 4(a) to 4(d). At the top of each table the *gldA* used in the examples and that is BLASTED is mentioned.

TABLE 4

Description	Identity (%)	Accession number
(a): BLAST Query - gldA from <i>Escherichia coli</i>		
glycerol dehydrogenase, NAD [<i>Escherichia coli</i> str. K-12 substr. MG1655]	100	NP_418380.4
glycerol dehydrogenase [<i>Escherichia coli</i> O127:H6 str. E2348/69]	99	YP_002331714.1
glycerol dehydrogenase [<i>Citrobacter youngae</i>]	94	WP_006686227.1
glycerol dehydrogenase [<i>Citrobacter freundii</i>]	92	WP_003840533.1
(b): BLAST Query - gldA from <i>Klebsiella pneumoniae</i>		
glycerol dehydrogenase [<i>Klebsiella pneumoniae</i> 342]	100	YP_002236495.1
glycerol dehydrogenase [<i>Citrobacter freundii</i>]	93	WP_003024745.1
Glycerol dehydrogenase (EC 1.1.1.6) [<i>Enterobacter aerogenes</i> EA1509E]	92	YP_004590977.1
glycerol dehydrogenase [<i>Escherichia coli</i>]	91	WP_016241524.1
glycerol dehydrogenase [<i>Enterococcus aerogenes</i>]	87	See examples herein strains with CAS15
glycerol dehydrogenase [<i>Yersinia aldovae</i>]	74	WP_004701845.1
glycerol dehydrogenase [<i>Enterobacteriaceae bacterium</i> LSJC7]	61	WP_017375113.1
glycerol dehydrogenase [<i>Citrobacter youngae</i>]	60	WP_006686227.1
(c): BLAST Query - gldA from <i>Enterococcus aerogenes</i>		
glycerol dehydrogenase [<i>Enterobacter aerogenes</i> KCTC 2190]	100	YP_004591726.1
Glycerol dehydrogenase (EC 1.1.1.6) [<i>Enterobacter aerogenes</i> EA1509E]	99	YP_007390021.1
glycerol dehydrogenase [<i>Klebsiella pneumoniae</i>]	92	WP_004203683.1
glycerol dehydrogenase [<i>Escherichia coli</i>]	88	WP_001322519.1
		See examples herein strains with CAS13
glycerol dehydrogenase [<i>Enterobacter cloacae</i> subsp. <i>cloacae</i> ATCC 13047]	87	YP_003615506.1
(d): BLAST Query - gldA from <i>Yersinia aldovae</i>		
glycerol dehydrogenase [<i>Yersinia aldovae</i>]	100	WP_004701845.1
glycerol dehydrogenase [<i>Yersinia intermedia</i>]	95	WP_005189747.1
glycerol dehydrogenase [<i>Serratia liquefaciens</i> ATCC 27592]	81	YP_008232202.1
glycerol dehydrogenase [<i>Escherichia coli</i>]	76	WP_016241524.1
		See examples herein strains with CAS13.
hypothetical protein EAE_03845 [<i>Enterobacter aerogenes</i> KCTC 2190]	75	YP_004590977.1
glycerol dehydrogenase [<i>Aeromonas hydrophila</i>]	65	WP_017410769.1

Dihydroxyacetone Kinase

[0169] The recombinant yeast cell further functionally expresses a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity.

[0170] A protein having dihydroxyacetone kinase activity is herein also referred to as “dihydroxyacetone kinase protein”, “dihydroxyacetone kinase enzyme” or simply as “dihydroxyacetone kinase”. The dihydroxyacetone kinase is abbreviated herein as DAK.

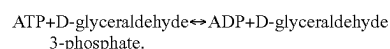
[0171] Preferences for a dihydroxyacetone kinase and the nucleic sequences encoding for such are as described in WO2015028582, incorporated herein by reference.

[0172] The protein having dihydroxy kinase activity may suitably belong to the enzyme categories of E.C. 2.7.1.28 and/or E.C. 2.7.1.29. The recombinant yeast cell thus suitably functionally expresses a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 and/or E.C. 2.7.1.29).

[0173] A dihydroxyacetone kinase is preferably herein understood as an enzyme that catalyzes the chemical reaction (EC 2.7.1.29):



and/or the chemical reaction (EC 2.7.1.28):



[0174] Other names in common use for a dihydroxyacetone kinase include glycerone kinase, ATP:glycerone phosphotransferase and (phosphorylating) acetol kinase. It is further understood that glycerone and dihydroxyacetone are the same molecule. A dihydroxyacetone kinase protein may be further defined by its amino acid sequence. Likewise a dihydroxyacetone kinase protein may be further defined by a nucleotide sequence encoding the dihydroxyacetone kinase protein. As explained in detail above under definitions, a certain dihydroxyacetone kinase protein that is defined by a nucleotide sequence encoding the enzyme,

includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the dihydroxyacetone kinase protein.

[0175] Preferably the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a native protein having dihydroxyacetone kinase activity. More preferably, the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is a native nucleic acid sequence.

[0176] Yeast comprises two native isozymes of dihydroxyacetone kinase (DAK1 and DAK2). These native dihydroxyacetone kinase enzymes are preferred according to the invention. Preferably the host cell is a *Saccharomyces cerevisiae* cell and preferably the above native dihydroxyacetone kinase enzymes are the native dihydroxyacetone kinase enzymes of a *Saccharomyces cerevisiae* yeast cell. The amino acid sequences of the native dihydroxyacetone kinase proteins of *Saccharomyces cerevisiae*, DAK1 and DAK2, have been illustrated respectively by SEQ ID NO: 18 and SEQ ID NO: 19.

[0177] It is also possible for the recombinant yeast cell to functionally express a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity, where the nucleic acid sequence is a heterologous nucleic acid sequence. In an embodiment the recombinant yeast cell comprises a heterologous gene encoding a dihydroxyacetone kinase. Suitable heterologous genes include the genes encoding dihydroxyacetone kinases from *Saccharomyces kudriavzevii*, *Zygosaccharomyces bailii*, *Kluyveromyces fragilis*, *Candida glabrata*, *Yarrowia lipolytica*, *Klebsiella pneumoniae*, *Enterobacter aerogenes*, *Escherichia coli*, *Yarrowia lipolytica*, *Schizosaccharomyces pombe*, *Botryotinia fuckeliana*, and *Exophiala dermatitidis*. Preferred heterologous proteins having dihydroxyacetone kinase activity include those derived from respectively *Klebsiella pneumoniae*, *Yarrowia lipolytica* and *Schizosaccharomyces pombe*, as illustrated respectively by SEQ ID NO: 20, SEQ ID NO: 21 and SEQ ID NO: 22.

[0178] The recombinant yeast cell may or may not comprise a genetic modification that causes overexpression of a dihydroxyacetone kinase, for example by overexpression of a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity. The nucleotide sequence encoding the dihydroxyacetone kinase may be native or heterologous to the cell. Nucleic acid sequences that may be used for overexpression of dihydroxyacetone kinase in the cells of the invention are for example the dihydroxyacetone kinase genes from *S. cerevisiae* (DAK1) and (DAK2) as e.g. described by Molin et al., "Dihydroxy-acetone kinases in *Saccharomyces cerevisiae* are involved in detoxification of dihydroxyacetone" (2003), J. Biol. Chem., vol. 278: pages 1415-1423, incorporated herein by reference.

[0179] The native nucleic acid sequences encoding dihydroxyacetone kinase proteins in *Saccharomyces cerevisiae*, DAK1 and DAK2, have been illustrated respectively by SEQ ID NO: 23 and SEQ ID NO: 24.

[0180] Preferably the recombinant yeast cell does comprise a genetic modification that increases the specific activity of any dihydroxyacetone kinase in the cell. For example, the recombinant yeast cell may comprise one or more native and/or heterologous nucleic acid sequence encoding one or more native and/or heterologous dihydroxyacetone kinase protein(s), such as DAK1 and/or DAK2, that is/are overexpressed. A native dihydroxyacetone kinase, such as DAK1

and/or DAK2, may for example be overexpressed via one or more genetic modifications resulting in more copies of the gene encoding for the dihydroxy acetone kinase than present in the non-genetically modified cell, and/or a non-native promoter may be applied.

[0181] Preferably the recombinant yeast cell is a recombinant yeast cell, wherein the expression of the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is under control of a promoter. The promoter can for example be a promoter that is native to another gene in the host cell.

[0182] For overexpression of the nucleotide sequence encoding the dihydroxyacetone kinase, the nucleotide sequence (to be overexpressed) can also be placed in an expression construct wherein it is operably linked to suitable expression regulatory regions/sequences to ensure overexpression of the dihydroxyacetone kinase enzyme upon transformation of the expression construct into the host cell of the invention (see above). Suitable promoters for (over)expression of the nucleotide sequence coding for the enzyme having dihydroxyacetone kinase activity include promoters that are preferably insensitive to catabolite (glucose) repression and/or that are active under anaerobic conditions. A dihydroxyacetone kinase that is overexpressed, is preferably overexpressed by at least a factor 1.1, 1.2, 1.5, 2, 5, 10 or 20 as compared to a strain which is genetically identical except for the genetic modification causing the overexpression. Preferably, the dihydroxyacetone kinase is overexpressed under anaerobic conditions by at least a factor 1.1, 1.2, 1.5, 2, 5, 10 or 20 as compared to a strain which is genetically identical except for the genetic modification causing the overexpression. It is to be understood that these levels of overexpression may apply to the steady state level of the enzyme's activity (specific activity in the cell), the steady state level of the enzyme's protein as well as to the steady state level of the transcript coding for the enzyme in the cell. Overexpression of the nucleotide sequence in the host cell produces a specific dihydroxyacetone kinase activity of at least 0.002, 0.005, 0.01, 0.02 or 0.05 U min⁻¹ (mg protein)⁻¹, determined in cell extracts of the transformed host cells at 30° C. as described e.g. in the Examples of WO2013/081456.

[0183] A most preferred dihydroxyacetone kinase protein is the dihydroxyacetone kinase protein encoded by the DAK1 gene from *Saccharomyces cerevisiae*. SEQ ID NO: 18 shows the amino acid sequence of a suitable dihydroxyacetone kinase protein, encoded by the DAK1 gene from *Saccharomyces cerevisiae*. SEQ ID NO: 23 illustrates the nucleic acid sequence of the DAK1 gene itself.

[0184] If the recombinant yeast cell comprises one or more overexpressed nucleic acid sequences encoding for a dihydroxyacetone kinase, the recombinant yeast cell therefore most preferably comprises one or more overexpressed nucleotide sequence encoding a dihydroxyacetone kinase derived from *Saccharomyces cerevisiae*, as exemplified by the nucleic acid sequence shown in SEQ ID NO: 23.

[0185] In a preferred embodiment the dihydroxy acetone kinase is encoded by an endogenous gene, e.g. a DAK1 gene, which endogenous gene is preferably placed under control of a constitutive promoter.

[0186] Preferably the protein having dihydroxy acetone kinase activity thus comprises or consists of:

[0187] an amino acid sequence of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22; or

[0188] a functional homologue of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22; or

[0189] a functional homologue of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22.

The protein having an amino acid sequence of SEQ ID NO: 18 and functional homologues thereof are most preferred.

[0190] Preferable the nucleic acid sequence encoding the protein having dihydroxy acetone kinase activity comprises or consists of:

[0191] a nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24; or

[0192] a functional homologue of SEQ ID NO: 23 or SEQ ID NO: 24, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24; or

[0193] a functional homologue of SEQ ID NO: 23 or SEQ ID NO: 24, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24.

[0194] The nucleic acid sequence (e.g. the gene) encoding for the dihydroxy acetone kinase protein may suitably be incorporated in the genome of the recombinant yeast cell.

[0195] Examples of suitable dihydroxyacetone kinases are listed in table 5(a) to 5(d). At the top of each table the DAK's used in the examples and that is BLASTED is mentioned.

TABLE 5

Description	Identity (%)	Accession number
(a): BLAST Query - DAK1 from <i>Saccharomyces cerevisiae</i>		
Dak1p [<i>Saccharomyces cerevisiae</i> S288c]	100	NP_013641.1
dihydroxyacetone kinase [<i>Saccharomyces cerevisiae</i> YJM789]	99	EDN64325.1
DAK1-like protein [<i>Saccharomyces kudriavzevii</i> IFO 1802]	95	EJT44075.1
ZYBA0S11-03576g1_1 [<i>Zygosaccharomyces bailii</i> CLIB 213]	77	CDF91470.1
hypothetical protein [<i>Kluyveromyces lactis</i> NRRL Y-1140]	70	XP_451751.1
hypothetical protein [<i>Candida glabrata</i> CBS 138]	63	XP_449263.1
Dak2p [<i>Saccharomyces cerevisiae</i> S288c]	44	NP_116602.1
DAK1 [<i>Yarrowia lipolytica</i>]	41	See examples herein strains with CAS23
(b): BLAST Query - dhaK from <i>Klebsiella pneumoniae</i>		
dihydroxyacetone kinase subunit DhaK [<i>Klebsiella pneumoniae</i> 342]	100	YP_002236493.1
dihydroxyacetone kinase subunit K [<i>Klebsiella pneumoniae</i>]	99	WP_004149886.1
dihydroxyacetone kinase subunit K [<i>Enterobacter aerogenes</i>]	96	WP_020077889.1
dihydroxyacetone kinase subunit DhaK [<i>Escherichia coli</i> IAI39]	88	YP_002407536.1
dihydroxyacetone kinase, DhaK subunit [<i>Escherichia coli</i>]	87	WP_001398949.1
(c): BLAST Query - DAK1 from <i>Yarrowia lipolytica</i>		
YALI0F09273p [<i>Yarrowia lipolytica</i>]	100	XP_505199.1
dihydroxyacetone kinase [<i>Schizosaccharomyces pombe</i>]	46	AAC83220.1
dihydroxyacetone kinase Dak1 [<i>Schizosaccharomyces pombe</i> 972h-]	45	NP_593241.1

TABLE 5-continued

Description	Identity (%)	Accession number
dihydroxyacetone kinase [<i>Saccharomyces cerevisiae</i> RM11-1a]	44	EDV12567.1
Dak2p [<i>Saccharomyces cerevisiae</i> JAY291]	44	EEU04233.1
BN860_19306g1_1 [<i>Zygosaccharomyces bailii</i> CLIB 213]	44	CDF87998.1
Dak1p [<i>Saccharomyces cerevisiae</i> CEN.PK113-7D]	42	EIW08612.1
		See examples herein strains with CAS21
(d): BLAST Query - DAK1 from <i>Schizosaccharomyces pombe</i>		
dihydroxyacetone kinase Dak1 [<i>Schizosaccharomyces pombe</i> 972h-]	100	NP_593241.1
putative dihydroxyacetone kinase protein [<i>Botryotinia fuckeliana</i> BeDW1]	48	EMR88164.1
Dihydroxyacetone kinase 1 [<i>Fusarium oxysporum</i> f. sp. <i>cubense</i> race 1]	48	ENH64704.1
Dak1p [<i>Saccharomyces cerevisiae</i> CEN.PK113-7D]	46	EIW08612.1
Dak2p [<i>Saccharomyces cerevisiae</i> JAY291]	44	EEU04233.1
dihydroxyacetone kinase [<i>Exophiala dermatitidis</i> NIH/UT8656]	42	EHY55064.1

Redox Sink

[0196] Preferably the recombinant yeast cell can further comprise one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink.

[0197] For example, these one or more genetic modifications can be one or more genetic modifications for the functional expression of one or more, optionally heterologous, nucleic acid sequences encoding for one or more NAD⁺/NADH dependent proteins that function in a metabolic pathway to convert NADH to NAD⁺. Several examples of such metabolic pathways exist, as illustrated further below.

[0198] For example, the “one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink” can be chosen from the group consisting of:

[0199] a) one or more genetic modifications comprising or consisting of:

[0200] a nucleic acid sequence encoding a protein comprising phosphoketolase activity (EC 4.1.2.9 or EC 4.1.2.22, PKL); and/or

[0201] a nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8); and/or

[0202] a nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

and/or

[0203] b) one or more genetic modifications comprising or consisting of:

[0204] a nucleic acid sequence encoding for a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or

[0205] a nucleic acid sequences encoding for a protein having phosphoribulokinase (PRK) activity; and

[0206] optionally, a nucleic acid sequence encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity. and/or

[0207] c) one or more genetic modifications comprising or consisting of:

[0208] a nucleic acid sequence encoding a protein comprising NADH dependent acetylating acetaldehyde dehydrogenase activity.

[0209] For example, WO2014/081803 describes a recombinant microorganism expressing a heterologous phosphoketolase, phosphotransacetylase or acetate kinase and bifunctional acetaldehyde-alcohol dehydrogenase, incorporated herein by reference; and WO2015/148272 describes a recombinant *S. cerevisiae* strain expressing a heterologous phosphoketolase, phosphotransacetylase and acetylating acetaldehyde dehydrogenase, incorporated herein by reference. Further WO2018172328A1 describes a recombinant cell that may comprise one or more (heterologous) genes coding for an enzyme having phosphoketolase activity. The phosphoketolase (PKL) routes described in WO2014/081803, WO2015/148272 and WO2018172328A1, all incorporated herein by reference, provide preferred metabolic pathways to convert NADH to NAD⁺ and the NADH dependent phosphoketolase described therein is a preferred NADH dependent protein for application in the current invention.

Rubisco

[0210] As indicated above, the recombinant yeast cell may or may not functionally express one or more heterologous nucleic acid sequences encoding for ribulose-1,5-phosphate carboxylase/oxygenase (EC4.1.1.39; Rubisco), and optionally one or more molecular chaperones for Rubisco.

[0211] More preferably the recombinant yeast cell functionally expresses:

[0212] a heterologous nucleic acid sequence encoding a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or

[0213] a heterologous nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity; and/or

[0214] optionally one or more heterologous nucleic acid sequence encoding one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0215] The protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity is herein also referred to as “ribulose-1,5-biphosphate carboxylase oxygenase”, “ribulose-1,5-biphosphate carboxylase oxygenase protein”, “ribulose-1,5-biphosphate carboxylase oxygenase enzyme”, “Rubisco enzyme”, “Rubisco protein” or simply “Rubisco”. A ribulose-1,5-biphosphate carboxylase oxygenase may be further defined by its amino acid sequence. Likewise a ribulose-1,5-biphosphate carboxylase oxygenase may be further defined by a nucleotide sequence encoding the ribulose-1,5-biphosphate carboxylase oxygenase. As explained in detail above under definitions, a certain ribulose-1,5-biphosphate carboxylase oxygenase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridizing to such nucleotide sequence encoding the ribulose-1,5-biphosphate carboxylase oxygenase. Preferences for the Rubisco protein and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0216] The Rubisco protein may suitably be selected from the group of eukaryotic and prokaryotic Rubisco proteins. The Rubisco protein is preferably from a non-phototrophic organism. For example, the Rubisco protein may be from a chemolithoautotrophic microorganism. Good results have been achieved with a bacterial Rubisco protein. Preferably, the Rubisco protein originates from a *Thiobacillus*, in particular, *Thiobacillus denitrificans*, which is chemolithoautotrophic.

[0217] The Rubisco protein may be a single-subunit Rubisco protein or a Rubisco protein having more than one subunit. Preferably the Rubisco protein is a single-subunit Rubisco protein. Good results have been obtained with a Rubisco protein that is a so-called form-II Rubisco protein. Especially good results were achieved with a Rubisco protein encoded by a *cbbM* gene, also referred to as *CbbM*.

[0218] A preferred Rubisco protein is the Rubisco protein encoded by the *cbbM* gene from *Thiobacillus denitrificans*. SEQ ID NO: 25 shows the amino acid sequence of a suitable Rubisco protein, encoded by the *cbbM* gene from *Thiobacillus denitrificans*. SEQ ID NO: 26 illustrates the nucleic acid sequence of the *cbbM* gene from *Thiobacillus denitrificans*, codon optimized for *S. cerevisiae*.

[0219] Preferably the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity thus comprises or consists of:

[0220] an amino acid sequence of SEQ ID NO: 25; or

[0221] a functional homologue of SEQ ID NO: 25, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 25; or

[0222] a functional homologue of SEQ ID NO: 25, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 25, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 25.

[0223] Preferable the nucleic acid sequence encoding the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity comprises or consists of:

[0224] a nucleic acid sequence of SEQ ID NO: 26; or

[0225] a functional homologue of SEQ ID NO: 26, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 26; or

[0226] a functional homologue of SEQ ID NO: 26, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 26, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 26.

[0227] Examples of other suitable Rubisco polypeptides and their origin are given in Table 1 of WO2014/129898, incorporated herein by reference, and in Table 6 below, with reference to the sequence identity with the amino acid sequence of SEQ ID NO:25.

[0228] The nucleic acid sequence (e.g. the gene) encoding for the ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898 and by the article of Guadalupe-Medina et al., “Carbon dioxide fixation by Calvin-Cycle enzymes improves ethanol yield in yeast”, published in Biotechnol, Biofuels, 2013, vol. 6, p. 125, both herein incorporated by reference.

TABLE 6

Natural Rubisco polypeptides suitable for expression		
Source	Accession no.	MAX ID (%)
<i>Thiobacillus denitrificans</i>	AAA99178.2	100
<i>Sideroxydans lithotrophicus</i> ES-1	YP_003522651.1	94
<i>Thiothrix nivea</i> DSM 5205	ZP_10101642.1	91
<i>Halothiobacillus neapolitanus</i> c2	YP_003262978.1	90
<i>Acidithiobacillus ferrooxidans</i> ATCC 53993	YP_002220242.1	88
<i>Rhodospirillum rubrum</i> T118	YP_522655.1	86
<i>Thiorhodococcus dresvii</i> AZ1	ZP_08824342.1	85
uncultured prokaryote	AGE14067.1	82

[0229] As indicated above, the Rubisco protein is suitably functionally expressed in the recombinant yeast cell, at least during use in a fermentation process.

[0230] The nucleic acid sequence encoding for the Rubisco protein can be present in one, two or more copies with the recombinant yeast cell. Without wishing to be bound by any kind of theory it is believed that the robustness of the recombinant yeast cell is best served when the nucleic acid sequence (e.g. the gene) encoding for the Rubisco protein is present in the recombinant yeast cell in less than 12 copies, more preferably less than 8 copies. Preferably the recombinant yeast cell therefore comprises in the range from equal to or more than 1 copy, more preferably equal to or

more than 2 copies, to equal to or less than 7 copies, more preferably equal to or less than 6 copies of a nucleic acid sequence (e.g. a gene) encoding for a Rubisco protein. The recombinant yeast cell may for example comprise one, two, three, four, five, six or seven copies of a nucleic acid sequence encoding for ribulose-1,5-bisphosphate carboxylase oxygenase (Rubisco).

[0231] To increase the likelihood that the Rubisco protein is expressed at sufficient levels and in active form in the transformed (recombinant) host cells of the invention, the nucleic acid sequence encoding the Rubisco protein and other proteins as described herein (see below), are preferably adapted to optimise their codon usage to that of the host cell in question. The adaptiveness of a nucleic acid sequence encoding an enzyme to the codon usage of a host cell may be expressed as codon adaptation index (CAI). The codon adaptation index is herein defined as a measurement of the relative adaptiveness of the codon usage of a gene towards the codon usage of highly expressed genes in a particular host cell or organism. The relative adaptiveness (w) of each codon is the ratio of the usage of each codon, to that of the most abundant codon for the same amino acid. The CAI index is defined as the geometric mean of these relative adaptiveness values. Non-synonymous codons and termination codons (dependent on genetic code) are excluded. CAI values range from 0 to 1, with higher values indicating a higher proportion of the most abundant codons (see Sharp and Li, “The codon adaptation index—a measure of directional synonymous codon usage bias, and its potential applications”, (1987), published in *Nucleic Acids Research* vol. 15, pages 1281-1295; also see: Jansen et al., “Revisiting the codon adaptation index from a whole-genome perspective: analyzing the relationship between gene expression and codon occurrence in yeast using a variety of models”, (2003), *Nucleic Acids Res.* Vol. 31(8), pages 2242-51). An adapted nucleic acid sequence preferably has a CAI of at least 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8 or 0.9. Preferably, the sequences have been codon optimized for expression in the fungal host cell in question, such as for example *Saccharomyces cerevisiae* cells.

[0232] Preferably the functionally expressed Rubisco protein has an activity, defined by the rate of ribulose-1,5-bisphosphate-dependent ^{14}C -bicarbonate incorporation by cell extracts of at least $1 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$, in particular an activity of at least $2 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$, more in particular an activity of at least $4 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$. The upper limit for the activity is not critical. In practice, the activity may be about $200 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$ or less, in particular $25 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$, more in particular $15 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$ or less, e.g. about $10 \text{ nmol} \cdot \text{min}^{-1} \cdot (\text{mg protein})^{-1}$ or less. The conditions for an assay for determining this Rubisco activity are as found in the Examples (e.g. Example 4) of WO2014/129898, incorporated herein by reference.

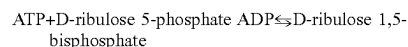
Phosphoribulokinase

[0233] Preferably recombinant yeast cell is also functionally expressing a heterologous nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity (EC2.7.1.19; PRK).

[0234] The protein having phosphoribulokinase (PRK) activity is herein also referred to as “phosphoribulokinase protein”, “phosphoribulokinase enzyme”, “phosphoribulokinase”, “PRK enzyme”, “PRK protein” or simply “PRK”.

Preferences for the PRK protein and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0235] A functionally expressed phosphoribulokinase (PRK, (EC 2.7.1.19)) according to the invention is capable of catalyzing the chemical reaction:



Thus, the two substrates of this enzyme are ATP and D-ribulose 5-phosphate; its two products are ADP and D-ribulose 1,5-bisphosphate.

[0236] The PRK protein belongs to the family of transferases, specifically those transferring phosphorus-containing groups (phosphotransferases) with an alcohol group as acceptor. The systematic name of this enzyme class is ATP:D-ribulose-5-phosphate 1-phosphotransferase. Other names in common use include phosphopentokinase, ribulose-5-phosphate kinase, phosphopentokinase, phosphoribulokinase (phosphorylating), 5-phosphoribulose kinase, ribulose phosphate kinase, PKK, PRuK, and PRK. The PRK enzyme participates in carbon fixation. A phosphoribulokinase (PRK) protein may be further defined by its amino acid sequence. Likewise a phosphoribulokinase (PRK) protein may be further defined by a nucleotide sequence encoding the phosphoribulokinase (PRK). As explained in detail above under definitions, a certain phosphoribulokinase (PRK) that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the phosphoribulokinase (PRK).

[0237] The PRK can be from a prokaryote or a eukaryote. Good results have been achieved with a PRK originating from a eukaryote. Preferably the PRK protein originates from a plant selected from Caryophyllales, in particular from *Amaranthaceae*, more in particular from *Spinacia*.

[0238] A preferred PRK protein is the PRK protein from *Spinacia*. SEQ ID NO: 27 shows the amino acid sequence of such PRK protein from *Spinacia*. SEQ ID NO: 28 illustrates the nucleic acid sequence of the prk gene from *Spinacia oleracea*—codon optimized for *S. cerevisiae*.

[0239] Preferably the protein having phosphoribulokinase (PRK) activity thus comprises or consists of:

[0240] an amino acid sequence of SEQ ID NO: 27; or

[0241] a functional homologue of SEQ ID NO: 27, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 27; or

[0242] a functional homologue of SEQ ID NO: 27, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 27, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 27.

[0243] Preferable the nucleic acid sequence encoding the protein having phosphoribulokinase (PRK) activity comprises or consists of:

[0244] a nucleic acid sequence of SEQ ID NO: 28; or

[0245] a functional homologue of SEQ ID NO: 28, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 28; or

[0246] a functional homologue of SEQ ID NO: 28, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 28, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 28.

[0247] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphoribulokinase (PRK) activity may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898, herein incorporated by reference.

[0248] Examples of suitable PRK polypeptides and their origin are given in Table 2 of WO2014/129898, incorporated herein by reference, and in Table 7 below, with reference to the sequence identity with the amino acid sequence of SEQ ID NO:27.

TABLE 7		
Natural PRK polypeptides suitable for expression with identity to PRK from <i>Spinacia</i>		
Source	Accession no.	MAX ID (%)
<i>Spinacia oleracea</i>	P09559.1	100
<i>Medicago truncatula</i>	XP_003612664.1	88
<i>Arabidopsis thaliana</i>	NP_174486.1	87
<i>Vitis vinifera</i>	XP_002263724.1	86
<i>Closterium peracerosum</i>	BAL03266.1	82
<i>Zea mays</i>	NP_001148258.1	78

[0249] The nucleic acid sequences encoding for the PRK protein may be under the control of a promoter (the “PRK promoter”) that enables higher expression under anaerobic conditions than under aerobic conditions. Examples of such promoters are described in WO2017/216136A1 and WO2018/228836, both herein incorporated by reference. More preferably such promoter has a PRK expression ratio anaerobic/aerobic of 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. Further preferences are as described in WO2018/228836, incorporated herein by reference.

Rubisco Chaperones

[0250] Optionally, the recombinant yeast cell further comprises one or more, preferably heterologous, nucleic acid sequences encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0251] Suitably such molecular chaperones are also referred herein as “chaperone protein”, “chaperonin” or simply “chaperone”. Preferences for the chaperones and the

nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0252] Preferably the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0253] Chaperonins are proteins that provide favorable conditions for the correct folding of other proteins, thus preventing aggregation. Newly made proteins usually must fold from a linear chain of amino acids into a three-dimensional form. Chaperonins belong to a large class of molecules that assist protein folding, called molecular chaperones. The energy to fold proteins is supplied by adenosine triphosphate (ATP). A review article about chaperones that is useful herein is written by Yébenes et al., “*Chaperonins: two rings for folding*” (2011), Trends in Biochemical Sciences, Vol. 36, No. 8, pages 424-432, incorporated herein by reference.

[0254] The chaperone or chaperones may be prokaryotic chaperones or eukaryotic chaperones. In addition, the chaperones may be homologous or heterologous. For example, the recombinant yeast cell may comprises one or more nucleic acid sequence encoding one or more homologous or heterologous, prokaryotic or eukaryotic, molecular chaperones, which—when expressed—are capable of functionally interacting with an enzyme in the recombinant yeast cell, in particular with at least one of Rubisco and PRK.

[0255] Suitably the chaperone or chaperones are derived from a bacterium, more preferably from *Escherichia*, in particular *E. coli*. Preferred chaperones are GroEL and GroEs from *E. coli*. Other preferred chaperones are chaperones from *Saccharomyces*, in particular *Saccharomyces cerevisiae* Hsp10 and Hsp60.

[0256] If the chaperones are naturally expressed in an organelle such as a mitochondrion (examples are Hsp60 and Hsp10 of *Saccharomyces cerevisiae*) relocation to the cytosol can be achieved e.g. by modifying the native signal sequence of the chaperonins. In eukaryotes the proteins Hsp60 and Hsp10 are structurally and functionally nearly identical to GroEL and GroES, respectively. Thus, it is contemplated that Hsp60 and Hsp10 from any recombinant yeast cell may serve as a chaperone for the Rubisco. This is described for example by Zeilstra-Ryalls et al., “*The universally conserved GroE (Hsp60) chaperonins*”, (1991), Annu Rev Microbiol. vol. 45, pages 301-325; and Horwich et al., “*Two Families of Chaperonin: Physiology and Mechanism*” (2007), Annu. Rev. Cell. Dev. Biol. Vol. 23, pages 115-145, both herewith incorporated by reference.

[0257] Good results have been achieved with a recombinant yeast cell comprising both the heterologous chaperones GroEL and GroES.

[0258] As an alternative to GroES a functional homologue of GroES may be present, in particular a functional homologue comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of GroES, respectively the amino sequence of SEQ ID NO: 31.

[0259] SEQ ID NO:31 provides a preferred translated protein sequence, based on GroES of *Escherichia coli*. SEQ ID NO: 32 provides a synthetic nucleic acid sequence, based

on GroES from *Escherichia coli*, codon optimized for expression in *Saccharomyces cerevisiae*.

[0260] Examples of suitable natural chaperones polypeptide homologous to GroES are given in Table 8.

TABLE 8

Natural chaperones homologous to GroES polypeptides suitable for expression
>gi 115388105 ref XP_001211558.1 :2-101 10 kDa heat shock protein, mitochondrial [<i>Aspergillus terreus</i> NIH2624]
>gi 116196854 ref XP_001224239.1 :1-102 conserved hypothetical protein [<i>Chaetomium globosum</i> CBS 148.51]
>gi 119175741 ref XP_001240050.1 :3-102 hypothetical protein CIMG_09671 [<i>Coccidioides immitis</i> RS]
>gi 119471607 ref XP_001258195.1 :12-111 chaperonin, putative [<i>Neosartorya fischeri</i> NRRL181]
>gi 121699818 ref XP_001268174.1 :8-106 chaperonin, putative [<i>Aspergillus clavatus</i> NRRL 1]
>gi 126274604 ref XP_001387607.1 :2-102 predicted protein [<i>Scheffersomyces stipitis</i> CBS 6054]
>gi 146417701 ref XP_001484818.1 :5-106 conserved hypothetical protein [<i>Meyerozyma guilliermondii</i> ATCC 6260]
>gi 154303611 ref XP_001552212.1 :1-102 10 kDa heat shock protein, mitochondrial [<i>Botryotinia fuckeliana</i> B05.10]
>gi 156049571 ref XP_001590752.1 :1-102 hypothetical protein SS1G_08492 [<i>Sclerotinia sclerotiorum</i> 1980]
>gi 156840987 ref XP_001643870.1 :1-103 hypothetical protein KpoL_495p10 [<i>Vanderwaltozyma polyspora</i> DSM 70924]
>gi 169608295 ref XP_001797567.1 :1-101 hypothetical protein SNOG_07218 [<i>Phaeosphaeria nodorum</i> SN15]
>gi 171688384 ref XP_001909132.1 :1-102 hypothetical protein [<i>Podospora anserina</i> S mat+]
>gi 189189366 ref XP_001931022.1 :71-168 10 kDa chaperonin [<i>Pyrenophora tritici-repentis</i> Pt-1C-BFP]
>gi 19075598 ref NP_588098.1 :1-102 mitochondrial heat shock protein Hsp10 (predicted) [<i>Schizosaccharomyces pombe</i> 972h-]
>gi 212530240 ref XP_002145277.1 :3-100 chaperonin, putative [<i>Talaromyces marneffeii</i> ATCC 18224]
>gi 212530242 ref XP_002145278.1 :3-95 chaperonin, putative [<i>Talaromyces marneffeii</i> ATCC 18224]
>gi 213404320 ref XP_002172932.1 :1-102 mitochondrial heat shock protein Hsp10 [<i>Schizosaccharomyces japonicus</i> yFS275]
>gi 225557301 gb EEH05587.1 :381-478 pre-mRNA polyadenylation factor flp1 [<i>Ajellomyces capsulatus</i> G186AR]
>gi 225684092 gb EEH22376.1 :3-100 heat shock protein [<i>Paracoccidioides brasiliensis</i> Pb03]
>gi 238490530 ref XP_002376502.1 :2-104 chaperonin, putative [<i>Aspergillus flavus</i> NRRL3357]
>gi 238878220 gb EEQ41858.1 :1-106 10 kDa heat shock protein, mitochondrial [<i>Candida albicans</i> WO-1]
>gi 240280207 gb EER43711.1 :426-523 pre-mRNA polyadenylation factor flp1 [<i>Ajellomyces capsulatus</i> H143]
>gi 241950445 ref XP_002417945.1 :1-103 10 kDa chaperonin, putative; 10 kDa heat shock protein mitochondrial (hsp10), putative [<i>Candida dubliniensis</i> CD36]
>gi 242819222 ref XP_002487273.1 :90-182 chaperonin, putative [<i>Talaromyces stipitatus</i> ATCC]
>gi 254566327 ref XP_002490274.1 :1-102 Putative protein of unknown function [<i>Komagataella pastoris</i> GS115]
>gi 254577241 ref XP_002494607.1 :1-103 ZYRO0A05434p [<i>Zygosaccharomyces rouxii</i>]
>gi 255717999 ref XP_002555280.1 :1-103 KLTH0G05588p [<i>Lachancea thermotolerans</i>]
>gi 255956581 ref XP_002569043.1 :2-101 Pc21g20560 [<i>Penicillium chrysogenum</i> Wisconsin 54-1255]
>gi 258572664 ref XP_002545094.1 :16-108 chaperonin GroS [<i>Uncinocarpus reesii</i> 1704]
>gi 261190594 ref XP_002621706.1 :3-100 chaperonin [<i>Ajellomyces dermatitidis</i> SLH14081]
>gi 295664909 ref XP_002793006.1 :3-100 10 kDa heat shock protein, mitochondrial [<i>Paracoccidioides</i> sp. 'lutzi'Pb01]
>gi 296412657 ref XP_002836039.1 :76-177 hypothetical protein [<i>Tuber melanosporum</i> Mel28]
>gi 302307854 ref NP_984626.2 :2-102 AEL235Wp [<i>Ashbya gossypii</i> ATCC 10895]
>gi 302894117 ref XP_003045939.1 :1-102 predicted protein [<i>Nectria haematococca</i> mpVI 77-13-4]
>gi 303318351 ref XP_003069175.1 :3-100 10 kDa heat shock protein, mitochondrial, putative [<i>Coccidioides posadasii</i> C735 delta SOWgp]
>gi 310795300 gb EFQ30761.1 :1-102 chaperonin 10 kDa subunit [<i>Glomerella graminicola</i> M1.001]
>gi 315053085 ref XP_003175916.1 :12-109 chaperonin GroS [<i>Arthroderma gypseum</i> CBS 118893]
>gi 317032114 ref XP_001394060.2 :334-433 heat shock protein [<i>Aspergillus niger</i> CBS 513.88]
>gi 317032116 ref XP_001394059.2 :2-101 heat shock protein [<i>Aspergillus niger</i> CBS 513.88]
>gi 320583288 gb EFW97503.1 :6-106 chaperonin, putative heat shock protein, putative [<i>Ogataea parapolymorpha</i> DL-1]
>gi 320591507 gb EFX03946.1 :1-102 heat shock protein [<i>Grosmannia clavigera</i> kw1407]
>gi 322700925 gb EFY92677.1 :1-102 chaperonin [<i>Metarhizium acridum</i> CQMa 102]
>gi 325096696 gb EGC50006.1 :409-506 pre-mRNA polyadenylation factor flp1 [<i>Ajellomyces capsulatus</i> H88]

TABLE 8-continued

Natural chaperones homologous to GroES polypeptides suitable for expression
>gi 326471604 gb EGD95613.1 :14-111 chaperonin 10 Kd subunit [<i>Trichophyton tonsurans</i> CBS112818]
>gi 327293056 reflXP_003231225.1 :3-100 chaperonin [<i>Trichophyton rubrum</i> CBS 118892]
>gi 330942654 reflXP_003306155.1 :37-136 hypothetical protein PTT_19211 [<i>Pyrenophora teres</i> f. <i>teres</i> 0-1]
>gi 336268042 reflXP_003348786.1 :47-147 hypothetical protein SMAC_01809 [<i>Sordaria macrospora</i> khell]
>gi 340519582 gb EGR49820.1 :1-109 predicted protein [<i>Trichoderma reesei</i> QM6a]
>gi 340960105 gb EGS21286.1 :3-103 putative mitochondrial 10 kDa heat shock protein [<i>Chaetomium thermophilum</i> var. <i>thermophilum</i> DSM 1495]
>gi 342883802 gb EGU84224.1 : 1-102 hypothetical protein FOXB_05181 [<i>Fusarium oxysporum</i> Fo5176]
>gi 344302342 gb EGW32647.1 :2-102 hypothetical protein SPAPADRAFT_61712 [<i>Spathaspora passalidarum</i> NRRL Y-27907]
>gi 345570750 gb EGX53571.1 :1-102 hypothetical protein AOL_s00006g437 [<i>Arthrobotrys oligospora</i> ATCC 24927]
>gi 346321154 gb EGX90754.1 :1-102 chaperonin [<i>Cordyceps militaris</i> CM01]
>gi 346970393 gb EGY13845.1 :1-102 heat shock protein [<i>Verticillium dahliae</i> VdLs. 17]
>gi 354548296 emb CCE45032.1 :1-106 hypothetical protein CPAR2_700360 [<i>Candida parapsilosis</i>]
>gi 358385052 gb EHK22649.1 :1-102 hypothetical protein TRIVIDRAFT_230640 [<i>Trichoderma virens</i> Gv 29-8]
>gi 358393422 gb EHK42823.1 :1-101 hypothetical protein TRIATDRAFT_258186 [<i>Trichoderma atroviride</i> IMI 206040]
>gi 361126733 gb EHK98722.1 :1-97 putative 10 kDa heat shock protein, mitochondrial [<i>Glare lozoyensis</i> 74030]
>gi 363753862 reflXP_003647147.1 :2-102 hypothetical protein Ecym_5593 [<i>Eremothecium cymbalariae</i> DBVPG#7215]
>gi 365758401 gb EHN00244.1 :1-106 Hsp10p [<i>Saccharomyces cerevisiae</i> x <i>Saccharomyces kudriavzevii</i> VIN7]
>gi 365987664 reflXP_003670663.1 :1-103 hypothetical protein NDAI_0F01010 [<i>Naumovozyma dairenensis</i> CBS 421]
>gi 366995125 reflXP_003677326.1 :1-103 hypothetical protein NCAS_0G00860 [<i>Naumovozyma castellii</i> CBS 4309]
>gi 366999797 reflXP_003684634.1 :1-103 hypothetical protein TPHA_0C00430 [<i>Tetrapisispora phaffii</i> CBS 4417]
>gi 367009030 reflXP_003679016.1 :1-103 hypothetical protein TDEL_0A04730 [<i>Torulaspora delbruekii</i>]
>gi 367023138 reflXP_003660854.1 :1-104 hypothetical protein MYCTH_59302 [<i>Myceliophthora thermophila</i> ATCC 42464]
>gi 367046344 reflXP_003653552.1 :1-102 hypothetical protein THITE_2116070 [<i>Thielavia terrestris</i> NRRL8126]
>gi 378726440 gb EHY52899.1 :9-109 chaperonin GroES [<i>Exophiala dermatitidis</i> NIH/UT8656]
>gi 380493977 emb CCF33483.1 :1-102 chaperonin 10 kDa subunit [<i>Colletotrichum higginsianu</i>]
>gi 385305728 gb EIF49680.1 :1-102 10 kDa heat shock mitochondrial [<i>Dekkera bruxellensis</i> AWR11499]
>gi 389628546 reflXP_003711926.1 :1-102 hsp10-like protein [<i>Magnaporthe oryzae</i> 70-15]
>gi 396462608 reflXP_003835915.1 :1-101 similar to 10 kDa heat shock protein [<i>Leptosphaeria maculans</i> JN3]
>gi 398392541 reflXP_003849730.1 :1-102 hypothetical protein MYCGRDRAFT_105721 [<i>Zymoseptoria tritici</i> IPO323]
>gi 400597723 gb EJP65453.1 :24-124 chaperonin 10 kDa subunit [<i>Beauveria bassiana</i> ARSEF 2860]
>gi 401623646 gb EJS41738.1 :1-106 hsp10p [<i>Saccharomyces arboricola</i> H-6]
>gi 401842164 gb EJT44422.1 :1-92 HSP10-like protein [<i>Saccharomyces kudriavzevii</i> IFO 1802]
>gi 402084027 gb EJT79045.1 :1-102 hsp10-like protein [<i>Gaeumannomyces graminis</i> var. <i>tritici</i>]
>gi 403215209 emb CCK69709.1 :1-104 hypothetical protein KNAG_0C06130 [<i>Kazachstania naganishii</i> CBS 8797]
>gi 406604629 emb CCH43969.1 :4-100 hypothetical protein BN7_3524 [<i>Wickerhamomyces ciferrii</i>]
>gi 406867021 gb EKD20060.1 :56-156 hypothetical protein MBM_02012 [<i>Marssonina brunnea</i> f.sp. 'multigermutubi' MB_m1]
>gi 407926227 gb EKG19196.1 :74-174 GroES-like protein [<i>Macrophomina phaseolina</i> MS6]
>gi 408398157 gb EKJ77291.1 :11-111 hypothetical protein FPSE_02566 [<i>Fusarium pseudograminearum</i> CS3096]
>gi 410082063 reflXP_003958610.1 :1-103 hypothetical protein KAFR_0H00660 [<i>Kazachstania africana</i> CBS2517]
>gi 425777664 gb EKV15823.1 :58-157 Chaperonin, putative [<i>Penicillium digitatum</i> Pd1]
>gi 440639680 gb ELR09599.1 :1-102 chaperonin GroES [<i>Geomyces destructans</i> 20631-21]
>gi 444323906 reflXP_004182593.1 :1-105 hypothetical protein TBLA_0J00760 [<i>Tetrapisisporablattae</i> CBS 6284]

TABLE 8-continued

Natural chaperones homologous to GroES polypeptides suitable for expression
>gi 448083208 ref XP_004195335.1 :2-101 Piso0_005888 [<i>Milleromyces farinosa</i> CBS 7064]
>gi 448087837 ref XP_004196425.1 :2-102 Piso0_005888 [<i>Milleromyces farinosa</i> CBS 7064]
>gi 448534948 ref XP_003870866.1 :1-106 Hsp10 protein [<i>Candida orthopsilosis</i> Co 90-125]
>gi 449295977 gb EMC91998.1 :1-102 hypothetical protein BAUCODRAFT_39148 [<i>Baudoinia compn</i>]
>gi 46123659 ref XP_386383.1 :3-103 hypothetical protein FG06207.1 [<i>Gibberella zeae</i> PH-1]
>gi 50289455 ref XP_447159.1 :1-103 hypothetical protein [<i>Candida glabrata</i> CBS 138]
>gi 50308731 ref XP_454370.1 :1-103 hypothetical protein [<i>Kluyveromyces lactis</i> NRRL Y-1140]
>gi 50411066 ref XP_457014.1 :1-106 DEHA2B01122p [<i>Debaryomyces hansenii</i> CBS767]
>gi 50545998 ref XP_500536.1 :1-102 YALI0B05610p [<i>Yarrowia lipolytica</i>]
>gi 51013895 gb AAT93241.1 :1-106 YOR020C [<i>Saccharomyces cerevisiae</i>]
>gi 6324594 ref NP_014663.1 :1-106 Hsp10p [<i>Saccharomyces cerevisiae</i> S288c]
>gi 67523953 ref XP_660036.1 :2-101 hypothetical protein AN2432.2 [<i>Aspergillus nidulans</i> FGSC A4]
>gi 70992219 ref XP_750958.1 :12-106 chaperonin [<i>Aspergillus fumigatus</i> Af293]
>gi 85079266 ref XP_956315.1 :1-104 hypothetical protein NCU04334 [<i>Neurospora crassa</i> OR74A]

[0261] As an alternative to GroEL a functional homologue of GroEL may be present, in particular a functional homologue comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of GroEL, respectively the amino sequence of SEQ ID NO: 29.

[0262] SEQ ID NO:29 provides a preferred translated protein sequence, based on GroEL of *Escherichia coli*. SEQ ID NO: 30 provides a synthetic nucleic acid sequence, based on GroEL from *Escherichia coli*, codon optimized for expression in *Saccharomyces cerevisiae*.

[0263] Suitable natural chaperones polypeptides homologous to GroEL are given in Table 9.

TABLE 9

Natural chaperones homologous to GroEL polypeptides suitable for expression
>gi 115443330 ref XP_001218472.1 heat shock protein 60, mitochondrial precursor [<i>Aspergillus terreus</i> NIH2624]
>gi 114188341 gb EAU30041.1 heat shock protein 60, mitochondrial precursor [<i>Aspergillus terreus</i> NIH2624]
>gi 119480793 ref XP_001260425.1 antigenic mitochondrial protein HSP60, putative [<i>Neosartorya fischeri</i> NRRL 181] >gi 119408579 gb EAW18528.1 antigenic mitochondrial protein HSP60, putative [<i>Neosartorya fischeri</i> NRRL 181]
>gi 126138730 ref XP_001385888.1 hypothetical protein PICST_90190 [<i>Scheffersomyces stipitis</i> CBS 6054] >gi 126093166 gb ABN67859.1 mitochondrial groEL-type heat shock protein [<i>Scheffersomyces stipitis</i> CBS 6054]
>gi 145246630 ref XP_001395564.1 heat shock protein 60 [<i>Aspergillus niger</i> CBS 513.88]
>gi 134080285 emb CAK46207.1 unnamed protein product [<i>Aspergillus niger</i>]
>gi 350636909 gb EHA25267.1 hypothetical protein ASPNIDRAFT_54001 [<i>Aspergillus niger</i> ATCC 1015]
>gi 146413148 ref XP_001482545.1 heat shock protein 60, mitochondrial precursor [<i>Meyerozyma guilliermondii</i> ATCC 6260]
>gi 154277022 ref XP_001539356.1 heat shock protein 60, mitochondrial precursor [<i>Ajellomyces capsulatus</i> NAm1] >gi 150414429 gb EDN09794.1 heat shock protein 60, mitochondrial precursor [<i>Ajellomyces capsulatus</i> NAm1]
>gi 154303540 ref XP_001552177.1 heat shock protein 60 [<i>Botryotinia fuckeliana</i> B05.10]
>gi 347840915 emb CCD55487.1 similar to heat shock protein 60 [<i>Botryotinia fuckeliana</i>]
>gi 156063938 ref XP_001597891.1 heat shock protein 60, mitochondrial precursor [<i>Sclerotinia sclerotiorum</i> 1980] >gi 154697421 gb EDN97159.1 heat shock protein 60, mitochondrial precursor [<i>Sclerotinia sclerotiorum</i> 1980 UF-70]
>gi 156844469 ref XP_001645297.1 hypothetical protein KpoL_1037p35 [<i>Vanderwaltozyma polyspora</i> DSM 70294] >gi 156115957 gb EDO17439.1 hypothetical protein KpoL_1037p35 [<i>Vanderwaltozyma polyspora</i> DSM 70294]
>gi 16416029 emb CAB91379.2 probable heat-shock protein hsp60 [<i>Neurospora crassa</i>]
>gi 350289516 gb EGZ70741.1 putative heat-shock protein hsp60 [<i>Neurospora tetrasperma</i> FGSC 2509]
>gi 169626377 ref XP_001806589.1 hypothetical protein SNOG_16475 [<i>Phaeosphaeria nodorum</i> SN15] >gi 111055053 gb EAT76173.1 hypothetical protein SNOG_16475 [<i>Phaeosphaeria nodorum</i> SN15]
>gi 169783766 ref XP_001826345.1 heat shock protein 60 [<i>Aspergillus oryzae</i> RIB40]
>gi 238493601 ref XP_002378037.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus flavus</i> NRRL3357] >gi 83775089 dbj BAE65212.1 unnamed protein product [<i>Aspergillus oryzae</i> RIB40] >gi 220696531 gb EED52873.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus flavus</i> NRRL3357] >gi 391869413 gb EIT78611.1 chaperonin, Cpn60/Hsp60p [<i>Aspergillus oryzae</i> 3.042]

TABLE 9-continued

Natural chaperones homologous to GroEL polypeptides suitable for expression
>gi 189190432 ref XP_001931555.1 heat shock protein 60, mitochondrial precursor [Pyrenophora tritici-repentis Pt-1C-BFP] >gi 187973161 gb EDU40660.1 heat shock protein 60, mitochondrial precursor [Pyrenophora tritici-repentis Pt-1C-BFP] >gi 190348913 gb EDK41467.2 heat shock protein 60, mitochondrial precursor [Meyerozyma guilliermondii ATCC 6260] >gi 225554633 gb EEH02929.1 hsp60-like protein [Ajellomyces capsulatus G186AR] >gi 238880068 gb EEQ43706.1 heat shock protein 60, mitochondrial precursor [Candida albicans WO-1] >gi 239613490 gb EEQ90477.1 chaperonin GroL [Ajellomyces dermatitidis ER-3] >gi 240276977 gb EER40487.1 hsp60-like protein [Ajellomyces capsulatus H143] >gi 241958890 ref XP_002422164.1 heat shock protein 60, mitochondrial precursor, putative [Candida dubliniensis CD36] >gi 223645509 emb CAX40168.1 heat shock protein 60, mitochondrial precursor, putative [Candida dubliniensis CD36] >gi 254572906 ref XP_002493562.1 Tetradecameric mitochondrial chaperonin [Komagataella pastoris GS115] >gi 238033361 emb CAY71383.1 Tetradecameric mitochondrial chaperonin [Komagataella pastoris GS115] >gi 254579947 ref XP_002495959.1 ZYRO0C07106p [Zygosaccharomyces rouxii] >gi 238938850 emb CAR27026.1 ZYRO0C07106p [Zygosaccharomyces rouxii] >gi 255712781 ref XP_002552673.1 KLTH0C10428p [Lachancea thermotolerans] >gi 238934052 emb CAR22235.1 KLTH0C10428p [Lachancea thermotolerans CBS 6340] >gi 255721795 ref XP_002545832.1 heat shock protein 60, mitochondrial precursor [Candida tropicalis MYA-3404] >gi 240136321 gb EER35874.1 heat shock protein 60, mitochondrial precursor [Candida tropicalis MYA-3404] >gi 255941288 ref XP_002561413.1 Pc16g11070 [Penicillium chrysogenum Wisconsin 54-1255] >gi 211586036 emb CAP93777.1 Pc16g11070 [Penicillium chrysogenum Wisconsin 54-1255] >gi 259148241 emb CAY81488.1 Hsp60p [Saccharomyces cerevisiae EC1118] >gi 260950325 ref XP_002619459.1 heat shock protein 60, mitochondrial precursor [Clavispora lusitanae ATCC 42720] >gi 238847031 gb EEQ36495.1 heat shock protein 60, mitochondrial precursor [Clavispora lusitanae ATCC 42720] >gi 261194577 ref XP_002623693.1 chaperonin GroL [Ajellomyces dermatitidis SLH14081] >gi 239588231 gb EEQ70874.1 chaperonin GroL [Ajellomyces dermatitidis SLH14081] >gi 327355067 gb EGE83924.1 chaperonin GroL [Ajellomyces dermatitidis ATCC 18188] >gi 296422271 ref XP_002840685.1 hypothetical protein [Tuber melanosporum Mel28] >gi 295636906 emb CAZ84876.1 unnamed protein product [Tuber melanosporum] >gi 296809035 ref XP_002844856.1 heat shock protein 60 [Arthroderma otae CBS 113480] >gi 238844339 gb EEQ34001.1 heat shock protein 60 [Arthroderma otae CBS 113480] >gi 302308696 ref NP_985702.2 AFR155Wp [Ashbya gossypii ATCC 10895] >gi 299790751 gb AAS53526.2 AFR155Wp [Ashbya gossypii ATCC 10895] >gi 374108933 gb AEY97839.1 FAFR155Wp [Ashbya gossypii FDAG1] >gi 302412525 ref XP_003004095.1 heat shock protein [Verticillium albo-atrum VaMs. 102] >gi 261356671 gb EEY19099.1 heat shock protein [Verticillium albo-atrum VaMs. 102] >gi 302505585 ref XP_003014499.1 hypothetical protein ARB_07061 [Arthroderma benhamiae CBS 112371] >gi 291178320 gb EFE34110.1 hypothetical protein ARB_07061 [Arthroderma benhamiae CBS 112371] >gi 302656385 ref XP_003019946.1 hypothetical protein TRV_05992 [Trichophyton verrucosum HKI 0517] >gi 291183723 gb EFE39322.1 hypothetical protein TRV_05992 [Trichophyton verrucosum HKI 0517] >gi 302915513 ref XP_003051567.1 predicted protein [Nectria haematococca mpVI 77-13-4] >gi 256732506 gb EEU45854.1 predicted protein [Nectria haematococca mpVI 77-13-4] >gi 310794550 gb EFQ30011.1 chaperonin GroL [Glomerella graminicola M1.001] >gi 315048491 ref XP_003173620.1 chaperonin GroL [Arthroderma gypseum CBS 118893] >gi 311341587 gb EFR00790.1 chaperonin GroL [Arthroderma gypseum CBS 118893] >gi 320580028 gb EFW94251.1 Tetradecameric mitochondrial chaperonin [Ogataea parapolyomorpha DL-1] >gi 320586014 gb EFW98693.1 heat shock protein mitochondrial precursor [Grossmannia clavigera kw1407] >gi 322692465 gb EFY84374.1 Heat shock protein 60 precursor (Antigen HIS-62) [Metarhizium acridum CQMa 102] >gi 322705285 gb EFY96872.1 Heat shock protein 60 (Antigen HIS-62) [Metarhizium anisopliae ARSEF 23] >gi 323303806 gb EGA57589.1 Hsp60p [Saccharomyces cerevisiae FostersB] >gi 323307999 gb EGA61254.1 Hsp60p [Saccharomyces cerevisiae FostersO] >gi 323332364 gb EGA73773.1 Hsp60p [Saccharomyces cerevisiae AWRI796] >gi 326468648 gb EGD92657.1 heat shock protein 60 [Trichophyton tonsurans CBS 112818] >gi 326479866 gb EGE03876.1 chaperonin GroL [Trichophyton equinum CBS 127.97] >gi 330915493 ref XP_003297052.1 hypothetical protein PTT_07333 [Pyrenophora teres f. teres 0-1] >gi 311330479 gb EFQ94847.1 hypothetical protein PTT_07333 [Pyrenophora teres f. teres 0-1] >gi 336271815 ref XP_003350665.1 hypothetical protein SMAC_02337 [Sordaria macrospora k-hell] >gi 380094827 emb CCC07329.1 unnamed protein product [Sordaria macrospora k-hell] >gi 336468236 gb EGO56399.1 hypothetical protein NEUTE1DRAFT_122948 [Neurospora tetrasperma FGSC 2508]

TABLE 9-continued

Natural chaperones homologous to GroEL polypeptides suitable for expression
>gi 340522598 gb EGR52831.1 hsp60 mitochondrial precursor-like protein [<i>Trichoderma reesei</i> QM6a]
>gi 341038907 gb EGS23899.1 mitochondrial heat shock protein 60-like protein [<i>Chaetomium thermophilum</i> var. <i>thermophilum</i> DSM 1495]
>gi 342886297 gb EGU86166.1 hypothetical protein FOXB_03302 [<i>Fusarium oxysporum</i> Fo5176]
>gi 344230084 gb EGV61969.1 chaperonin GroL [<i>Candida tenuis</i> ATCC 10573]
>gi 344303739 gb EGW33988.1 hypothetical protein SPAPADRAFT_59397 [<i>Spathaspora passalidarum</i> NRRL Y-27907]
>gi 345560428 gb EGX43553.1 hypothetical protein AOL_s00215g289 [<i>Arthrotrichum oligospora</i> ATCC 24927]
>gi 346323592 gb EGX93190.1 heat shock protein 60 (Antigen HIS-62) [<i>Cordyceps militaris</i> CM01]
>gi 346975286 gb EGY18738.1 heat shock protein [<i>Verticillium dahliae</i> VdLs.17]
>gi 354545932 emb CCE42661.1 hypothetical protein CPAR2_203040 [<i>Candida parapsilosis</i>]
>gi 358369894 dbj GAA86507.1 heat shock protein 60, mitochondrial precursor [<i>Aspergillus kawachii</i> IFO 4308]
>gi 358386867 gb EHK24462.1 hypothetical protein TRIVIDRAFT_79041 [<i>Trichoderma virens</i> Gv29-8]
>gi 358399658 gb EHK48995.1 hypothetical protein TRIATDRAFT_297734 [<i>Trichoderma atroviride</i> IMI 206040]
>gi 363750488 ref XP_003645461.1 hypothetical protein Ecym_3140 [<i>Eremothecium cymbalariae</i> DBVPG#7215]
>gi 356889095 gb AET38644.1 Hypothetical protein Ecym_3140 [<i>Eremothecium cymbalariae</i> DBVPG#7215]
>gi 365759369 gb EHN01160.1 Hsp60p [<i>Saccharomyces cerevisiae</i> x <i>Saccharomyces kudriavzevii</i> VIN7]
>gi 365764091 gb EHN05616.1 Hsp60p [<i>Saccharomyces cerevisiae</i> x <i>Saccharomyces kudriavzevii</i> VIN7]
>gi 365985626 ref XP_003669645.1 hypothetical protein NDAI_0D00880 [<i>Naumovozyma dairenensis</i> CBS 421]
>gi 343768414 emb CCD24402.1 hypothetical protein NDAI_0D00880 [<i>Naumovozyma dairenensis</i> CBS 421]
>gi 366995970 ref XP_003677748.1 hypothetical protein NCAS_0H00890 [<i>Naumovozyma castellii</i> CBS 4309]
>gi 342303618 emb CCC71399.1 hypothetical protein NCAS_0H00890 [<i>Naumovozyma castellii</i> CBS 4309]
>gi 367005154 ref XP_003687309.1 hypothetical protein TPHA_0J00520 [<i>Tetrapispora phaffii</i> CBS 4417]
>gi 357525613 emb CCE64875.1 hypothetical protein TPHA_0J00520 [<i>Tetrapispora phaffii</i> CBS 4417]
>gi 367017005 ref XP_003683001.1 hypothetical protein TDEL_0G04230 [<i>Torulaspora delbrueckii</i>]
>gi 359750664 emb CCE93790.1 hypothetical protein TDEL_0G04230 [<i>Torulaspora delbrueckii</i>]
>gi 367035486 ref XP_003667025.1 hypothetical protein MYCTH_2097570 [<i>Myceliophthora thermophila</i> ATCC 42464]
>gi 347014298 gb AEO61780.1 hypothetical protein MYCTH_2097570 [<i>Myceliophthora thermophila</i> ATCC 42464]
>gi 367055018 ref XP_003657887.1 hypothetical protein THITE_127923 [<i>Thielavia terrestris</i> NRRL 8126]
>gi 347005153 gb AEO71551.1 hypothetical protein THITE_127923 [<i>Thielavia terrestris</i> NRRL 8126]
>gi 378728414 gb EHY54873.1 heat shock protein 60 [<i>Exophiala dermatitidis</i> NIH/UT8656]
>gi 380494593 emb CCF33032.1 heat shock protein 60 [<i>Colletotrichum higginsianum</i>]
>gi 385305893 gb EIF49836.1 heat shock protein 60 [<i>Dekkera bruxellensis</i> AWR11499]
>gi 389638386 ref XP_003716826.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> 70-15]
>gi 351642645 gb EHA50507.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> 70-15]
>gi 440474658 gb ELQ43388.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> Y34]
>gi 440480475 gb ELQ61135.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> P131]
>gi 393243142 gb EJD50658.1 chaperonin GroL [<i>Auricularia delicata</i> TFB-10046 SS5]
>gi 396494741 ref XP_003844378.1 similar to heat shock protein 60 [<i>Leptosphaeria maculans</i> JN3]
>gi 312220958 emb CBY00899.1 similar to heat shock protein 60 [<i>Leptosphaeria maculans</i> JN3]
>gi 398393428 ref XP_003850173.1 chaperone ATPase HSP60 [<i>Zymoseptoria tritici</i> IPO323]
>gi 339470051 gb EGP85149.1 hypothetical protein MYCGRDRAFT_75170 [<i>Zymoseptoria tritici</i> IPO323]
>gi 401624479 gb EJS42535.1 hsp60p [<i>Saccharomyces arboricola</i> H-6]
>gi 401842294 gb EJT44530.1 HSP60-like protein [<i>Saccharomyces kudriavzevii</i> IFO 1802]
>gi 402076594 gb EJT72017.1 heat shock protein 60 [<i>Gaeumannomyces graminis</i> var. <i>tritici</i> R3-111a-1]
>gi 403213867 emb CCK68369.1 hypothetical protein KNAG_0A07160 [<i>Kazachstania naganishii</i> CBS 8797]
>gi 406606041 emb CCH42514.1 Heat shock protein 60, mitochondrial [<i>Wickerhamomyces ciferrii</i>]
>gi 406863285 gb EKD16333.1 heat shock protein 60 [<i>Marssonina brunnea</i> f. sp. 'multigermtubi' MB_m1]
>gi 407922985 gb EKG16075.1 Chaperonin Cpn60 [<i>Macrophomina phaseolina</i> MS6]

TABLE 9-continued

Natural chaperones homologous to GroEL polypeptides suitable for expression
>gi 408399723 gb EKJ78816.1 hypothetical protein FPSE_00959 [<i>Fusarium pseudograminearum</i> CS3096] >gi 410083028 ref XP_003959092.1 hypothetical protein KAFR_0101760 [<i>Kazachstania africana</i> CBS 2517] >gi 372465682 emb CCF59957.1 hypothetical protein KAFR_0101760 [<i>Kazachstania africana</i> CBS 2517] >gi 444315528 ref XP_004178421.1 hypothetical protein TBLA_0B00580 [<i>Tetrapisispora blattae</i> CBS 6284] >gi 387511461 emb CCH58902.1 hypothetical protein TBLA_0B00580 [<i>Tetrapisispora blattae</i> CBS 6284] >gi 448090588 ref XP_004197110.1 PISO_004347 [<i>Millerozyma farinosa</i> CBS 7064] >gi 448095015 ref XP_004198141.1 PISO_004347 [<i>Millerozyma farinosa</i> CBS 7064] >gi 359378532 emb CCE84791.1 PISO_004347 [<i>Millerozyma farinosa</i> CBS 7064] >gi 359379563 emb CCE83760.1 PISO_004347 [<i>Millerozyma farinosa</i> CBS 7064] >gi 448526196 ref XP_003869293.1 Hsp60 heat shock protein [<i>Candida orthopsilosis</i> Co 90-125] >gi 380353646 emb CCG23157.1 Hsp60 heat shock protein [<i>Candida orthopsilosis</i>] >gi 46123737 ref XP_386422.1 HS60_AJECA Heat shock protein 60, mitochondrial precursor (Antigen HIS-62) [<i>Gibberella zeae</i> PH-1] >gi 50292099 ref XP_448482.1 hypothetical protein [<i>Candida glabrata</i> CBS 138] >gi 49527794 emb CAG61443.1 unnamed protein product [<i>Candida glabrata</i>] >gi 50310975 ref XP_455510.1 hypothetical protein [<i>Kluyveromyces lactis</i> NRRL Y-1140] >gi 496444646 emb CAG98218.1 KLLA0F09449p [<i>Kluyveromyces lactis</i>] >gi 50422027 ref XP_459575.1 DEHA2E05808p [<i>Debaryomyces hansenii</i> CBS767] >gi 49655243 emb CAG87802.1 DEHA2E05808p [<i>Debaryomyces hansenii</i> CBS767] >gi 50555023 ref XP_504920.1 YALI0F02805p [<i>Yarrowia lipolytica</i>] >gi 49650790 emb CAG77725.1 YALI0F02805p [<i>Yarrowia lipolytica</i> CLIB122] >gi 6323288 ref NP_013360.1 Hsp60p [<i>Saccharomyces cerevisiae</i> S288c] >gi 123579 sp P19882.1 HSP60_YEAST RecName: Full=Heat shock protein 60, mitochondrial; AltName: Full=CPN60; AltName: Full=P66; AltName: Full=Stimulator factor 1 66 kDa component; Flags: Precursor >gi 171720 gb AAA34690.1 heat shock protein 60 (HSP60) [<i>Saccharomyces cerevisiae</i>] >gi 577181 gb AAB67380.1 Hsp60p: Heat shock protein 60 [<i>Saccharomyces cerevisiae</i>] >gi 151941093 gb EDN59473.1 chaperonin [<i>Saccharomyces cerevisiae</i> YJM789] >gi 190405319 gb EDV08586.1 chaperonin [<i>Saccharomyces cerevisiae</i> RM11-1a] >gi 207342889 gb EDZ70518.1 YLR259Cp-like protein [<i>Saccharomyces cerevisiae</i> AWRI1631] >gi 256271752 gb EEU06789.1 Hsp60p [<i>Saccharomyces cerevisiae</i> JAY291] >gi 285813676 tpg DAA09572.1 TPA: chaperone ATPase HSP60 [<i>Saccharomyces cerevisiae</i> S288c] >gi 323353818 gb EGA85673.1 Hsp60p [<i>Saccharomyces cerevisiae</i> VL3] >gi 349579966 dbj GAA25127.1 K7_Hsp60p [<i>Saccharomyces cerevisiae</i> Kyokai no. 7] >gi 392297765 gb EIW08864.1 Hsp60p [<i>Saccharomyces cerevisiae</i> CEN.PK113-7D] >gi 226279 prf 1504305A mitochondrial assembly factor >gi 68485963 ref XP_713100.1 heat shock protein 60 [<i>Candida albicans</i> SC5314] >gi 68486010 ref XP_713077.1 heat shock protein 60 [<i>Candida albicans</i> SC5314] >gi 6016258 sp O74261.1 HSP60_CANAL RecName: Full=Heat shock protein 60, mitochondrial; AltName: Full=60 kDa chaperonin; AltName: Full=Protein Cpn60; Flags: Precursor >gi 3552009 gb AAC34885.1 heat shock protein 60 [<i>Candida albicans</i>] >gi 46434552 gb EAK93958.1 heat shock protein 60 [<i>Candida albicans</i> SC5314] >gi 46434577 gb EAK93982.1 heat shock protein 60 [<i>Candida albicans</i> SC5314] >gi 71001164 ref XP_755263.1 antigenic mitochondrial protein HSP60 [<i>Aspergillus fumigatus</i> Af293] >gi 68852901 gb EAL93225.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus fumigatus</i> Af293] >gi 159129345 gb EDP54459.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus fumigatus</i> A1163] >gi 90970323 gb ABE02805.1 heat shock protein 60 [<i>Rhizophagus intraradices</i>]

[0264] The recombinant yeast cell preferably comprises, respectively functionally expresses, a GroES chaperone and a GroEL chaperone. Preferably a 10 kDa chaperone (“GroES”) from Table 8 is combined with a matching 60 kDa chaperone (“GroEL”) from Table 9 of the same organism genus or species for expression in the recombinant yeast cell.

[0265] For instance: >gi|189189366|ref|XP_001931022.1|: 71-168 10 kDa chaperonin [*Pyrenophora tritici-repentis*] expressed together with matching >gi|189190432|ref|XP_001931555.1|heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP]. All other combinations from Table 8 and 9 similarly made with same organism source are also available to the skilled person for expression. Furthermore, one may combine a chaperone from Table 8 from one organism with a chaperone from Table 9 from another organism, or one may combine GroES with a chaperone from Table 9, or one may combine GroEL with a chaperone from Table 8.

[0266] Preferably the molecular chaperone(s) thus comprise or consist of:

[0267] an amino acid sequence of SEQ ID NO: 29 and/or SEQ ID NO: 31; or

[0268] one or more functional homologue(s) of SEQ ID NO: 29 and/or SEQ ID NO: 31, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31; or

[0269] one or more functional homologue(s) of SEQ ID NO: 29 and/or SEQ ID NO: 31, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31, more preferably one or more functional homologue(s)

that has/have no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31.

[0270] Preferable the nucleic acid sequence(s) encoding the molecular chaperones comprise or consist of:

[0271] a nucleic acid sequence of SEQ ID NO: 30 and/or SEQ ID NO: 32; or

[0272] one or more functional homologue(s) of SEQ ID NO: 30 and/or SEQ ID NO: 32, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32; or

[0273] one or more functional homologue(s) of SEQ ID NO: 30 and/or SEQ ID NO: 32, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32, more preferably one or more functional homologue(s) of that has/have no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32.

[0274] The nucleic acid sequence(s) encoding for the molecular chaperones may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898, herein incorporated by reference.

Phosphoketolase

[0275] As indicated above, the recombinant yeast cell can comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0276] The recombinant cell may comprise one or more heterologous genes coding for a protein having phosphoketolase activity. Such a protein having phosphoketolase activity is herein also referred to as “phosphoketolase protein”, “phosphoketolase enzyme” or simply as “phosphoketolase”. Phosphoketolase is further herein abbreviated as “PKL” or “XFP”.

[0277] As used herein, a phosphoketolase catalyzes at least the conversion of D-xylulose 5-phosphate to D-glyceraldehyde 3-phosphate and acetyl phosphate. The phosphoketolase is involved in at least one of the following the reactions:

EC 4.1.2.9:

[0278]

D-xylulose-5-phosphate+phosphate $\rightleftharpoons$ acetyl phosphate+D-glyceraldehyde 3-phosphate+H₂O (IV)

D-ribulose-5-phosphate+phosphate $\rightleftharpoons$ acetyl phosphate+D-glyceraldehyde 3-phosphate+H₂O (V)

EC 4.1.2.22:

[0279]

D-fructose 6-phosphate+phosphate $\rightleftharpoons$ acetyl phosphate+D-erythrose 4-phosphate+H₂O (VI)

[0280] A suitable enzymatic assay to measure phosphoketolase activity is described e.g. in Sonderegger et al., “*Metabolic Engineering of a Phosphoketolase Pathway for Pentose Catabolism in Saccharomyces cerevisiae*”, (2004), Applied & Environmental Microbiology, vol. 70(5), pages 2892-2897, incorporated herein by reference.

[0281] Preferably the protein having phosphoketolase (PKL) activity comprises or consists of:

[0282] an amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36; or

[0283] a functional homologue of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36; or

[0284] a functional homologue of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36.

[0285] Suitable nucleic acid sequences coding for an phosphoketolase protein may in be found in an organism selected from the group of *Aspergillus niger*, *Neurospora crassa*, *L. casei*, *L. plantarum*, *L. plantarum*, *B. adolescentis*, *B. bifidum*, *B. gallicum*, *B. animalis*, *B. lactis*, *L. pentosum*, *L. acidophilus*, *P. chrysogenum*, *A. nidulans*, *A. clavatus*, *L. mesenteroides*, and *O. oenii*.

[0286] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphoketolase (PKL) activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0287] The recombinant cell may comprise one or more (heterologous) genes coding for an enzyme having phosphoketolase activity.

Phosphotransacetylase

[0288] As indicated above, the recombinant yeast cell can comprise a, preferably heterologous, nucleic acid sequence

encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0289] As used herein, a phosphotransacetylase catalyzes at least the conversion of acetyl phosphate to acetyl-CoA.

[0290] The recombinant cell may comprise one or more heterologous genes coding for a protein having phosphotransacetylase activity. Such a protein having phosphotransacetylase activity is herein also referred to as “phosphotransacetylase protein”, “phosphotransacetylase enzyme” or simply as “phosphotransacetylase”. phosphotransacetylase is further herein abbreviated as “PTA”.

[0291] Preferably the protein having phosphotransacetylase (PTA) activity comprises or consists of:

[0292] an amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40; or

[0293] a functional homologue of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40; or

[0294] a functional homologue of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40.

[0295] Suitable nucleic acid sequences coding for an enzyme having phosphotransacetylase may in be found in an organism selected from the group of *B. adolescentis*, *B. subtilis*, *C. cellulolyticum*, *C. phytofermentans*, *B. bifidum*, *B. animalis*, *L. mesenteroides*, *Lactobacillus plantarum*, *M. thermophila*, and *O. oeniis*.

[0296] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphotransacetylase (PTA) activity may suitably be incorporated in the genome of the recombinant yeast cell.

Acetate Kinase

[0297] As indicated above, the recombinant yeast cell can comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0298] As used herein, an acetate kinase catalyzes at least the conversion of acetate to acetyl phosphate.

[0299] The recombinant cell may comprise one or more, preferably heterologous, genes coding for a protein having acetate kinase activity (EC 2.7.2.12). Such a protein having acetate kinase activity is herein also referred to as “acetate kinase protein”, “acetate kinase enzyme” or simply as “acetate kinase”. Acetate kinase is further herein abbreviated as “ACK”.

[0300] Preferably the protein having acetate kinase (ACK) activity comprises or consists of:

[0301] an amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42; or

[0302] a functional homologue of SEQ ID NO: 41 or SEQ ID NO: 42, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42; or

[0303] a functional homologue of SEQ ID NO: 41 or SEQ ID NO: 42, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42.

[0304] The nucleic acid sequence (e.g. the gene) encoding for the protein having acetate kinase (ACK) activity may suitably be incorporated in the genome of the recombinant yeast cell.

Acetylating Acetaldehyde Dehydrogenase

[0305] As indicated above, the recombinant yeast cell can advantageously comprise and functionally express a, preferably heterologous, nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10).

[0306] If an acetylating acetaldehyde dehydrogenase is present, more preferably, the recombinant yeast cell functionally expresses:

[0307] a, preferably heterologous, nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and

[0308] a, suitably endogenous or heterologous, nucleic acid sequence encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC 1.1.1.2); and

[0309] a, suitably endogenous or heterologous, nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0310] Acetylating acetaldehyde dehydrogenase is an enzyme that catalyzes the conversion of acetyl-Coenzyme A to acetaldehyde (EC 1.2.1.10). This conversion can be represented by the equilibrium reaction formula:



[0311] A protein having acetylating acetaldehyde dehydrogenase activity is herein also referred to as “acetylating acetaldehyde dehydrogenase protein”, “acetylating acetaldehyde dehydrogenase enzyme” or simply “acetylating acetaldehyde dehydrogenase”. Preferences for a acetylating acetaldehyde dehydrogenase and the nucleic sequences encoding for such are as described in WO2011/010923 and WO2019/063507, incorporated herein by reference.

[0312] The nucleic acid sequence encoding a protein having NAD⁺-dependent acetylating acetaldehyde dehydrogenase activity (EC1.2.1.10) is preferably a heterologous nucleic acid sequence. The encoded NAD⁺-dependent acetylating acetaldehyde dehydrogenase may therefore preferably be a heterologous NAD⁺-dependent acetylating acetaldehyde dehydrogenase.

[0313] It is possible for the protein having acetylating acetaldehyde dehydrogenase activity to be monofunctional or bifunctional.

[0314] The nucleic acid sequence encoding the NAD⁺ dependent acetylating acetaldehyde dehydrogenase may in principle originate from any organism comprising a nucleic acid sequence encoding said dehydrogenase. Known acetylating acetaldehyde dehydrogenases that can catalyse the NADH-dependent reduction of acetyl-Coenzyme A to acetaldehyde may in general be divided in three types of NAD⁺ dependent acetylating acetaldehyde dehydrogenase functional homologues:

[0315] 1) Bifunctional proteins that catalyse the reversible conversion of acetyl-CoA to acetaldehyde, and the subsequent reversible conversion of acetaldehyde to ethanol. These type of proteins advantageously have both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity. An example of this type of proteins is the AdhE protein in *E. coli* (Gen Bank No: NP_415757). AdhE appears to be the evolutionary product of a gene fusion. The NH₂-terminal region of the AdhE protein is highly homologous to aldehyde:NAD⁺ oxidoreductases, whereas the COOH-terminal region is homologous to a family of Fe²⁺ dependent ethanol:NAD⁺ oxidoreductases (see Membrillo-Hernandez et al., “*Evolution of the adhE Gene Product of Escherichia coli from a Functional Reductase to a Dehydrogenase*”, (2000) J. Biol. Chem. 275: pages 33869-33875, herein incorporated by reference). The *E. coli* AdhE is subject to metal-catalyzed oxidation and therefore oxygen-sensitive (see Tamarit et al. “*Identification of the Major Oxidatively Damaged Proteins in Escherichia coli Cells Exposed to Oxidative Stress*” (1998) J. Biol. Chem. 273: pages 3027-3032, herein incorporated by reference).

[0316] 2) Proteins that catalyse the reversible conversion of acetyl-Coenzyme A to acetaldehyde in strictly or facultative anaerobic micro-organisms but do not possess alcohol dehydrogenase activity. An example of this type of proteins has been reported in *Clostridium kluyveri* (see Smith et al. “*Purification, Properties, and Kinetic Mechanism of Coenzyme A-Linked Aldehyde Dehydrogenase from Clostridium kluyveri*” (1980) Arch. Biochem. Biophys. Vol. 203: pages 663-675, incorporated herein by reference). An acetylating acetaldehyde dehydrogenase has been annotated in the genome of *Clostridium kluyveri* DSM 555 (GenBank No: EDK33116). A homologous protein AcdH is identified in the genome of *Lactobacillus plantarum* (Gen-

Bank No: NP_784141). Another example of this type of proteins is the said gene product in *Clostridium beijerinckii* NRRL B593 (see Toth et al. “*The ald Gene, Encoding a Coenzyme A-Acylating Aldehyde Dehydrogenase, Distinguishes Clostridium beijerinckii and Two Other Solvent-Producing Clostridia from Clostridium acetobutylicum*”, (1999), Appl. Environ. Microbiol. Vol. 65: pages 4973-4980, GenBank No: AAD31841, incorporated herein by reference).

[0317] 3) Proteins that are part of a bifunctional aldolase-dehydrogenase complex involved in 4-hydroxy-2-ketovaleate catabolism. Such bifunctional enzymes catalyze the final two steps of the meta-cleavage pathway for catechol, an intermediate in many bacterial species in the degradation of phenols, toluates, naphthalene, biphenyls and other aromatic compounds (Powlowski and Shingler “*Genetics and biochemistry of phenol degradation by Pseudomonas sp. CF600*” (1994) Biodegradation Vol. 5, pages 219-236, herein incorporated by reference). 4-Hydroxy-2-ketovaleate is first converted by 4-hydroxy-2-ketovaleate aldolase to pyruvate and acetaldehyde, subsequently acetaldehyde is converted by acetylating acetaldehyde dehydrogenase to acetyl-CoA. An example of this type of acetylating acetaldehyde dehydrogenase is the DmpF protein in *Pseudomonas* sp CF600 (GenBank No: CAA43226) (Shingler et al., “*Nucleotide Sequence and Functional Analysis of the Complete Phenol/3,4-Dimethylphenol Catabolic Pathway of Pseudomonas sp. Strain CF600*”, (1992), J. Bacteriol., Vol. 174, pages 711-724, incorporated herein by reference). The *E. coli* MphF protein (Ferrandez et al., “*Genetic Characterization and Expression in Heterologous Hosts of the 3-(3-Hydroxyphenyl) Propionate Catabolic Pathway of Escherichia coli K-12*” (1997) J. Bacteriol. 179: pages 2573-2581, GenBank No: NP_414885, incorporated herein by reference) is homologous to the DmpF protein in *Pseudomonas* sp. CF600.

[0318] In a preferred embodiment, the protein having acetylating acetaldehyde dehydrogenase activity is bifunctional and comprises both NAD⁺ dependent acetylating acetaldehyde dehydrogenase (EC 1.2.1.10) activity and NAD⁺ dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC 1.1.1.2).

[0319] A suitable nucleic acid sequence may in particular be found in an organism selected from the group of *Escherichia*, in particular *E. coli*; *Mycobacterium*, in particular *Mycobacterium marinum*, *Mycobacterium ulcerans*, *Mycobacterium tuberculosis*; *Carboxydotherrmus*, in particular *Carboxydotherrmus hydrogenoformans*; *Entamoeba*, in particular *Entamoeba histolytica*; *Shigella*, in particular *Shigella sonnei*; *Burkholderia*, in particular *Burkholderia pseudo mallei*, *Klebsiella*, in particular *Klebsiella pneumoniae*; *Azotobacter*, in particular *Azotobacter vinelandii*; *Azoarcus* sp; *Cupriavidus*, in particular *Cupriavidus taiwanensis*; *Pseudomonas*, in particular *Pseudomonas* sp. CF600; *Pelomaculum*, in particular *Pelotomaculum thermopropionicum*. Preferably, the nucleic acid sequence encoding the NAD⁺ dependent acetylating acetaldehyde dehydrogenase originates from *Escherichia*, more preferably from *E. coli*.

[0320] Particularly suitable is an mhpF gene from *E. coli*, or a functional homologue thereof. This gene is described in Ferrandez et al., “*Genetic Characterization and Expression in Heterologous Hosts of the 3-(3-Hydroxyphenyl) Propi-*

onate Catabolic Pathway of *Escherichia coli* K-12' (1997) J. Bacteriol. 179: pages 2573-2581. Good results have been obtained with *S. cerevisiae*, wherein an mhpF gene from *E. coli* has been incorporated. In a further advantageous embodiment the nucleic acid sequence encoding an (acetylating) acetaldehyde dehydrogenase is from *Pseudomonas*, in particular dmpF, e.g. from *Pseudomonas* sp. CF600.

[0321] Further, an acetylating acetaldehyde dehydrogenase (or nucleic acid sequence encoding such activity) may for instance be selected from the group of *Escherichia coli* adhE, *Entamoeba histolytica* adh2, *Staphylococcus aureus* adhE, *Piromyces* sp. E2 adhE, *Clostridium kluyveri* EDK33116, *Lactobacillus plantarum* acdH, *Escherichia coli* eutE, *Listeria innocua* acdH, and *Pseudomonas putida* YP 001268189.

[0322] Preferably the protein having NAD⁺-dependent acetylating acetaldehyde dehydrogenase activity comprises or consists of:

[0323] an amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48; or

[0324] a functional homologue of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of

SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48; or

[0325] a functional homologue of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48.

[0326] Most preferably the acetylating acetaldehyde dehydrogenase protein is a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity.

[0327] The nucleic acid sequence (e.g. the gene) encoding for the protein having acetylating acetaldehyde dehydrogenase activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0328] Examples of suitable enzymes are further illustrated below in tables 10(a) to 10(e) for BLAST of the listed enzymes, giving suitable alternative alcohol/acetaldehyde dehydrogenases.

TABLE 10

Description	Identity (%)	Accession number
(a) BLAST Query - adhE from <i>Escherichia coli</i>		
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Escherichia coli</i> O157:H7 str. Sakai]	100	NP_309768.1
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Escherichia coli</i> UTI89]	99	YP_540449.1
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Enterobacter</i> sp. 638]	95	YP_001177024.1
(b) BLAST Query - acdH from <i>Lactobacillus plantarum</i>		
acetaldehyde dehydrogenase [<i>Lactobacillus plantarum</i> WCFS1]	100	YP_004888365.1
acetaldehyde dehydrogenase [<i>Lactobacillus pentosus</i> IG1]	95	CCC16763.1
aldehyde-alcohol dehydrogenase [<i>Enterococcus cecorum</i>]	58	WP_016251441.1
aldehyde-alcohol dehydrogenase 2 [<i>Enterococcus faecalis</i>]	57	WP_016623694.1
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Lactobacillus zeae</i>]	55	WP_010493695.1
alcohol dehydrogenase [<i>Bacillus thuringiensis</i>]	54	WP_003280110.1
bifunctional acetaldehyde-CoA/alcohol dehydrogenase, partial [<i>Listeria monocytogenes</i>]	53	WP_009931954.1
(c) BLAST Query - eutE from <i>Escherichia coli</i>		
aldehyde oxidoreductase, ethanolamine utilization protein [<i>Escherichia coli</i> str. K-12 substr. MG1655]	100	NP_416950.1
ethanolamine utilization; acetaldehyde dehydrogenase [<i>Escherichia coli</i> O157:H7 str. EDL933]	99	NP_289007.1
aldehyde dehydrogenase [<i>Escherichia albertii</i>]	99	WP_001075674.1
(d) BLAST Query - Lin1129 from <i>Listeria innocua</i>		
aldehyde dehydrogenase [<i>Listeria innocua</i>]	100	NP_470466.1
>emb CAC96360.1 lin1129 [<i>Listeria innocua</i> Clip11262]		
ethanolamine utilization protein EutE [<i>Listeria innocua</i>]	99	WP_003761764.1
aldehyde dehydrogenase [<i>Listeria monocytogenes</i>]	95	AGR09081.1
hypothetical protein [<i>Enterococcus malodoratus</i>]	64	WP_010739890.1

TABLE 10-continued

Description	Identity (%)	Accession number
aldehyde dehydrogenase [<i>Yersinia aldovae</i>]	59	WP_004699364.1
aldehyde dehydrogenase EutE [<i>Klebsiella pneumoniae</i>]	58	WP_004205473.1
(e) BLAST Query - adhE from <i>Staphylococcus aureus</i>		
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Staphylococcus aureus</i> subsp. <i>aureus</i> Mu50]	100	NP_370672.1
aldehyde dehydrogenase family protein [<i>Staphylococcus aureus</i> CA-347]	99	YP_008127042.1
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Staphylococcus epidermidis</i>]	85	WP_002495347.1
aldehyde-alcohol dehydrogenase 2 [<i>Enterococcus faecalis</i>]	75	WP_016623694.1

Acetyl-Coenzyme A Synthetase

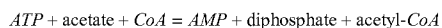
[0329] If the recombinant yeast cell functionally expresses a protein having acetylating acetaldehyde dehydrogenase activity, preferably the recombinant yeast cell is further functionally expressing:

[0330] a nucleic acid sequence encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC 1.1.1.2); and/or

[0331] a nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0332] A protein having acetyl-Coenzyme A synthetase activity can herein also be referred to as “acetyl-Coenzyme A synthetase protein”, “acetyl-Coenzyme A synthetase enzyme” or simply “acetyl-Coenzyme A synthetase” or even “acetyl CoA synthetase”. The protein is further abbreviated herein as “ACS”.

[0333] The acetyl-Coenzyme A synthetase, also known as acetate-CoA ligase or acetyl-activating enzyme, catalyses the formation of acetyl-CoA from acetate, coenzyme A (CoA) and ATP as shown below:



[0334] It is understood that the recombinant yeast cell may naturally comprise an endogenous gene encoding an acetyl-Coenzyme A synthetase protein. In the alternative, or in addition thereto, the recombinant yeast cell may comprise a heterologous nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0335] For example, the recombinant yeast cell according to the invention may comprise an acetyl-Coenzyme A synthetase, which may be present in the wild-type cell, as is for instance the case with *S. cerevisiae* which contains two acetyl-Coenzyme A synthetase isoenzymes encoded by the ACS1 (amino acid sequence illustrated as SEQ ID NO: 49) and ACS2 (amino acid sequence illustrated as SEQ ID NO: 50) genes (van den Berg et al (1996) J. Biol. Chem. 271: pages 28953-28959, incorporated herein by reference), or a

host cell may be provided with one or more heterologous gene(s) encoding this activity, e.g. the ACS1 and/or ACS2 gene of *S. cerevisiae* or a functional homologue thereof may be incorporated into a cell lacking acetyl-Coenzyme A synthetase isoenzyme activity.

[0336] Preferably the protein having NAD⁺-dependent acetyl-Coenzyme A synthetase activity comprises or consists of:

[0337] an amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50; or

[0338] a functional homologue of SEQ ID NO: 49 or SEQ ID NO: 50 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50; or

[0339] a functional homologue of SEQ ID NO: 49 or SEQ ID NO: 50 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50.

[0340] Preferably the recombinant yeast cell is a recombinant yeast cell wherein the, endogenous or heterologous, acetyl-Coenzyme A synthetase protein, is overexpressed, most preferably by using a suitable promoter as described for example in WO2011/010923, incorporated herein by reference. Any heterologous nucleic acid sequence (e.g. the gene) encoding for the protein having acetyl-Coenzyme A synthetase activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0341] Examples of suitable proteins having acetyl-Coenzyme A synthetase activity are listed in table 11. At the top of table 11 the ACS2 used in the examples and that is BLASTED is mentioned.

TABLE 11

BLAST Query - ACS2 from <i>Saccharomyces cerevisiae</i>		
Description	Identity (%)	Accession number
acetate-CoA ligase ACS2 [<i>Saccharomyces cerevisiae</i> S288c]	100	NP_013254.1
acetyl CoA synthetase [<i>Saccharomyces cerevisiae</i> YJM789]	99	EDN59693.1

TABLE 11-continued

BLAST Query - ACS2 from <i>Saccharomyces cerevisiae</i>		
Description	Identity (%)	Accession number
acetate-CoA ligase [<i>Kluyveromyces lactis</i> NRRL Y-1140]	85	XP_453827.1
acetate-CoA ligase [<i>Candida glabrata</i> CBS 138]	83	XP_445089.1
acetate-CoA ligase [<i>Scheffersomyces stipitis</i> CBS 6054]	68	XP_001385819.1
acetyl-coenzyme A synthetase FacA [<i>Aspergillus fumigatus</i> A1163]	63	EDP50475.1
acetate-CoA ligase facA- <i>Penicillium chrysogenum</i> [<i>Penicillium chrysogenum</i> Wisconsin 54-1255]	62	XP_002564696.1

Alcohol Dehydrogenase

[0342] If the recombinant yeast cell functionally expresses a protein having acetylating acetaldehyde dehydrogenase activity, preferably the recombinant yeast cell is further functionally expressing:

[0343] a nucleic acid sequence encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2); and/or

[0344] a nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0345] A protein having alcohol dehydrogenase activity is herein also referred to as “alcohol dehydrogenase protein”, “alcohol dehydrogenase enzyme” or simply “alcohol dehydrogenase”. The protein is further abbreviated herein as “ADH”.

[0346] The alcohol dehydrogenase enzyme catalyses the conversion of acetaldehyde into ethanol.

[0347] It is understood that the recombinant yeast cell may naturally comprise an endogenous nucleic acid sequence encoding an alcohol dehydrogenase protein. In the alternative, or in addition thereto, the recombinant yeast cell may comprise a heterologous nucleic acid sequence encoding a protein having alcohol dehydrogenase activity

[0348] For example, the recombinant yeast cell may naturally comprise a gene encoding alcohol dehydrogenase, as is the case with *S. cerevisiae* (Amino acid sequences of the native *S. cerevisiae* alcohol dehydrogenases ADH1, ADH2, ADH3, ADH4 and ADH5 are illustrated respectively as SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 and SEQ ID NO: 55), see Lutstorf and Megnet, “Multiple Forms of Alcohol Dehydrogenase in *Saccharomyces cerevisiae*”, (1968), Arch. Biochem. Biophys., vol. 126, pages 933-944, incorporated herein by reference, or Ciriacy, “Genetics of Alcohol Dehydrogenase in *Saccharomyces cerevisiae* I. Isolation and genetic analysis of *adh* mutants”, (1975), Mutat. Res. 29, pages 315-326, incorporated herein by reference).

[0349] Preferably, however, the recombinant yeast cell comprises alcohol dehydrogenase activity within a, suitably heterologous, bifunctional enzyme having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity as described herein above. That is, most preferably the alcohol dehydrogenase protein is a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity. When the recombinant yeast cell comprises a heterologous nucleic acid sequence encoding a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity, any native

nucleic acid sequences encoding for any native protein encoding alcohol dehydrogenase activity may or may not be disrupted and/or deleted.

[0350] The recombinant yeast cell may therefore advantageously be a recombinant yeast cell functionally expressing:

[0351] one or more heterologous nucleic acid sequence(s) encoding a bifunctional protein having NAD⁺-dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2); and

[0352] one or more, native or heterologous, nucleic acid sequence(s) encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1),

wherein optionally one or more native nucleic acid sequence(s) encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2) are disrupted or deleted.

[0353] Alternatively the recombinant yeast cell may advantageously be a recombinant yeast cell functionally expressing:

[0354] one or more, native or heterologous, nucleic acid sequence(s) encoding a monofunctional protein having NAD⁺-dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and

[0355] one or more, native or heterologous, nucleic acid sequence(s) encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1); and

[0356] one or more, native or heterologous, nucleic acid sequence(s) encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2).

[0357] Preferences for the bifunctional protein are provided above and are as listed for the acetylating acetaldehyde dehydrogenase protein. If the protein is not bifunctional, the NAD⁺-dependent alcohol dehydrogenase protein is preferably a protein having NAD⁺-dependent alcohol dehydrogenase activity that comprises or consists of:

[0358] an amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55; or

[0359] a functional homologue of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55; or

[0360] a functional homologue of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55.

[0361] Any heterologous nucleic acid sequence (e.g. the gene) encoding for the protein having NAD⁺-dependent alcohol dehydrogenase activity may suitably be incorporated in the genome of the recombinant yeast cell.

Deletion or Disruption of Glycerol 3-Phosphate Phosphohydrolase and/or Glycerol 3-Phosphate Dehydrogenase

[0362] The recombinant yeast cell further may or may not comprise a deletion or disruption of one or more endogenous nucleotide sequence encoding a glycerol 3-phosphate phosphohydrolase gene and/or encoding a glycerol 3-phosphate dehydrogenase gene.

[0363] Preferably enzymatic activity needed for the NADH-dependent glycerol synthesis in the yeast cell is reduced or deleted. The reduction or deletion of the enzymatic activity of glycerol 3-phosphate phosphohydrolase and/or glycerol 3-phosphate dehydrogenase can be achieved by modifying one or more genes encoding a NAD-dependent glycerol 3-phosphate dehydrogenase (GPD) and/or one or more genes encoding a glycerol phosphate phosphatase (GPP), such that the enzyme is expressed considerably less than in the wild-type or such that the gene encodes a polypeptide with reduced activity. Such modifications can be carried out using commonly known biotechnological techniques, and may in particular include one or more knock-out mutations or site-directed mutagenesis of promoter regions or coding regions of the structural genes encoding GPD and/or GPP. Alternatively, yeast strains that are defective in glycerol production may be obtained by random mutagenesis followed by selection of strains with reduced or absent activity of GPD and/or GPP. *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes are shown in WO2011010923, and are disclosed in SEQ ID NO: 24-27 of that application.

[0364] Preferably the recombinant yeast is a recombinant yeast that further comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase (GPD) gene. The one or more of the glycerol phosphate phosphatase (GPP) genes may or may not be deleted or disrupted.

[0365] More preferably the recombinant yeast is a recombinant yeast that comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase 1 (GPD1) gene. The glycerol-3-phosphate dehydrogenase 2 (GPD2) gene may or may not be deleted or disrupted.

[0366] Most preferably the recombinant yeast is a recombinant yeast that comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase 1 (GPD1) gene, whilst the glycerol-3-phosphate dehydrogenase 2 (GPD2) gene remains active and/or intact. Preferably therefore, only one of the *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes is

disrupted and deleted, whereas most preferably only GPD1 is chosen from the group consisting of GPD1, GPD2, GPP1 and GPP2 genes to be disrupted or deleted.

[0367] Without wishing to be bound to any kind of theory it is believed that a recombinant yeast according to the invention wherein the GPD1 gene, but not the GPD2 gene, is deleted or disrupted, can be advantageous when applied in a fermentation process where the glucose at the start of or during the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L.

[0368] Preferably at least one gene encoding a GPD and/or at least one gene encoding a GPP is entirely deleted, or at least a part of the gene is deleted that encodes a part of the enzyme that is essential for its activity. Good results can be achieved with a *S. cerevisiae* cell, wherein the open reading frames of the GPD1 gene and/or of the GPD2 gene have been inactivated. Inactivation of a structural gene (target gene) can be accomplished by a person skilled in the art by synthetically synthesizing or otherwise constructing a DNA fragment consisting of a selectable marker gene flanked by DNA sequences that are identical to sequences that flank the region of the host cell's genome that is to be deleted. Suitably, good results can be obtained with the inactivation of the GPD1 and GPD2 genes in *Saccharomyces cerevisiae* by integration of the marker genes kanMX and hphMX4. Subsequently this DNA fragment is transformed into a host cell. Transformed cells that express the dominant marker gene are checked for correct replacement of the region that was designed to be deleted, for example by a diagnostic polymerase chain reaction or Southern hybridization.

[0369] Thus, in the recombinant yeast cells of the invention, glycerol 3-phosphate phosphohydrolase activity in the cell and/or glycerol 3-phosphate dehydrogenase activity in the cell can be advantageously reduced.

Glucoamylase

[0370] Preferably, the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding for a glucoamylase (EC 3.2.1.20 or 3.2.1.3).

[0371] A protein having glucoamylase activity is herein also referred to as "glucoamylase enzyme", "glucoamylase protein" or simply "glucoamylase". Glucoamylase has herein been abbreviated as "GA".

[0372] Glucoamylase, also referred to as amyloglucosidase, alpha-glucosidase, glucan 1,4-alpha glucosidase, maltase glucoamylase, and maltase-glucoamylase, catalyses at least the hydrolysis of terminal 1,4-linked alpha-D-glucose residues from non-reducing ends of amylose chains to release free D-glucose. A glucoamylase may be further defined by its amino acid sequence. Likewise a glucoamylase may be further defined by a nucleotide sequence encoding the glucoamylase. As explained in detail above under definitions, a certain glucoamylase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the glucoamylase.

[0373] Preferably the protein having glucoamylase activity comprises or consists of:

[0374] an amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58; or

[0375] a functional homologue of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58; or

[0376] a functional homologue of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58.

[0377] The polypeptide of SEQ ID NO: 56 encodes a "mature glucoamylase", referring to the enzyme in its final form after translation and any post-translational modifications, such as N-terminal processing, C-terminal truncation, glycosylation, phosphorylation, etc.

[0378] In an embodiment the nucleotide sequence encodes a polypeptide having an amino acid sequence of SEQ ID NO: 57 or a variant thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95, 98%, or 99% with the amino acid sequence of SEQ ID NO: 57. Amino acids 1-17 of the SEQ ID NO: 57 may encode for a native signal sequence.

[0379] In another embodiment the nucleotide sequence allowing the expression of a glucoamylase encodes a polypeptide having an amino acid sequence of SEQ ID NO: 58 or a variant thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95, 98%, or 99% with the amino acid sequence of SEQ ID NO: 58. Amino acids 1-19 of the SEQ ID NO: 58 may encode for a signal sequence.

[0380] A signal sequence (also referred to as signal peptide, targeting signal, localization signal, localization sequence, transit peptide, leader sequence or leader peptide) can be present at the N-terminus of a polypeptide (here, the glucoamylase) where it signals that the polypeptide is to be excreted, for example outside the cell and into the media.

[0381] The nucleic acid sequence (e.g. the gene) encoding for the protein having glucoamylase activity may suitably be incorporated in the genome of the recombinant yeast cell.

Nitrate Reductase

[0382] The recombinant yeast cell may also advantageously comprise, respectively functionally express, a nucleic acid sequences encoding an enzyme having NADH-dependent nitrate reductase activity and/or a nucleic acid sequences encoding an enzyme having NADH-dependent nitrite reductase activity. Details for the expression of such an alternative redox sink have been described in non-published US patent application U.S. 63/087,642 filed with

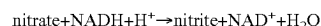
the United States Patent Office on 5 Oct. 2020, the contents of which are herewith incorporated by reference.

[0383] Nitrate reductase (NR) catalyzes the reduction of nitrate (NO_3^-) to nitrite (NO_2^-). Nitrite reductase catalyzes the reduction of nitrite to ammonia (NH_3). Nitrate reductase and/or nitrite reductase can be part of a so-called nitrogen assimilation pathway in certain cells. Cells comprising nitrate reductase activity and/or nitrite reductase activity include certain plant cells and bacterial cells and a few yeast cells. As indicated by Linder, the ability to assimilate inorganic nitrogen sources other than ammonia is thought to be rare among budding yeasts. Among the few fungi that are naturally capable to assimilate nitrate or nitrite are *Blastobotrys adeninivorans* (family Trichomonascaceae) *Candida boidinii* (family Pichiaceae), *Cyberlindnera jadinii* (family Phaffiomycetaceae), and *Ogataea polymorpha* (family Pichiaceae).

[0384] Preferably the recombinant yeast cell as described herein comprises at least one or more genes encoding a NADH-dependent nitrate reductase.

[0385] By a NADH-dependent nitrate reductase is herein understood a nitrate reductase that is exclusively depended on NADH as a co-factor or that is predominantly dependent on NADH as a cofactor. Preferably the NADH-dependent nitrate reductase has a ratio of catalytic efficiency for $\text{NADPH}/\text{NADP}^+$ as a cofactor $(k_{cat}/K_m)^{\text{NADP}^+}$ to NADH/NAD^+ as cofactor $(k_{cat}/K_m)^{\text{NAD}^+}$ i.e. a catalytic efficiency ratio $(k_{cat}/K_m)^{\text{NADP}^+}:(k_{cat}/K_m)^{\text{NAD}^+}$, of more than 1:1, more preferably of equal to or more than 2:1, still more preferably of equal to or more than 5:1, even more preferably of equal to or more than 10:1, yet even more preferably of equal to or more than 20:1, even still more preferably of equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the NADH-dependent nitrate reductase may have a catalytic efficiency ratio $(k_{cat}/K_m)^{\text{NADP}^+}:(k_{cat}/K_m)^{\text{NAD}^+}$ of equal to or less than 1,000,000,000:1 (i.e. $1 \cdot 10^9$). Most preferably the NADH-dependent nitrate reductase is exclusively depended on NADH/ NAD^+ as a co-factor. That is, most preferably the NADH-dependent nitrate reductase has an absolute requirement for NADH/ NAD^+ as a cofactor instead of NADPH/ NADP^+ as a cofactor.

[0386] Preferably the NADH-dependent nitrate reductase is a NADH-dependent nitrate reductase with enzyme classification EC 1.7.1.1. (i.e. with EC number EC 1.7.1.1) or enzyme classification EC.1.6.6.1 (i.e. with EC number 1.6.6.1). Suitably the NADH-dependent nitrate reductase, also referred to as NADH-dependent nitrate oxidoreductase, is an enzyme that catalyzes at least the following chemical reaction:



[0387] Suitable NADH-dependent nitrate reductases may include one or more NADH-dependent nitrate reductases as obtained or derived from *Agrostemma githago*, *Amaranthus hybridus*, *Amaranthus tricolor*, *Ankistrodesmus braunii*, *Arabidopsis thaliana*, *Aspergillus niger*, *Aspergillus nidulans*, *Auxenochlorella pyrenoidosa*, *Bradyrhizobium* sp., *Bradyrhizobium* sp. 750, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Candida boidinii*, *Candida utilis*, *Capsicum frutescens*, *Chenopodium album*, *Cyberlindnera jadinii*, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Capsicum frutescens*, *Chenopodium album*, *Chlamydomonas reinhardtii*, *Chlorella fusca*, *Chlorella* sp. *Chlo-*

rella sp. Berlin, *Chlorella vulgaris*, *Conticribra weissflogii*, *Cucumis sativus*, *Cucurbita maxima*, *Cucurbita pepo*, *Cucurbita* sp., *Dunaliella tertiolecta*, *Emiliania huxleyi*, *Emericella nidulans*, *Fusarium oxysporum*, *Fusarium oxysporum* JCM 11502, *Glyceria maxima*, *Glycine max*, *Gossypium hirsutum*, *Gracilaria chilensis*, *Gracilaria tenuistipitata*, *Helianthus annuus*, *Hordeum vulgare*, *Lactuca sativa*, *Lemna minor*, *Lupinus albus*, *Mycobacterium tuberculosis*, *Nicotiana plumbaginifolia*, *Nicotiana tabacum*, *Ogataea angusta*, *Ogataea polymorpha*, *Oryza sativa*, *Phaeocystis antarctica*, *Phragmites australis*, *Physcomitrella patens*, *Pisum arvense*, *Polytrichum commune*, *Pyropia yezoensis*, *Raphanus sativus*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus* E1F1, *Ricinus communis*, *Selaginella kraussiana*, *Sinapis alba*, *Skeletonema costatum*, *Skeletonema tropicum*, *Solanum lycopersicum*, *Spinacia oleracea*, *Suaeda maritima*, *Tetraselmis gracilis*, *Thalassia testudinum*, *Thalassiosira Antarctica*, *Thalassiosira pseudonana*, *Triticum aestivum*, *Triticum turgidum* subsp *durum*, *Ulva* sp. and/or *Zea mays*; and/or functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrate reductases; and/or functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrate reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrate reductases.

[0388] Preferred NADH-dependent nitrate reductases include the NADH-dependent nitrate reductases as obtained or derived from *Candida boidinii* (a nitrate reductase capable of utilizing both NADH and NADPH as electron donors), *Candida utilis* (a nitrate reductase capable of utilizing both NADH and NADPH as electron donors), *Fusarium oxysporum* (as described by Fujii et al. in their article titled "Denitrification by the Fungus *Fusarium oxysporum* Involves NADH-Nitrate Reductase" published in Biosci. Biotechnol. Biochem., 72 (2), pages 412-420, 2008, incorporated herein by reference), *Spinacia oleracea* and *Zea mays*.

[0389] Preferred NADH-dependent nitrate reductases hence include: NADH-dependent nitrate reductases comprising a polypeptide having an amino acid sequence of SEQ ID NO:74 and/or SEQ ID NO:75, as described herein; and/or functional homologues of SEQ ID NO:74 and/or SEQ ID NO:75 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of SEQ ID NO:74 and/or SEQ ID NO:75 respectively; and/or functional homologues of SEQ ID NO:74 and/or SEQ ID NO:75 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of SEQ ID NO:74 and/or SEQ ID NO:75 respectively. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20,

10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:74 and/or SEQ ID NO:75 respectively.

[0390] Preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrate reductase activity. More preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrate reductase activity selected from the group consisting of NADH-dependent nitrate reductases as obtained or derived from *Agrostemma githago*, *Amaranthus hybridus*, *Amaranthus tricolor*, *Ankistrodesmus braunii*, *Arabidopsis thaliana*, *Aspergillus niger*, *Aspergillus nidulans*, *Auxenochlorella pyrenoidosa*, *Bradyrhizobium* sp., *Bradyrhizobium* sp. 750, *Brassica juncea*, *Brassica*, *oleracea*, *Camellia sinensis*, *Candida boidinii*, *Candida utilis*, *Capsicum frutescens*, *Chenopodium album*, *Cyberlindnera jadinii*, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Capsicum frutescens*, *Chenopodium album*, *Chlamydomonas reinhardtii*, *Chlorella fusca*, *Chlorella* sp. *Chlorella* sp. Berlin, *Chlorella vulgaris*, *Conticribra weissflogii*, *Cucumis sativus*, *Cucurbita maxima*, *Cucurbita pepo*, *Cucurbita* sp., *Dunaliella tertiolecta*, *Emiliania huxleyi*, *Emericella nidulans*, *Fusarium oxysporum*, *Fusarium oxysporum* JCM 11502, *Glyceria maxima*, *Glycine max*, *Gossypium hirsutum*, *Gracilaria chilensis*, *Gracilaria tenuistipitata*, *Helianthus annuus*, *Hordeum vulgare*, *Lactuca sativa*, *Lemna minor*, *Lupinus albus*, *Mycobacterium tuberculosis*, *Nicotiana plumbaginifolia*, *Nicotiana tabacum*, *Ogataea angusta*, *Ogataea polymorpha*, *Oryza sativa*, *Phaeocystis antarctica*, *Phragmites australis*, *Physcomitrella patens*, *Pisum arvense*, *Polytrichum commune*, *Pyropia yezoensis*, *Raphanus sativus*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus* E1F1, *Ricinus communis*, *Selaginella kraussiana*, *Sinapis alba*, *Skeletonema costatum*, *Skeletonema tropicum*, *Solanum lycopersicum*, *Spinacia oleracea*, *Suaeda maritima*, *Tetraselmis gracilis*, *Thalassia Testudinum*, *Thalassiosira Antarctica*, *Thalassiosira pseudonana*, *Triticum aestivum*, *Triticum turgidum* subsp *durum*, *Ulva* sp. and *Zea mays*, and functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrate reductases; and functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrate reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrate reductases.

[0391] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of any of SEQ ID NO:74 and/or SEQ ID NO:75 or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:74 and/or SEQ ID NO:75. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:74 and/or SEQ ID NO:75 respectively.

[0392] The recombinant yeast cell may combine one or more genes encoding the above NADH-dependent nitrate reductase with one or more genes encoding an NADPH-dependent nitrite reductase. Preferably, however, the recombinant yeast cell combines one or more genes encoding the above NADH-dependent nitrate reductase with one or more genes encoding a NADH-dependent nitrite reductase.

[0393] Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as

can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:74, are listed in Table 12 below.

[0394] Table 12: Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:74, are listed in Table 12 below.

TABLE 12

Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 74, are listed in Table 12 below.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 74 [%]
K3YG56	Cluster: Nitrate reductase	PACMAD clade	94.06
P16081	Cluster: Nitrate reductase [NADH] 1	cellular organisms	90.54
UPI0004DEBCF9	Cluster: nitrate reductase [NADH]-like	<i>Zea mays</i>	88.57
A0A6G1CIJ1	Cluster: Uncharacterized protein	<i>Oryza meyeriana</i> var. <i>granulata</i>	85.7
P27967	Cluster: Nitrate reductase [NADH]	Pooideae	85.37
A0A3B6T8X1	Cluster: Nitrate reductase	Triticinae	86.11
A0A4S8IHP6	Cluster: Nitrate reductase	Musaceae	78.55
A0A2T7D896	Cluster: Nitrate reductase	<i>Panicum</i>	78.22
A0A199UYK8	Cluster: Nitrate reductase	<i>Ananas comosus</i>	77.45
A0A4S8J849	Cluster: Nitrate reductase	Mesangiospermae	77.67
A0A2H3Y7X8	Cluster: Nitrate reductase	Arecaceae	76.68
A0A4S8JVI1	Cluster: Nitrate reductase	<i>Musa balbisiana</i>	75.58
A0A200QET5	Cluster: Nitrate reductase	<i>Macleaya cordata</i>	72.39
F6H0M5	Cluster: Nitrate reductase	<i>Vitis</i>	72.61
A0A4Y7J3A8	Cluster: Nitrate reductase	<i>Papaver somniferum</i>	71.95
A0A218XU44	Cluster: Nitrate reductase	<i>Punica granatum</i>	71.73
A0A2C9PIN3	Cluster: Nitrate reductase	<i>Phellodendron amurense</i>	72.06
UPI000B8CEBA2	Cluster: nitrate reductase [NADH]-like	<i>Carica papaya</i>	71.51
A0A2R6RP67	Cluster: Nitrate reductase	<i>Actinidia chinensis</i> var. <i>chinensis</i>	70.63
A0A5J5BUB1	Cluster: Nitrate reductase	<i>Nyssa sinensis</i>	71.29
V4TCH9	Cluster: Nitrate reductase	Citrus	71.51
P36859	Cluster: Nitrate reductase [NADH]	<i>Petunia hybrida</i>	71.62
A0A0E0GDD5	Cluster: Nitrate reductase	Oryzinae	74.76
UPI001263D0A7	Cluster: nitrate reductase [NAD(P)H]-like	<i>Pistacia vera</i>	71.4
A0A061FFV7	Cluster: Nitrate reductase	Byttnerioideae	70.85
K3YPT2	Cluster: Nitrate reductase	Panicoideae	75.21

TABLE 12-continued

Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 74, are listed in Table 12 below.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 74 [%]
A0A1U7ZDA9	Cluster: Nitrate reductase	<i>Nelumbo nucifera</i>	72.17
A0A0P0VPX4	Cluster: Nitrate reductase	<i>Oryza</i>	74.17
A0A6A3CP56	Cluster: Nitrate reductase	<i>Hibiscus syriacus</i>	69.97
A0A067KLT8	Cluster: Nitrate reductase	<i>Jatropha curcas</i>	70.08
A0A2I4H0D3	Cluster: Nitrate reductase	<i>Juglans regia</i>	69.86
A0A5J9UYH6	Cluster: Uncharacterized protein (Fragment)	PACMAD clade	74.26
A0A5N6RHU6	Cluster: Nitrate reductase	fabids	70.63
A0A1U8KZC0	Cluster: Nitrate reductase	Malvaceae	70.74
UPI000C0393D8	Cluster: nitrate reductase [NADH]-like	<i>Durio zibethinus</i>	73.33
B9H7A1	Cluster: Nitrate reductase	Saliceae	70.41
P17569	Cluster: Nitrate reductase [NADH]	Cucurbitaceae	70.74
UPI000581598E	Cluster: nitrate reductase [NADH] 2-like	<i>Sesamum indicum</i>	70.41
UPI000F7C3BDF	Cluster: nitrate reductase [NADH] 2 isoform X1	<i>Abrus precatorius</i>	74.12
P08509	Cluster: Nitrate reductase [NADH] 2	Solanaceae	70.08
P39867	Cluster: Nitrate reductase [NADH], clone PBNBR1405	Brassicaceae	71.65
UPI000C1D548F	Cluster: nitrate reductase [NADH] 2-like	<i>Olea europaea</i> var. <i>syvestris</i>	71.22
P39866	Cluster: Nitrate reductase [NADH] 2	Phaseoleae	73.88
P27968	Cluster: Nitrate reductase [NAD(P)H]	Triticeae	72.82
A8IJ84	Cluster: Nitrate reductase	<i>Beta vulgaris</i>	71.14
A0A2U1KIT7	Cluster: Nitrate reductase	<i>Artemisia annua</i>	69.64
A0A165XNP4	Cluster: Nitrate reductase	<i>Daucus carota</i> subsp. <i>sativus</i>	69.31
B9RJB1	Cluster: Nitrate reductase	Mesangiospermae	74
P43101	Cluster: Nitrate reductase [NADH]	Asteraceae	69.86
I1LW34	Cluster: Nitrate reductase	<i>Glycine</i> subgen. <i>Soja</i>	74.49
UPI000980E182	Cluster: nitrate reductase [NADH]-like	<i>Ananas comosus</i>	75.54
A0A4P1R4X1	Cluster: Nitrate reductase	<i>Lupinus angustifolius</i>	71.22
Q8H1T7	Cluster: Nitrate reductase	<i>Tilia platyphyllos</i>	69.42
I1IER4	Cluster: Nitrate reductase	<i>Brachypodium distachyon</i>	73.47
A0A1U7Z2Y3	Cluster: Nitrate reductase	<i>Nelumbo nucifera</i>	72.45

TABLE 12-continued

Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 74, are listed in Table 12 below.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 74 [%]
A0A068VGB5	Cluster: Nitrate reductase	<i>Coffea</i>	68.87
A0A2T7F896	Cluster: Nitrate reductase	<i>Panicum</i>	73.27
A0A2Z7C1Z2	Cluster: Nitrate reductase	<i>Doroceras hygroscopicum</i>	69.31
UPI00053C7726	Cluster: nitrate reductase [NADH] 2	<i>Tarenaya hassleriana</i>	69.64
A0A445BT68	Cluster: Nitrate reductase (NADH)	<i>Arachis</i>	72.55
P39869	Cluster: Nitrate reductase [NADH]	<i>Lotus japonicus</i>	73.76
UPI0010FB6302	Cluster: nitrate reductase [NADH] 2	<i>Cajanus cajan</i>	73.64
Q93XS1	Cluster: Nitrate reductase	<i>Prunus</i>	73.76
A0A022Q6Q6	Cluster: Nitrate reductase	<i>Erythranthe guttata</i>	69.2
A0A1U8NV83	Cluster: Nitrate reductase	<i>Gossypium</i>	73.04
P27783	Cluster: Nitrate reductase [NAD(P)H]	fabids	69.53
A0A2T8KWM5	Cluster: Nitrate reductase	<i>Panicum hallii</i>	72.93
A0A2I4GBD5	Cluster: Nitrate reductase	<i>Juglans regia</i>	69.42
A0A5N6PSR2	Cluster: Nitrate reductase	<i>Mikania micrantha</i>	69.2
A0A2P6PI42	Cluster: Nitrate reductase	Mesangiospermae	73.88
E7ELN7	Cluster: Nitrate reductase	IRL clade	70.41
A0A4U5QSR8	Cluster: Nitrate reductase	<i>Populus</i>	73.64
A0A2K2BFU1	Cluster: Nitrate reductase	Saliceae	74.12
A0A1Z5RNB1	Cluster: Nitrate reductase	<i>Sorghum bicolor</i>	76.67
A0A0M4P4G3	Cluster: Nitrate reductase	rosids	73.28
M4CVT6	Cluster: Nitrate reductase	<i>Brassica campestris</i>	70.47
A0A251SAE9	Cluster: Nitrate reductase	<i>Helianthus annuus</i>	72.55
A0A2G5DUU4	Cluster: Nitrate reductase	<i>Aquilegia coerulea</i>	71.24
P23312	Cluster: Nitrate reductase [NADH]	Chenopodioideae	72.79
P11832	Cluster: Nitrate reductase [NADH] 1	Brassicaceae	71.22
P11035	Cluster: Nitrate reductase [NADH] 2	rosids	68.76
W1NJD0	Cluster: Nitrate reductase	Magnoliopsida	69.31
UPI000C048389	Cluster: nitrate reductase [NADH]-like	<i>Durio zibethinus</i>	71.36
A0A6A3C902	Cluster: Nitrate reductase	<i>Hibiscus syriacus</i>	73.16
P39870	Cluster: Inducible nitrate reductase [NADH] 2	<i>Glycine</i> subgen. <i>Soja</i>	73.52
UPI000F7C9CC8	Cluster: inducible nitrate reductase [NADH] 2	<i>Abrus precatorius</i>	72.79
A0A6A5MI86	Cluster: Nitrate reductase	<i>Lupinus</i>	70.31

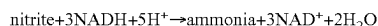
Nitrite Reductase

[0395] As indicated above, nitrite reductase catalyzes the reduction of nitrite to ammonia (NH₃).

[0396] Preferably the recombinant yeast cell as described herein comprises at least one or more genes encoding a NADH-dependent nitrite reductase.

[0397] By a NADH-dependent nitrite reductase is herein understood a nitrite reductase that is exclusively depended on NADH as a co-factor or that is predominantly dependent on NADH as a cofactor. Preferably the NADH-dependent nitrite reductase has a ratio of catalytic efficiency for NADPH/NADP⁺ as a cofactor $(k_{cat}/K_m)^{NADP^+}$ to NADH/NAD⁺ as cofactor $(k_{cat}/K_m)^{NAD^+}$, i.e. a catalytic efficiency ratio $(k_{cat}/K_m)^{NADP^+}:(k_{cat}/K_m)^{NAD^+}$, of more than 1:1, more preferably of equal to or more than 2:1, still more preferably of equal to or more than 5:1, even more preferably of equal to or more than 10:1, yet even more preferably of equal to or more than 20:1, even still more preferably of equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the NADH-dependent nitrite reductase may have a catalytic efficiency ratio $(k_{cat}/K_m)^{NADP^+}:(k_{cat}/K_m)^{NAD^+}$ of equal to or less than 1,000,000,000:1 (i.e. $1 \cdot 10^9$). Most preferably the NADH-dependent nitrite reductase is exclusively depended on NADH/NAD⁺ as a co-factor. That is, most preferably the NADH-dependent nitrite reductase has an absolute requirement for NADH/NAD⁺ as a cofactor instead of NADPH/NADP⁺ as a cofactor.

[0398] Preferably the NADH-dependent nitrite reductase is a NADH-dependent nitrite reductase with enzyme classification EC 1.7.1.15 (i.e. with EC number EC 1.7.1.15). Suitably the NADH-dependent nitrite reductase, also referred to as NADH-dependent nitrite oxidoreductase, is an enzyme that catalyzes at least the following chemical reaction:



The person skilled in the art will understand that the ammonia may also be present and/or referred to as so-called ammonium hydroxide NH₄OH

[0399] Suitable NADH-dependent nitrite reductases may include one or more NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), *Arcobacter ellisii*, *Arcobacter pacificus*, *Bacillus subtilis*, *Bacillus subtilis* JH642, *Cupriavidus taiwanensis*, *Escherichia coli*, *Ralstonia taiwanensis*, *Ralstonia syzygii*, *Ralstonia solanacearum*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus*, *Paraburkholderia ribeironis*; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrite reductases; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrite reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrite reductases.

[0400] *Escherichia coli* utilizes several distinct enzymes in its nitrite assimilation pathway. The nirD gene encodes a NADH-dependent nitrite reductase (NADH) small subunit, whilst the nirB gene encodes a NADH-dependent nitrite reductase (NADH) large subunit.

[0401] Preferred NADH-dependent nitrite reductases include the NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), a nitrite reductase capable of utilizing both NADH and NADPH as electron donors, and/or *Escherichia coli*. At high nitrate and/or nitrite concentrations, the nitrite reductase encoded by the nirB gene of *Escherichia coli* is especially preferred.

[0402] Preferred NADH-dependent nitrite reductases hence include: NADH-dependent nitrite reductases comprising a polypeptide having an amino acid sequence of SEQ ID NO:76 (*E. coli* nitrite reductase small subunit encoded by nirD) and/or SEQ ID NO:77 (*E. coli* nitrite reductase large subunit encoded by nirB) and/or SEQ ID NO:78 (*Emericella nidulans* nitrate reductase encoded by niiA), as described herein; and/or functional homologues of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 respectively; and/or functional homologues of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 respectively. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 respectively.

[0403] Preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrite reductase activity. More preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrite reductase activity selected from the group consisting of NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), *Arcobacter ellisii*, *Arcobacter pacificus*, *Bacillus subtilis*, *Bacillus subtilis* JH642, *Cupriavidus taiwanensis*, *Escherichia coli*, *Ralstonia taiwanensis*, *Ralstonia syzygii*, *Ralstonia solanacearum*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus*, *Paraburkholderia ribeironis*; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrite reductases; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrite reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or

5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrite reductases.

[0404] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of any of SEQ ID NO:76 (*E. coli* nitrate reductase small subunit encoded by nirD) and/or SEQ ID NO:77 (*E. coli* nitrate reductase large subunit encoded by nirB) and/or SEQ ID NO:78 (*Emericella nidulans* nitrate reductase encoded by niiA), or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 respectively.

[0405] The recombinant yeast cell may combine one or more genes encoding one or more of the above NADH-

dependent nitrite reductases with one or more genes encoding an NADPH-dependent nitrate reductase. Preferably, however, the recombinant yeast cell combines one or more genes encoding one or more of the above NADH-dependent nitrite reductases with one or more genes encoding a NADH-dependent nitrate reductase.

[0406] Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/) as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:76 (small subunit encoded by nirD), are listed in Table 13 below.

[0407] Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/) as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:77 (large subunit encoded by nirB), are listed in Table 14 below.

TABLE 13

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/) as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 76 (small subunit encoded by nirD).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 76 [%]
E2QFM2	Cluster: Nitrite reductase	Gammaproteobacteria	99.07
A0A078LHD3	Cluster: Nitrite reductase small subunit	Enterobacteriaceae	94.44
P0A230	Cluster: Nitrite reductase (NADH) small subunit	Bacteria	93.52
A0A170JME2	Cluster: Nitrite reductase [NAD(P)H] small subunit	Enterobacteriaceae	91.67
A0A0F6TX22	Cluster: Nitrite reductase	Gammaproteobacteria	91.67
A0A6D0EP17	Cluster: Nitrite reductase small subunit NirD	Enterobacteriaceae	87.04
A0A198H1N5	Cluster: Nitrite reductase [NAD(P)H] small subunit	Bacteria	86.11
A0A447MQ21	Cluster: Nitrite reductase [NAD(P)H] small subunit	Enterobacteriaceae	84.26
UPI00077B73C7	Cluster: nitrite reductase small subunit NirD	<i>Enterobacter timonensis</i>	83.33
A0A094XNF5	Cluster: Nitrite reductase (NADH) small subunit	Enterobacterales	82.41
UPI00105F841D	Cluster: nitrite reductase small subunit NirD	<i>Enterobacter</i> sp. OV724	81.48
A0A1X0XB32	Cluster: Nitrite reductase small subunit	Enterobacterales	81.48
A0A1A9FFN3	Cluster: Nitrite reductase small subunit	<i>Lelliottia amnigena</i>	81.48
UPI0002D5E0A7	Cluster: nitrite reductase small subunit NirD	<i>Metakosakonia massiliensis</i>	80.56
UPI0004E24438	Cluster: nitrite reductase small subunit NirD	<i>Siccibacter colletis</i>	79.63

TABLE 13-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 76 (small subunit encoded by nirD).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 76 [%]
A0A514EKN7	Cluster: Nitrite reductase [NAD(P)H] small subunit	Enterobacteriaceae	77.78
A0A2P8VHS5	Cluster: Nitrite reductase small subunit NirD	Enterobacteriaceae	80.56
A0A4V3BP91	Cluster: Nitrite reductase (NADH) small subunit	cellular organisms	80.56
A0A3R9F800	Cluster: Nitrite reductase small subunit NirD	Enterobacteriaceae	79.63
R9VMQ4	Cluster: Nitrite reductase small subunit	Enterobacteriaceae	78.7
A0A285KH74	Cluster: Nitrite reductase (NADH) small subunit	<i>Enterobacter</i> sp. CC120223-11	80.37
A0A658U6S1	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Enterobacter</i>	77.78
A0A085A429	Cluster: Nitrite reductase [NAD(P)H] small subunit	Enterobacteriaceae	78.7
A0A1V0L823	Cluster: Nitrite reductase small subunit	<i>Kosakonia</i>	76.85
UPI001431C56A	Cluster: nitrite reductase small subunit NirD	<i>Cedecea</i> colo	75
A0A085G1A1	Cluster: Nitrite reductase [NAD(P)H] small subunit	Enterobacteriaceae	75
A0A089PL75	Cluster: Nitrite reductase	<i>Pluralibacter gergoviae</i>	76.85
A0A3C0H311	Cluster: Nitrite reductase small subunit	Enterobacteriaceae	77.78
A0A3N2DXC8	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	unclassified <i>Enterobacter</i>	75.93
A0A0L0ATR1	Cluster: Nitrite reductase	Enterobacteriaceae	78.5
K8AGI3	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Cronobacter condimenti</i> 1330	75
A0A116EAU0	Cluster: Nitrite reductase (NADH) small subunit	<i>Enterobacter</i> sp. kpr-6	75
A0A0D5X0Q1	Cluster: Nitrite reductase	Enterobacteriaceae	75
C9Y259	Cluster: Nitrite reductase [NAD(P)H] small subunit	Enterobacteriaceae	74.07
A0A4R0GWC0	Cluster: Nitrite reductase small subunit NirD	<i>Kosakonia</i> quasiasacchari	75
UPI00057BF926	Cluster: nitrite reductase small subunit NirD	<i>Enterobacter</i> sp. Bisph1	73.15
A0A495AHM0	Cluster: Nitrite reductase small subunit NirD	<i>Enterobacter</i> sp. R1(2018)	72.22
A0A447QKU3	Cluster: Nitrite reductase [NAD(P)H] small subunit	Yersiniaceae	75.47

TABLE 13-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 76 (small subunit encoded by nirD).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 76 [%]
I2B4C9	Cluster: Nitrite reductase (NAD(P)H)	<i>Shimwellia blattae</i> (strain ATCC 29907/ DSM 4481/JCM 1650/ NBRC 105725/CDC 9005-74)	73.83
UPI0012B7CAE2	Cluster: nitrite reductase small subunit NirD	<i>Erwinia</i> sp. CPCC 100877	72.22
UPI00073DB139	Cluster: nitrite reductase small subunit NirD	[<i>Erwinia</i>] <i>teleogrylli</i>	73.83
UPI000237D150	Cluster: nitrite reductase small subunit NirD	<i>Enterobacter mori</i>	80.56
A0A4P8YE33	Cluster: Nitrite reductase small subunit NirD	Enterobacteriaceae	72.22
UPI0005DC1E97	Cluster: nitrite reductase small subunit NirD	<i>Yersinia rohdei</i>	71.03
A0A0U1HQI4	Cluster: Nitrite reductase small subunit	<i>Yersinia</i>	71.03
A0A0K0HM15	Cluster: Nitrite reductase [NAD(P)H] small subunit	Enterobacterales	72.9
H8NP95	Cluster: Nitrite reductase small subunit	Yersiniaceae	69.16
A0A2S4QQ85	Cluster: Nitrite reductase small subunit	Enterobacteriaceae	69.44
A0A0A7RXK3	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Frischella perrara</i>	71.3
D4DYL5	Cluster: Nitrite reductase [NAD(P)H], small subunit	<i>Serratia odorifera</i> DSM 4582	73.58
A0A370QTS9	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Enterobacillus tribolii</i>	71.7
A0A1B9JUB0	Cluster: Nitrite reductase small subunit	<i>Gilliamella apicola</i>	69.16
A0A085U6Z8	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Yersinia ruckeri</i>	72.22
UPI00156B8D43	Cluster: UPI00156B8D43 related cluster	unknown	70.75
A0A0T9KRE5	Cluster: Nitrite reductase small subunit	<i>Yersinia</i>	70.37
A0A066TFS8	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Gilliamella</i>	69.16
A0A1B9M9X4	Cluster: Nitrite reductase small subunit	<i>Gilliamella</i>	68.22
A0A0E1NM95	Cluster: Nitrite reductase (NAD(P)H)	Yersiniaceae	68.52
A0A080K0G0	Cluster: Ferredoxin subunits of nitrite reductase and ring-hydroxylating dioxygenase	<i>Snodgrassella alvi</i> SCGC AB-598-O02	69.16
A0A0Q4N870	Cluster: Nitrite reductase small subunit	Enterobacterales	67.29
UPI0004A391D4	Cluster: nitrite reductase small subunit NirD	<i>Tatumella saanichensis</i>	69.81

TABLE 13-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 76 (small subunit encoded by nirD).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 76 [%]
A0A1B9ZVE6	Cluster: Nitrite reductase small subunit	<i>Gilliamella apicola</i>	67.29
A0A0T9JI92	Cluster: Nitrite reductase small subunit	<i>Yersinia</i>	68.52
A0A2N5EJM2	Cluster: Nitrite reductase small subunit NirD	<i>Chimaeribacter</i>	67.59
A0A2V4E983	Cluster: Nitrite reductase small subunit	<i>Gilliamella</i>	65.74
A0A419N5U0	Cluster: Nitrite reductase small subunit NirD	Yersiniaceae	67.29
A0A0F7H970	Cluster: Nitrite reductase	<i>Serratia</i>	70.75
A0A318NZT5	Cluster: Nitrite reductase small subunit NirD	Yersiniaceae	69.81
A0A250B1W9	Cluster: Nitrite reductase small subunit	<i>Gibbsiella quercinecans</i>	69.81
A0A6G9I849	Cluster: Nitrite reductase small subunit NirD	<i>Orbus</i> sp. IPMB12	65.74
A0A419M923	Cluster: Nitrite reductase small subunit NirD	<i>Rahmella inusitata</i>	67.92
A0A389M9P6	Cluster: Nitrite reductase (NADH) small subunit	Enterobacterales	67.29
A0A0N8GUB4	Cluster: Nitrite reductase (NADH) small subunit	Vibrionales	67.96
A0A443IEH9	Cluster: Nitrite reductase	<i>Pantoea</i> sp. LMG 27579	68.87
A0A1X1MLP8	Cluster: Nitrite reductase	<i>Vibrio</i> sp. qd031	69.9
A0A542BGK2	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Serratia fonticola</i>	68.87
A0A2U3B6U3	Cluster: Nitrite reductase small subunit	<i>Vibrio</i> sp. E4404	67.92
A0A240CDV1	Cluster: Nitrite reductase [NAD(P)H] small subunit	Gammaproteobacteria	67.92
A0A2S9I8J3	Cluster: Nitrite reductase small subunit	Bacteria	67.29
A0A495RE76	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Orbus hercynius</i>	67.59
A0A178KL05	Cluster: Nitrite reductase small subunit	<i>Photobacterium</i>	66.04
A0A198GJ99	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Cosenzaea myxofaciens</i> ATCC 19692	63.55
A0A2N0W0U3	Cluster: Nitrite reductase small subunit	Enterobacterales	66.36
A0A1B9JKC9	Cluster: Nitrite reductase small subunit	<i>Gilliamella apicola</i>	66.67
UPI0011BEE383	Cluster: nitrite reductase small subunit NirD	<i>Pantoea</i> sp. CCBC3-3-1	65.09
A0A085G554	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Ewingella americana</i>	67.29
A0A1B1NU77	Cluster: Nitrite reductase (NADH)	<i>Vibrio</i>	67.96

TABLE 13-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 76 (small subunit encoded by nirD).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 76 [%]
A0A2N7G6C9	Cluster: Nitrite reductase small subunit	<i>Vibrio</i> sp. 10N.261.55.A7	67.96
A0A349FFE1	Cluster: Nitrite reductase small subunit	<i>Vibrio</i> sp.	69.52
A0A1B9JS56	Cluster: Nitrite reductase small subunit	<i>Gilliamella apicola</i>	64.49
A0A366XH21	Cluster: Nitrite reductase small subunit	Vibrionales	66.67
UPI00062A46AC	Cluster: nitrite reductase small subunit NirD	<i>Tatumella morbirosei</i>	67.59
UPI00046A4834	Cluster: nitrite reductase small subunit NirD	<i>Pantoea</i> sp. IMH	68.87

TABLE 14

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 77 (large subunit encoded by nirB).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 77 [%]
W9B2S1	Cluster: NirB protein	Bacteria	93.27
A0A486K457	Cluster: Nitrite reductase [NAD(P)H]	Enterobacterales	93.15
A0A1B7KIV8	Cluster: Nitrite reductase [NAD(P)H] large subunit	Enterobacterales	85.41
A0A087L3N6	Cluster: Nitrite reductase	<i>Serratia</i>	84.6
A0A654DEI2	Cluster: Nitrite reductase, large subunit, nucleotide-and Fe/S-cluster binding	Enterobacterales	84.36
UPI00156BAB07	Cluster: UPI00156BAB07 related cluster	unknown	81.3
A0A2I0FWV2	Cluster: Nitrite reductase (NAD(P)H)	Enterobacterales	81.52
I2B4D0	Cluster: Nitrite reductase (NAD(P)H)	bacterium CwR94 <i>Shimwellia blattae</i> (strain ATCC 29907/ DSM 4481/JCM 1650/NBRC 105725/ CDC 9005-74)	85.24
A0A370QTZ7	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	<i>Enterobacillus tribolii</i>	83.39
UPI000DEFBD5D	Cluster: nitrite reductase large subunit	<i>Edaphovirga cremea</i>	83.53
A0A380ASL5	Cluster: Nitric oxide reductase FIRd-NAD(+) reductase	cellular organisms	84.36
A0A443IEI0	Cluster: Nitrite reductase	unclassified <i>Pantoea</i>	81.42
A0A221T4L0	Cluster: Nitrite reductase [NAD(P)H] large subunit	Enterobacterales	80.45
A0A2N5EJN0	Cluster: Nitrite reductase (NAD(P)H)	Enterobacterales	82.68

TABLE 14-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 77 (large subunit encoded by nirB).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 77 [%]
A0A377N6S4	Cluster: Benzene 1,2-dioxygenase system ferredoxin--NAD(+) reductase subunit	Enterobacterales	82.82
A0A4P8SFP8	Cluster: Nitrite reductase (NAD(P)H)	<i>Nissabacter</i> sp. SGAir0207	82.11
A0A0L7TEE8	Cluster: Nitrite reductase	<i>Erwinia iniecta</i>	79.62
A0A1W6B3D3	Cluster: Nitrite reductase large subunit	Erwiniaceae	79.88
A0A506V9E3	Cluster: Nitrite reductase large subunit	<i>Mixta</i>	79.41
UPI000468AD1D	Cluster: nitrite reductase large subunit	Erwiniaceae	81.02
A0A0N0Z921	Cluster: Nitrite reductase [NAD(P)H] large subunit	<i>Moellerella wisconsensis</i>	79.34
B4EXR2	Cluster: Nitrite reductase [NAD(P)H] large subunit	Enterobacterales	78.34
A0A198GKA3	Cluster: Nitrite reductase [NAD(P)H] large subunit	<i>Cosenzaea myxofaciens</i> ATCC 19692	78.11
A0A0F3LTP0	Cluster: Nitrite reductase	Enterobacterales	78.46
E1SGS9	Cluster: Nitrite reductase (NAD(P)H) subunit	Erwiniaceae	77.78
A0A1X1EZA9	Cluster: Nitrite reductase large subunit	<i>Pantoea cypripedii</i>	78.25
A0A3A9I4K8	Cluster: Nitrite reductase large subunit	Erwiniaceae	77.9
A0A2S9I8J4	Cluster: Nitrite reductase (NAD(P)H)	Erwiniaceae	79.48
A0A0R4FQI6	Cluster: Nitrite reductase, large subunit, nucleotide-binding	<i>Xenorhabdus nematophila</i>	76.42
UPI001265FD68	Cluster: nitrite reductase large subunit	<i>Erwinia endophytica</i>	77.49
UPI00146D663B	Cluster: nitrite reductase large subunit	<i>Rahnella</i> sp. SAP-1	78.32
A0A3S4CWL7	Cluster: Rubredoxin-NAD(+) reductase	Enterobacterales	78.65
A0A1Y2S8S4	Cluster: NADH oxidase	<i>Xenorhabdus vietnamensis</i>	76.78
A0A0A3YW59	Cluster: Nitrite reductase	<i>Erwinia typographi</i>	79.31
A0A506PS78	Cluster: Nitrite reductase large subunit	<i>Pantoea</i>	76.12
D8MY05	Cluster: Nitrite reductase [NAD(P)H] large subunit	<i>Erwinia</i>	79.12
E6WGY1	Cluster: Nitrite reductase (NAD(P)H), large subunit	unclassified <i>Pantoea</i>	75.62
A0A3R8ZPT1	Cluster: Nitrite reductase large subunit	unclassified <i>Erwinia</i>	75.3
A0A5M7NEQ9	Cluster: Nitrite reductase large subunit	Enterobacterales	75.3
A0A2V2BEC9	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	<i>Pantoea</i>	76.36
A0A1X0W567	Cluster: Nitrite reductase large subunit	<i>Rouxiella</i>	77.88
A0A6G9I851	Cluster: Nitrite reductase large subunit	<i>Orbus</i> sp. IPMB12	77.66

TABLE 14-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 77 (large subunit encoded by nirB).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 77 [%]
UPI0011BF5221	Cluster: nitrite reductase large subunit	<i>Pantoea</i> sp. CCBC3-3-1	76.66
A0A556RIT3	Cluster: Nitrite reductase large subunit	<i>Gilliamella</i>	74.91
A0A1R4J5I6	Cluster: Nitrite reductase [NAD(P)H] large subunit	<i>Vibrio</i>	72.82
A0A1B9MVL8	Cluster: Nitrite reductase large subunit	<i>Gilliamella</i>	75.03
UPI0013596BEF	Cluster: nitrite reductase large subunit	unclassified	72.59
A0A1B9L8J9	Cluster: Nitrite reductase large subunit	<i>Psychromonas</i> <i>Gilliamella apicola</i>	74.79
A0A095TED5	Cluster: Nitrite reductase	<i>Tatumella</i>	74.67
A0A2S7VXZ8	Cluster: Nitrite reductase large subunit	<i>Photobacterium</i>	73.6
A0A495REZ8	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	<i>Orbus hercynius</i>	75.15
A0A2T3QB92	Cluster: Nitric oxide reductase FIRd-NAD(+) reductase	<i>Photobacterium damsela</i>	75.18
A0A1X0WF55	Cluster: Nitrite reductase large subunit	Gammaproteobacteria	76.51
A0A2I8TVB9	Cluster: Nitrite reductase large subunit	<i>Vibrio</i>	75.33
A0A2T3NPM2	Cluster: Nitrite reductase large subunit	<i>Photobacterium sanctipauli</i>	74.85
R9PFS3	Cluster: Nitrite reductase	<i>Agarivorans</i>	73.18
A0A0B7J736	Cluster: Nitrite reductase, large subunit, NAD(P)H-binding	<i>Photobacterium</i>	73.6
U4KF90	Cluster: Nitrite reductase [NAD(P)H] large subunit	<i>Vibrio nigripulchritudo</i>	73.78
A0A1Y6IN13	Cluster: Nitrite reductase [NAD(P)H]	<i>Vibrio mangrovi</i>	73.54
A0A1L9L1Z8	Cluster: Nitrite reductase [NAD(P)H]	Vibrionaceae	73.78
A0A0A7RZE6	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	<i>Frischella perrara</i>	73.24
A0A1Q9GSK0	Cluster: Nitrite reductase large subunit	<i>Photobacterium</i>	74.97
UPI000200DAF9	Cluster: nitrite reductase large subunit	<i>Vibrio furnissii</i>	74.97
A0A378PW16	Cluster: Nitric oxide reductase FIRd-NAD(+) reductase	Vibrionaceae	72.82
A0A1B9JKB8	Cluster: Nitrite reductase large subunit	<i>Gilliamella apicola</i>	72.65
D4ZBU5	Cluster: Nitrite reductase [NAD(P)H], large subunit	<i>Shewanella violacea</i> (strain JCM 10179/ CIP 106290/LMG 19151/DSS12)	72.4
UPI00117F52C3	Cluster: nitrite reductase large subunit	<i>Vibrio furnissii</i>	74.97
UPI0001B93571	Cluster: nitrite reductase large subunit	<i>Vibrio furnissii</i>	74.97
A0A1B1NUU8	Cluster: Nitrite reductase (NADH)	Vibrionaceae	72.94
A0A178KL92	Cluster: Nitrite reductase large subunit	<i>Photobacterium</i>	73.39

TABLE 14-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 77 (large subunit encoded by nirB).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 77 [%]
A0A1T4RZM3	Cluster: Nitrite reductase (NADH) large subunit	<i>Vibrio cincinnatiensis</i>	74.37
UPI0012ADB8F8	Cluster: nitrite reductase large subunit	<i>Vibrio furnissii</i>	74.97
A0A418YIX2	Cluster: Nitrite reductase large subunit	<i>Motilimonas</i> sp.	73
Q6LS87	Cluster: Putative nitrite reductase (NAD(P)H), large subunit	<i>Photobacterium</i>	73.67
A0A4R3I7X5	Cluster: Nitrite reductase (NADH) large subunit	<i>Reinekea marinisedimentorum</i>	73.72
UPI0003F87F52	Cluster: nitrite reductase large subunit	<i>Psychromonas</i>	72.02
UPI00037B4F3D	Cluster: nitrite reductase large subunit	<i>Uliginosibacterium gangwonense</i>	70.2
A0A2N7DBN5	Cluster: Nitrite reductase large subunit	<i>Vibrio</i> sp. 10N.286.49.B3	73.06
A0A484GE91	Cluster: Nitrite reductase large subunit	<i>Candidatus Schmidhempelia bombi</i> str. Bimp	72.16
A0A2M8H0G2	Cluster: Nitrite reductase (NAD(P)H)	unclassified <i>Vibrio</i>	73.84
U3CNX4	Cluster: Nitrite reductase large subunit	<i>Vibrio</i>	73.78
UPI0010A63CB5	Cluster: nitrite reductase large subunit	<i>Vibrio</i> sp. H11	73.12
A0A090IJG5	Cluster: Nitrite reductase (NAD(P)H) large subunit	<i>Moritella</i>	71.92
A0A2N0WYN5	Cluster: Nitrite reductase (NAD(P)H)	<i>Psychromonas</i>	72.87
UPI000B356675	Cluster: nitrite reductase large subunit	<i>Thaumasiovibrio subtropicus</i>	73.12
A0A1R4LRM2	Cluster: Nitrite reductase [NAD(P)H]	<i>Vibrio</i>	69.41
UPI00082A845F	Cluster: nitrite reductase large subunit	<i>Erwinia</i> sp. ErVv1	72.27
B8KAR6	Cluster: Nitrite reductase (NAD(P)H), large subunit	<i>Vibrio</i>	73.3
A0A1G8E1S6	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	<i>Vibrio xiamenensis</i>	72.59
A0A0C5WZ76	Cluster: Nitrite reductase (NAD(P)H), large subunit	<i>Photobacterium</i>	72.94
A0A2G4AWG1	Cluster: Nitrite reductase [NAD(P)H]	Vibrionaceae	73.18

Nitrate/Nitrite Transporter

[0408] Preferably, the recombinant yeast cell further comprises one or more genetic modifications that result in an increased transport of oxidized nitrogen source, such as nitrate or nitrite, into the yeast cell. More preferably the recombinant yeast cell further comprising one or more genes encoding a nitrate and/or nitrite transporter.

[0409] Suitable transporters may include the sulphite transporters Ssu1 and Ssu2 (as described by Cabrera et al in their article titled “Molecular Components of Nitrate and Nitrite Efflux in Yeast”, published February 2014 Volume 13

Number 2 Eukaryotic Cell p. 267-278, herein incorporated by reference); and the nitrate/nitrite transporter YNT1 derived from *Pichia angusta* (also referred to as *Hansenula polymorpha*) and/or a functional homologues of one or more of such nitrate/nitrite transporters comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of the aforementioned nitrate/nitrite transporters; and/or functional homologues of one or more of such nitrate/nitrite transporters comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of

such aforementioned nitrate/nitrite transporters, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned nitrate/nitrite transporter YNT1.

[0410] Preferably the recombinant yeast cell comprises a nucleic acid sequence encoding the nitrate/nitrite transporter YNT1 derived from *Pichia angusta* and/or a functional homologues of such nitrate/nitrite transporter YNT1 comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with nitrate/nitrite transporter YNT1; and/or functional homologues of such nitrate/nitrite transporter YNT1 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned nitrate/nitrite transporter YNT1, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned nitrate/nitrite transporter YNT1.

[0411] Preferred nitrate/nitrite transporter hence include: nitrate/nitrite transporters comprising a polypeptide having an amino acid sequence of SEQ ID NO:79, as described

herein; and/or functional homologues of SEQ ID NO:79 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with SEQ ID NO:79; and/or functional homologues of SEQ ID NO:79 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO:79. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:79.

[0412] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of SEQ ID NO:79 or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:79. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:79 respectively.

[0413] Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:79 are listed in Table 15 below.

TABLE 15

Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 79.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 79 [%]
W1QLT0	Cluster: Nitrate/nitrite transporter	<i>Ogataea</i>	95.66
W6MFQ5	Cluster: Nitrate/nitrite transporter	<i>Kuraishia capsulata</i> CBS 1993	56.89
A0A1E4TX12	Cluster: Nitrate/nitrite transporter	<i>Pachysolen tannophilus</i> NRRL Y-2460	56.55
A0A1E3NV96	Cluster: Nitrate/nitrite transporter	<i>Wickerhamomyces anomalus</i> (strain ATCC 58044/CBS 1984/NCYC 433/NRRL Y-366-8)	56.52
K0KNB9	Cluster: Nitrate/nitrite transporter	<i>Wickerhamomyces ciferrii</i> (strain F-60-10/ATCC 14091/CBS 111/JCM 3599/NBRC 0793/NRRL Y-1031)	56.46
A0A0H5C0H7	Cluster: Nitrate/nitrite transporter	<i>Cyberlindnera jadinii</i> (strain ATCC 18201/CBS 1600/CCRC 20928/JCM 3617/NBRC 0987/NRRL Y-1542)	54.44
A0A5P8N9I7	Cluster: Nitrate/nitrite transporter	<i>Cyberlindnera americana</i>	54.58
A0A061BB65	Cluster: CYFA0S24e00188g1_1	<i>Cyberlindnera fabianii</i>	52.72
A0A060T893	Cluster: Nitrate/nitrite transporter	<i>Blastobotrys adenivorans</i>	47.18
B3WFS7	Cluster: Nitrate/nitrite transporter	<i>Blastobotrys adenivorans</i>	47.79
A0A261XZN8	Cluster: Nitrate/nitrite transporter	<i>Bifiguratus adelaidae</i>	47.49

TABLE 15-continued

Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 79.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 79 [%]
A0A0E9N9H8	Cluster: Nitrate/nitrite transporter	<i>Saitoella complicata</i> (strain BCRC 22490/ CBS 7301/JCM 7358/ NBRC 10748/RRRL Y-17804)	46.17
A0A060T786	Cluster: Nitrate/nitrite transporter	<i>Blastobotrys adenivorans</i>	43.91
A0A0A1T938	Cluster: Nitrate/nitrite transporter	<i>Torrubiella hemipterigena</i>	42.57
A0A3D8R1D7	Cluster: Nitrate/nitrite transporter	<i>Coleophoma</i>	42.66
A0A370TMI7	Cluster: Nitrate/nitrite transporter	<i>Venustampulla echinocandica</i>	41.72
A0A3N4HFN0	Cluster: Nitrate/nitrite transporter	<i>Ascobolus immersus</i> RN42	42.16
A0A167RLM4	Cluster: Nitrate/nitrite transporter	<i>Cordyceps fumosorosea</i> ARSEF 2679	40.04
A0A179I4H5	Cluster: Nitrate/nitrite transporter	<i>Akanthomyces lecanii</i>	40.94
A0A0G2E479	Cluster: Nitrate/nitrite transporter	<i>Phaeomoniella chlamydospora</i>	42.33
A0A6A6CV94	Cluster: Nitrate/nitrite transporter	<i>Zasmidium cellare</i> ATCC 36951	42.91
S3CRP8	Cluster: Nitrate/nitrite transporter	<i>Glarea lozoyensis</i>	40.4
A0A1B8DXT4	Cluster: Nitrate/nitrite transporter	unclassified <i>Pseudogymnoascus</i>	42.74
A0A094B0S5	Cluster: Nitrate/nitrite transporter	<i>Pseudogymnoascus</i>	42.94
A0A6A6WTZ3	Cluster: Nitrate/nitrite transporter	<i>Melanomma pulvis-pyrius</i> CBS 109.77	42.51
G3JS45	Cluster: Nitrate/nitrite transporter	<i>Cordyceps militaris</i>	40.65
A0A0E9NCH7	Cluster: Nitrate/nitrite transporter	<i>Saitoella complicata</i> (strain BCRC 22490/ CBS 7301/JCM 7358/ NBRC 10748/RRRL Y-17804)	44.27
A0A0C3CTX3	Cluster: Nitrate/nitrite transporter	<i>Oidiiodendron maius</i> Zn	43.15
A0A6G1J7V7	Cluster: Nitrate transporter-like protein	<i>Lentithecium fluviatile</i> CBS 122367	42.63
A0A4U0XSZ7	Cluster: Nitrate/nitrite transporter	<i>Cryomyces minteri</i>	41.8
UPI000CE1C7E3	Cluster: nitrate transporter-like	<i>Quercus suber</i>	41.33
A0A6A5XQJ5	Cluster: Nitrate/nitrite transporter	<i>Aaosphaeria arxii</i> CBS 175.79	42.24
A0A1W5CWS1	Cluster: Nitrate/nitrite transporter	<i>Lasallia pustulata</i>	43.11
A0A6A6R7R9	Cluster: Nitrate/nitrite transporter	Mytiliniaceae	43.99
U1GKS4	Cluster: Nitrate/nitrite transporter	<i>Endocarpon pusillum</i> (strain Z07020/HMAS-L-300199)	42.43
D5GLI8	Cluster: Nitrate/nitrite transporter	<i>Tuber melanosporum</i> (strain Mel28)	41.75
A0A5J5EIC4	Cluster: Nitrate/nitrite transporter	<i>Sphaerospora brunnea</i>	41.7
UPI00144AC6B7	Cluster: nitrate transporter	<i>Lindgomyces ingoldianus</i>	42.6
A0A6A6RHP6	Cluster: Nitrate/nitrite transporter	<i>Massarina eburnea</i> CBS 473.64	42.6
A0A6A6JM23	Cluster: Nitrate/nitrite transporter	<i>Westerdykella ornata</i>	42.63
A0A317SYP8	Cluster: Nitrate/nitrite transporter	Tuber	41.34

TABLE 15-continued

Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 79.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 79 [%]
A0A232LMW8	Cluster: Nitrate/nitrite transporter	<i>Elaphomyces granulatus</i>	43.09
A0A1L7WDP7	Cluster: Nitrate/nitrite transporter	<i>Phialocephala subalpina</i>	42.04
A0A6A5UCT1	Cluster: Nitrate/nitrite transporter	<i>Byssothecium circinans</i>	43.38
A0A6A6BJV5	Cluster: Nitrate/nitrite transporter	<i>Aplosporella prunicola</i> CBS 121167	44.08
A0A6G1LEQ8	Cluster: Nitrate transporter	<i>Teratosphaeria nubilosa</i>	41.68
S3DBU1	Cluster: Nitrate/nitrite transporter	<i>Glarea lozoyensis</i>	43.51
A0A4U0TQA4	Cluster: Nitrate/nitrite transporter	<i>Hortaea thailandica</i>	40.68
A0A4U0X4U1	Cluster: Nitrate/nitrite transporter	<i>Friedmanniomyces</i>	40.28
A0A507R2R1	Cluster: Nitrate/nitrite transporter	<i>Monascus purpureus</i>	41.84
A0A2V1DKP2	Cluster: Nitrate/nitrite transporter	<i>Periconia macrospinoso</i>	41.46
A0A1L9RJV0	Cluster: Nitrate/nitrite transporter	<i>Aspergillus wentii</i> DTO 134E9	42.54
A0A167P4D7	Cluster: Nitrate/nitrite transporter	<i>Sporothrix insectorum</i> RCEF 264	41.12
UPI000CE20498	Cluster: nitrate transporter-like	<i>Quercus suber</i>	42.71
A0A4Z1P351	Cluster: Nitrate/nitrite transporter	<i>Venturia</i>	41.63
A0A6G1K4M7	Cluster: Nitrate transporter-like protein	<i>Pleomassaria siparia</i> CBS 279.74	42.16
M7U767	Cluster: Nitrate/nitrite transporter	Sclerotiniaceae	42.65
A0A0D2AUR8	Cluster: Nitrate/nitrite transporter	<i>Verruconis gallopava</i>	41.41
A0A6G1H5R5	Cluster: Nitrate transporter	<i>Aulographum hederar</i> CBS 113979	41.73
A0A1Y2TEF0	Cluster: Nitrate/nitrite transporter	unclassified <i>Hypoxyton</i>	41.77
A0A177BV64	Cluster: Nitrate/nitrite transporter	<i>Paraphaeosphaeria sporulosa</i>	41.75
A0A6A5WR45	Cluster: Nitrate/nitrite transporter	<i>Ammiculicola lignicola</i> CBS 123094	42.63
A0A6G1KNX3	Cluster: Nitrate transporter-like protein	<i>Pleomassaria siparia</i> CBS 279.74	41.41
W9WHT3	Cluster: Nitrate/nitrite transporter	Herpotrichiellaceae	42.14
A0A2T2P8J8	Cluster: Nitrate/nitrite transporter	<i>Corynespora cassicola</i> Philippines	43.06
A0A6A6T246	Cluster: Nitrate/nitrite transporter	<i>Lophiostoma macrostomum</i> CBS 122681	41.41
A0A2P8ABJ3	Cluster: Nitrate/nitrite transporter	<i>Elsinoe australis</i>	42.25
A0A1Y1ZNY9	Cluster: Nitrate/nitrite transporter	<i>Clohesyomyces aquaticus</i>	40.77
W2RSI4	Cluster: Nitrate/nitrite transporter	<i>Cyphelophora europaea</i> CBS 101466	41.24
A0A2V1E4S3	Cluster: Nitrate/nitrite transporter	<i>Periconia macrospinoso</i>	41.41
A0A6A5V026	Cluster: Nitrate/nitrite transporter	<i>Bimuria novae- zelandiae</i> CBS 107.79	41.15
A0A2K1QG31	Cluster: Nitrate/nitrite transporter	Elsinoaceae	42.25
A0A094A2I4	Cluster: Nitrate/nitrite transporter	<i>Pseudogymnascus</i> sp. VKM F-4281 (FW- 2241)	43.54
A0A6A6XX03	Cluster: Nitrate/nitrite transporter	<i>Melanomma pulvis- pyrius</i> CBS 109.77	42.07

TABLE 15-continued

Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 79.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID NO: 79 [%]
A0A2J5HKF9	Cluster: Nitrate/nitrite transporter	<i>Aspergillus</i>	40.56
A0A6A6IDB6	Cluster: Nitrate/nitrite transporter	<i>Trematosphaeria pertusa</i>	42.45
A0A6A5ST79	Cluster: Nitrate/nitrite transporter	<i>Clathrospora elyinae</i>	41.85
A0A3N4JV10	Cluster: Nitrate/nitrite transporter	<i>Choioomyces venosus</i> 120613-1	42.06
A0A6A5XFW9	Cluster: Nitrate/nitrite transporter	<i>Aaosphaeria arxii</i> CBS 175.79	41.43
A0A553IBL5	Cluster: Nitrate/nitrite transporter	<i>Xylaria flabelliformis</i>	40.24
R7Z451	Cluster: Nitrate/nitrite transporter	<i>Coniosporium apollinis</i> (strain CBS 100218)	40.61
A0A139I046	Cluster: Nitrate/nitrite transporter	<i>Pseudocercospora musae</i>	40.24
A0A2T2NBP3	Cluster: Nitrate/nitrite transporter	<i>Corynespora cassicola</i> Philippines	41.84
A0A177E3H1	Cluster: Nitrate/nitrite transporter	<i>Alternaria</i>	42.21
B0XPC3	Cluster: Nitrate/nitrite transporter	<i>Neosartorya fumigata</i>	42.89
A0A150V198	Cluster: Nitrate/nitrite transporter	<i>Acidomyces richmondensis</i> BFW	42.08
A0A3N4KQT8	Cluster: Nitrate/nitrite transporter	<i>Morchella conica</i> CCBAS932	40.44
A0A1Y2DSQ4	Cluster: Nitrate/nitrite transporter	<i>Pseudomassariella vexata</i>	41.05
A0A2V1BSJ8	Cluster: Nitrate/nitrite transporter	<i>Helotiales incertae sedis</i>	42.04
A0A1V1TKZ5	Cluster: Nitrate/nitrite transporter	Fungi	40.24
K1WJ63	Cluster: Nitrate/nitrite transporter	<i>Marssonina brunnea</i> f. sp. <i>multigermtubi</i> (strain MB_m1)	42.06
A0A6A6E950	Cluster: Nitrate/nitrite transporter	<i>Zopfia rhizophila</i> CBS 207.26	41.92
A0A2K1R263	Cluster: Nitrate/nitrite transporter	<i>Sphaceloma murrayae</i>	41.83
A0A2S7Q2X6	Cluster: Nitrate/nitrite transporter	unclassified <i>Rutstroemia</i>	42.32

Co-Factors

[0414] Preferably the recombinant yeast cell further comprises suitable co-factors to enhance the activity of the above mentioned NADH-dependent nitrate reductase and/or NADH-dependent nitrite reductase. Preferred cofactors include flavin adenine dinucleotide (FAD), heme prosthetic groups, and/or molybdenum cofactor (MoCo). Preferably the recombinant yeast cell may therefore further comprise one or more genes encoding enzymes for the synthesis of one or more of flavin adenine dinucleotide (FAD), heme prosthetic groups, and/or molybdenum cofactor (MoCo). For example, the recombinant yeast cell may comprise one or more genes encoding for an enzyme having FAD synthase activity. Preferred co-factors are as exemplified in non-published US patent application U.S. 63/087,642 filed with the United States Patent Office on 5 Oct. 2020, the contents of which are herewith incorporated by reference.

Recombinant Expression

[0415] The recombinant yeast cell is a recombinant cell. That is to say, a recombinant yeast cell comprises, or is

transformed with or is genetically modified with a nucleotide sequence that does not naturally occur in the cell in question. Techniques for the recombinant expression of enzymes in a cell, as well as for the additional genetic modifications of a recombinant yeast cell are well known to those skilled in the art. Typically such techniques involve transformation of a cell with nucleic acid construct comprising the relevant sequence. Such methods are, for example, known from standard handbooks, such as Sambrook and Russel (2001) “*Molecular Cloning: A Laboratory Manual*”, (3rd edition), published by Cold Spring Harbor Laboratory Press, or F. Ausubel et al., eds., “*Current protocols in molecular biology*”, Green Publishing and Wiley Interscience, New York (1987). Methods for transformation and genetic modification of fungal host cells are known from e.g. EP-A-0635574, WO98/46772, WO 99/60102, WO00/37671, WO90/14423, EP-A-0481008, EP-A-0635574 and U.S. Pat. No. 6,265,186.

Fermentation Process

[0416] The invention further provides a process for the production of ethanol, comprising converting a carbon source, preferably a carbohydrate or another organic carbon source, using a recombinant yeast cell as described in this specification, thereby forming ethanol.

[0417] The feed for this fermentation process suitably comprises one or more fermentable carbon sources. The fermentable carbon source preferably comprises or is consisting of one or more fermentable carbohydrates. More preferably, the fermentable carbon source comprises one or more mono-saccharides, disaccharides and/or polysaccharides. For example, the fermentable carbon source may comprise one or more carbohydrates selected from the group consisting of glucose, fructose, sucrose, maltose, xylose, arabinose, galactose, mannose and trehalose. The fermentable carbon source, preferably comprising or consisting of one or more carbohydrates, may suitably be obtained from starch, cellulose, hemicellulose lignocellulose, and/or pectin. Suitably the fermentable carbon source may be in the form of a, preferably aqueous, slurry, suspension, or a liquid.

[0418] The concentration of fermentable carbohydrate, such as for example glucose, during fermentation is preferably equal to or more than 80 g/L. That is, the initial concentration of glucose at the start of the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L. The start of the fermentation may be the moment when the fermentable carbohydrate is brought into contact with the recombinant cell of the invention.

[0419] The fermentable carbon source may be prepared by contacting starch, lignocellulose, and/or pectin with an enzyme composition, wherein one or more mono-saccharides, disaccharides and/or polysaccharides are produced, and wherein the produced mono-saccharides, disaccharides and/or polysaccharides are subsequently fermented to give a fermentation product.

[0420] Before enzymatic treatment, the lignocellulosic material may be pretreated. The pretreatment may comprise exposing the lignocellulosic material to an acid, a base, a solvent, heat, a peroxide, ozone, mechanical shredding, grinding, milling or rapid depressurization, or a combination of any two or more thereof. This chemical pretreatment is often combined with heat-pretreatment, e.g. between 150-220° C. for 1 to 30 minutes. Subsequently the pretreated material can be subjected to enzymatic hydrolysis to release sugars that may be fermented according to the invention. This may be executed with conventional methods, e.g. contacting with cellulases, for instance cellobiohydrolase(s), endoglucanase(s), beta-glucosidase(s) and optionally other enzymes. The conversion with the cellulases may be executed at ambient temperatures or at higher temperatures, at a reaction time to release sufficient amounts of sugar(s). The result of the enzymatic hydrolysis is hydrolysis product comprising C5/C6 sugars, herein designated as the sugar composition.

[0421] In one embodiment the fermentable carbohydrate is, or is comprised by a biomass hydrolysate, such as a corn

stover or corn fiber hydrolysate. Such biomass hydrolysate may in its turn comprise, or be derived from corn stover and/or corn fiber.

[0422] By a “hydrolysate” is herein understood a polysaccharide-comprising material (such as corn stover, corn starch, corn fiber, or lignocellulosic material, which polysaccharides have been depolymerized through the addition of water to form mono and oligosaccharide sugars. Hydrolysates may be produced by enzymatic or acid hydrolysis of the polysaccharide-containing material.

[0423] A biomass hydrolysate may be a lignocellulosic biomass hydrolysate. Lignocellulose herein includes hemicellulose and hemicellulose parts of biomass. Also lignocellulose includes lignocellulosic fractions of biomass. Suitable lignocellulosic materials may be found in the following list: orchard primings, chaparral, mill waste, urban wood waste, municipal waste, logging waste, forest thinnings, short-rotation woody crops, industrial waste, wheat straw, oat straw, rice straw, barley straw, rye straw, flax straw, soy hulls, rice hulls, rice straw, corn gluten feed, oat hulls, sugar cane, corn stover, corn stalks, corn cobs, corn husks, switch grass, miscanthus, sweet sorghum, canola stems, soybean stems, prairie grass, gamagrass, foxtail; sugar beet pulp, citrus fruit pulp, seed hulls, cellulosic animal wastes, lawn clippings, cotton, seaweed, algae (including macroalgae and microalgae), trees, softwood, hardwood, poplar, pine, shrubs, grasses, wheat, wheat straw, sugar cane bagasse, corn, corn husks, corn hobs, corn kernel, fiber from kernels, products and by-products from wet or dry milling of grains, municipal solid waste, waste paper, yard waste, herbaceous material, agricultural residues, forestry residues, municipal solid waste, waste paper, pulp, paper mill residues, branches, bushes, canes, corn, corn husks, an energy crop, forest, a fruit, a flower, a grain, a grass, a herbaceous crop, a leaf, bark, a needle, a log, a root, a sapling, a shrub, switch grass, a tree, a vegetable, fruit peel, a vine, sugar beet pulp, wheat midlings, oat hulls, hard or soft wood, organic waste material generated from an agricultural process, forestry wood waste, or a combination of any two or more thereof. Algae, such as macroalgae and microalgae have the advantage that they may comprise considerable amounts of sugar alcohols such as sorbitol and/or mannitol. Lignocellulose, which may be considered as a potential renewable feedstock, generally comprises the polysaccharides cellulose (glucans) and hemicelluloses (xylans, heteroxylans and xyloglucans). In addition, some hemicellulose may be present as glucomannans, for example in wood-derived feedstocks. The enzymatic hydrolysis of these polysaccharides to soluble sugars, including both monomers and multimers, for example glucose, cellobiose, xylose, arabinose, galactose, fructose, mannose, rhamnose, ribose, galacturonic acid, glucuronic acid and other hexoses and pentoses occurs under the action of different enzymes acting in concert. In addition, pectins and other pectic substances such as arabinans may make up considerably proportion of the dry mass of typically cell walls from non-woody plant tissues (about a quarter to half of dry mass may be pectins). Lignocellulosic material may be pretreated. The pretreatment may comprise exposing the lignocellulosic material to an acid, a base, a solvent, heat, a peroxide, ozone, mechanical shredding, grinding, milling or rapid depressurization, or a combination of any two or more thereof. This chemical pretreatment is often combined with heat-pretreatment, e.g. between 150-220° C. for 1 to 30 minutes.

[0424] The process for the production of ethanol may comprise an aerobic propagation step and an anaerobic fermentation step. More preferably the process according to the invention is a process comprising an aerobic propagation step wherein the population of the recombinant yeast cell is increased; and an anaerobic fermentation step wherein the carbon source is converted to ethanol by using the recombinant yeast cell population.

[0425] By propagation is herein understood a process of recombinant yeast cell growth that leads to increase of an initial recombinant yeast cell population. Main purpose of propagation is to increase the population of the recombinant yeast cell using the recombinant yeast cell's natural reproduction capabilities as living organisms. That is, propagation is directed to the production of biomass and is not directed to the production of ethanol. The conditions of propagation may include adequate carbon source, aeration, temperature and nutrient additions. Propagation is an aerobic process, thus the propagation tank must be properly aerated to maintain a certain level of dissolved oxygen. Adequate aeration is commonly achieved by air inductors installed on the piping going into the propagation tank that pull air into the propagation mix as the tank fills and during recirculation. The capacity for the propagation mix to retain dissolved oxygen is a function of the amount of air added and the consistency of the mix, which is why water is often added at a ratio of between 50:50 to 90:10 mash to water. "Thick" propagation mixes (80:20 mash-to-water ratio and higher) often require the addition of compressed air to make up for the lowered capacity for retaining dissolved oxygen. The amount of dissolved oxygen in the propagation mix is also a function of bubble size, so some ethanol plants add air through spargers that produce smaller bubbles compared to air inductors. Along with lower glucose, adequate aeration is important to promote aerobic respiration during propagation, making the environment during propagation different from the anaerobic environment during fermentation.

[0426] By an anaerobic fermentation process is herein understood a fermentation step run under anaerobic conditions.

[0427] The anaerobic fermentation is preferably run at a temperature that is optimal for the cell. Thus, for most recombinant yeast cells, the fermentation process is performed at a temperature which is less than about 50° C., less than about 42° C., or less than about 38° C. For recombinant yeast cell or filamentous fungal host cells, the fermentation process is preferably performed at a temperature which is lower than about 35, about 33, about 30 or about 28° C. and at a temperature which is higher than about 20, about 22, or about 25° C.

[0428] The ethanol yield, based on xylose and/or glucose, in the process according to the invention is preferably at least about 50, about 60, about 70, about 80, about 90, about 95 or about 98%. The ethanol yield is herein defined as a percentage of the theoretical maximum yield.

[0429] The process according to the invention, and the propagation step and/or fermentation step suitably comprised therein can be carried out in batch, fed-batch or continuous mode. A separate hydrolysis and fermentation (SHF) process or a simultaneous saccharification and fermentation (SSF) process may also be applied.

[0430] The recombinant yeast and process according to the invention advantageously allow for a more robust process. Advantageously the process, or any anaerobic fermenta-

tion during the process can be carried out in the presence of high concentrations of carbon source. The process, respectively any anaerobic fermentation step therein, is therefore preferably carried out in the presence of a glucose concentration of 25 g/L or more, 30 g/L or more, 35 g/L or more, 40 g/L or more, 45 g/L or more, 50 g/L or more, 55 g/L or more, 60 g/L or more, 65 g/L or more, 70 g/L or more, 75 g/L or more, 80 g/L or more, 85 g/L or more, 90 g/L or more, 95 g/L or more, 100 g/L or more, 110 g/L or more, 120 g/L or more or may for example be in the range of 25 g/L-250 g/L, 30 g/L-200 g/L, 40 g/L-200 g/L, 50 g/L-200 g/L, 60 g/L-200 g/L, 70 g/L-200 g/L, 80 g/L-200 g/L, or 90 g/L-200 g/L.

[0431] For the recovery of the fermentation product existing technologies are used. For different fermentation products different recovery processes are appropriate. Existing methods of recovering ethanol from aqueous mixtures commonly use fractionation and adsorption techniques. For example, a beer still can be used to process a fermented product, which contains ethanol in an aqueous mixture, to produce an enriched ethanol-containing mixture that is then subjected to fractionation (e.g., fractional distillation or other like techniques). Next, the fractions containing the highest concentrations of ethanol can be passed through an adsorber to remove most, if not all, of the remaining water from the ethanol. In an embodiment in addition to the recovery of fermentation product, the yeast may be recycled.

[0432] All patent and literature references cited in the present specification are hereby incorporated by reference in their entirety.

[0433] The following examples are offered for illustrative purposes only, and are not intended to limit the scope of the present invention in any way

EXAMPLES

General Molecular Biology Techniques

[0434] Unless indicated otherwise, the methods used are standard biochemical techniques. Examples of suitable general methodology textbooks include Sambrook et al., *Molecular Cloning, a Laboratory Manual* (1989) and Ausubel et al., *Current Protocols in Molecular Biology* (1995), John Wiley & Sons, Inc.

HPLC Analysis

[0435] HPLC analysis is typically conducted as described in "*Determination of sugars, byproducts and degradation products in liquid fraction in process sample*"; Laboratory Analytical Procedure (LAP, Issue date: Dec. 8, 2006; by A. Sluiter, B. Hames, R. Ruiz, C. Scarlata, J. Sluiter, and D. Templeton; Technical Report (NREL/TP-51042623); January 2008; National Renewable Energy Laboratory.

[0436] After fermentation, samples for HPLC analysis were separated from yeast biomass and insoluble components (corn mash) by passing the clear supernatant after centrifugation through a 0.2 µm pore size filter.

Strains and DNA Sequences Used in the Examples

[0437] Table 16 provides an overview of the genotypes of the strain

[0438] Table 17 provides an overview of the nucleic acid sequences referred to in these examples.

TABLE 16

<i>S. cerevisiae</i> strains used in the examples		
Strain name	Genotype	Source
Parental strain	Wildtype	Commercial <i>Saccharomyces cerevisiae</i> strain, available from Lesaffre
Intermediate strain IX15	PRK, cbbM, GroES, GroEL	Following strain construction described in WO2013/144257A1 and
Reference strain RX16	PRK, cbbM, GroES, GroEL DAK1, gldA, ACT1.pro-T5	Example 2
Example strain NX17	PRK, cbbM, GroES, GroEL DAK1, gldA, ANB1.pro-T5	Example 3
Example strain NX18	PRK, cbbM, GroES, GroEL DAK1, gldA, HEM13.pro-T5	Example 4

TABLE 17

DNA sequences used in the examples		
SEQ ID NO:	Short description	Example(s)
SEQ ID NO: 59	25-Sc_EFT2.pro-Sc_DAK1.orf-Sc_ENO1.ter-2A ("fragment A").	2
SEQ ID NO: 60	2A-Sc_HHF2.pro-Ec_gldA.orf-Sc_CTC1.ter-2B ("fragment B").	2
SEQ ID NO: 61	2B-Sc_ACT1.pro_0001-Zrou_T5.orf-Sc_TEF2.ter_0001-2C ("fragment C").	2
SEQ ID NO: 62	25-Sc_MYO4.pro-Sc_DAK1.orf-Sc_GPM1.ter-2A ("fragment D").	3, 4
SEQ ID NO: 63	2A-Sc_HHF2.pro-Ec_gldA.orf-Sc_EFM1.ter-2B ("fragment E").	3, 4
SEQ ID NO: 64	2B-Sc_ANB1.pro_0001-Zrou_T5.orf-Sc_TEF1.ter_0001-2C ("fragment F").	3
SEQ ID NO: 65	2B-Sc_HEM13.pro_0001-Zrou_T5.orf-Sc_TEF1.ter_0001-2C ("fragment G").	4
SEQ ID NO: 66	Connector 25: ATCCGTACACGTTTGTGCCGAACCCGAAGTATTGCAACA GGTCGAAGTTC	2, 3, 4
SEQ ID NO: 67	Connector 2A: ACCCGATCCCGAAGAAAGTCCAACTCCAGTTCGATTCCC TGCTTTCGAGT	2, 3, 4
SEQ ID NO: 68	Connector 2B: TCCTCGATCGGCCAGTTGGGAGCAAGGTGGTACTTCT GTCGTCGTATTT	2, 3, 4
SEQ ID NO: 69	Connector 2C: ATTTGGAACAGCAGGCGAGGTGTATTGGTGCAGAAGA GGTGTTCGAG	2, 3, 4
SEQ ID NO: 70	Sc_INT7.03_FLANK5	2
SEQ ID NO: 71	Sc_INT7.03_FLANK3	2
SEQ ID NO: 72	INT28_FLANK5	3, 4
SEQ ID NO: 73	INT28_FLANK3	3, 4

Starter Strains

[0439] Strains were prepared using Ethanol Red® as starting strain. Ethanol Red® is a commercial *Saccharomyces cerevisiae* strain, available from Lesaffre.

[0440] A strain construction approach that can be followed is described in WO2013/144257A1 and WO2015/028582, incorporated herein by reference.

[0441] Expression cassettes from various genes of interest can be recombined in vivo into a pathway at a specific locus upon transformation of this yeast (U.S. Pat. No. 9,738,890 B2). The promoter, ORF and terminator sequences are assembled into expression cassettes with Golden Gate technology, as described for example by Engler et al., “*Generation of Families of Construct Variants Using Golden Gate Shuffling*”, (2011), published in chapter 11 of Chaofu Lu et al. (eds.), *cDNA Libraries: Methods and Applications, Methods in Molecular Biology*, vol. 729, pages 167-180, incorporated herein by reference, and ligated into BsaI-digested backbone vectors that decorated the expression cassettes with the connectors for the in vivo recombination step. The expression cassettes including connectors are amplified by PCR. In addition, a 5'- and a 3'-DNA fragment of the up- and downstream part of the integration locus was amplified using PCR and decorated by a connector sequence. Upon transformation of yeast cells with these DNA fragments, in vivo recombination and integration into the genome takes place at the desired location. CRISPR-Cas9 technology is used to make a unique double stranded break at the integration locus to target the pathway to this specific locus (see DiCarlo et al., “*Genome engineering in Saccharomyces cerevisiae using CRISPR-Cas systems*”, (2013), *Nucleic Acids Res* Vol. 41, pages 4336-4343, incorporated herein by reference) and WO16110512 and US2019309268. The gRNA was expressed from a multi-copy yeast shuttling vector that contains a natMX marker which confers resistance to the yeast cells against the antibiotic substance nourseothricin (NTC). The backbone of this plasmid is based on pRS305 (see Sikorski and Hieter, “*A System of Shuttle Vectors and Yeast Host Strains Designed for Efficient Manipulation of DNA in Saccharomyces cerevisiae*”, (1989), *Genetics*, vol. 122, pages 19-27, incorporated herein by reference), including a functional 2 micron ORI sequence. The *Streptococcus pyogenes* CRISPR-associated protein 9 (Cas9) was expressed from a pRS414 plasmid (see Sikorski and Hieter, 1989, as indicated above) with kanMX marker which confers resistance to the yeast cells against the antibiotic substance geneticin (G418). The guide RNA and protospacer sequences were designed with a gRNA designer tool (known by a person skilled in the art and for example described in <https://www.atum.bio/eCommerce/cas9/input>).

Example 1: Construction of “Rubisco” Strain (Intermediate Strain IX15)

[0442] In the current example the starter strain was transformed with the cbbM gene encoding the single subunit of ribulose-1,5-biphosphate-carboxylase (RuBisCO) from *Thiobacillus denitrificans*, genes encoding chaperonins GroEL and GroES from *E. coli* to aid in the proper folding of the RuBisCO protein in the cytosol of *S. cerevisiae*, a gene encoding phosphoribulokinase (prk) from *S. oleacea* as described by Guadalupe-Medina et al., “*Carbon dioxide fixation by Calvin-Cycle enzymes improves ethanol yield in yeast*”, published in *Biotechnol, Biofuels*, 2013, vol. 6, page

125 onwards, incorporated herein by reference. This resulted in reference strain IX15 which contained cbbM, prk, groEL and groES (see Table 16 for detailed genotypes).

Example 2: Construction of Reference Strain RX16

[0443] A reference strain RX16 was construed, comprising a glycerol transporter derived from *Z. rouxii* “Zrou_T5”, preceded by a constitutive promoter according to the prior art “Sc_ACT1.pro_0001”.

[0444] Reference strain RX16 was constructed by transforming the intermediate strain IX15 obtained in example 1 with three expression cassettes;

[0445] Expression cassette “fragment A”: 25-EFT2p. Sc_DAK1.ENO1t-2A;

[0446] Expression cassette “fragment B”: 2A-HHF2p. Ec_gldA.CTC1t-2B; and

[0447] Expression cassette “fragment C”: 2B-Sc_ACT1.pro_0001-Zrou_T5.orf-Sc_TEF2.ter_0001-2C.

[0448] Expression cassette “fragment A”: The first cassette contained a DNA fragment named “fragment A” was compiled using Golden Gate Cloning and comprised the *S. cerevisiae* EFT2 promoter (Sc_EFT2.pro), *S. cerevisiae* DAK1 orf (Sc_DAK1.orf) and *S. cerevisiae* ENO1 terminator (Sc_ENO1.ter). The cassette was decorated with 50 bp connectors 25 and 2A. Connector 25 had a nucleic acid sequence as illustrated in: SEQ ID NO: 66. Connector 2A had a nucleic acid sequence as illustrated in: SEQ ID NO: 67. The nucleic acid sequence of the DNA fragment “fragment A” is illustrated in SEQ ID NO: 59.

[0449] Expression cassette “fragment B”: The second cassette contained a DNA fragment named “fragment B”, and comprised the *S. cerevisiae* HHF2 promoter (Sc_HHF2.pro), *E. coli* gldA orf (Ec_gldA.orf) and *S. cerevisiae* CTC1 terminator (Sc_CTC1.ter). The cassette was decorated with 50 bp connectors 2A and 2B. Connector 2A had a nucleic acid sequence as illustrated in: SEQ ID NO: 67. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. The nucleic acid sequence of the DNA fragment “fragment B” is illustrated in SEQ ID NO: 60.

[0450] Expression cassette “fragment C”: The third cassette contained a DNA fragment named “fragment C”, and comprised the *S. cerevisiae* ACT1 promoter (Sc_ACT1.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* TEF2 terminator (Sc_TEF2.ter_0001). The cassette was decorated with 50 bp connectors 2B and 2C. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. Connector 2C had a nucleic acid sequence as illustrated in: SEQ ID NO: 69. The nucleic acid sequence of the DNA fragment “fragment C” is illustrated in SEQ ID NO: 61.

[0451] The above three cassettes were integrated in intermediate strain IX15 in the locus INT7.03 located on a non-coding region on chromosome VII between coding sequences PUP2 (YGR253C) and ENO1 (YGR254W) of *S. cerevisiae* using CRISPR-Cas9 using the following sequences for homologous integration:

[0452] Sc_INT7.03_FLANK5 (illustrated by SEQ ID NO: 70); and

[0453] Sc_INT7.03_FLANK3 (illustrated by SEQ ID NO: 71).

[0454] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT7.03 locus of the

three expression cassettes. Plasmid free colonies were selected which resulted in new strain RX16 (see Table 16 for detailed genotypes).

Example 3: Construction of New NX17

[0455] New strain NX17 was constructed by transforming the intermediate strain IX15 obtained in example 1 with three expression cassettes:

[0456] Expression cassette “fragment D”: 25-Sc_MYO4.pro-Sc_DAK1.orf-Sc_GPM1.ter-2A;

[0457] Expression cassette “fragment E”: 2A-Sc_HHF2.pro-Ec_gldA.orf-Sc_EFM1.ter-2B; and

[0458] Expression cassette “fragment F”: 2B-Sc_ANB1.pro_0001-Zrou_T5.orf-Sc_TEF1.ter_0001-2C.

[0459] Expression cassette “fragment D”: The first cassette named “fragment D” was compiled using Golden Gate Cloning and comprised the *S. cerevisiae* MYO4 promoter (Sc_MYO4.pro), *S. cerevisiae* DAK1 orf (Sc_DAK1.orf) and *S. cerevisiae* GPM1 terminator (Sc_GPM1.ter). The cassette was decorated with 50 bp connectors 25 and 2A. Connector 25 had a nucleic acid sequence as illustrated in: SEQ ID NO: 66. Connector 2A had a nucleic acid sequence as illustrated in: SEQ ID NO: 67. The nucleic acid sequence of the DNA fragment “fragment D” is illustrated in SEQ ID NO: 62.

[0460] Expression cassette “fragment E”: The second cassette named “fragment E” comprised *S. cerevisiae* HHF2 promoter (Sc_HHF2.pro), *E. coli* gldA orf (Ec_gldA.orf) and *S. cerevisiae* EFM1 terminator (Sc_EFM1.ter). The cassette was decorated with 50 bp connectors 2A and 2B. Connector 2A had a nucleic acid sequence as illustrated in: SEQ ID NO: 67. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. The nucleic acid sequence of the DNA fragment “fragment E” is illustrated in SEQ ID NO: 63.

[0461] Expression cassette “fragment F”: The third cassette named “fragment F”, comprised the *S. cerevisiae* ANB1 promoter (Sc_ANB1.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* terminator (Sc_TEF1.ter_0001). The cassette was decorated with 50 bp connectors 2B and 2C. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. Connector 2C had a nucleic acid sequence as illustrated in: SEQ ID NO: 69. The nucleic acid sequence of the DNA fragment “fragment F” is illustrated in SEQ ID NO: 64.

[0462] The above three cassettes were integrated in intermediate strain IX15 in the INT28 locus using CRISPR-Cas9 using. These three cassettes were integrated in the locus INT28 located on a non-coding region on Chromosome IV between YDR345C (HXT3) and YDRT246C (SVF1) of *S. cerevisiae* using CRISPR-Cas9 using the following sequences for homologous integration:

[0463] INT28_FLANK5 (illustrated by SEQ ID NO: 72); and

[0464] INT28_FLANK3 (illustrated by SEQ ID NO: 73)

[0465] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT28 locus of the three expression cassettes. Plasmid free colonies were selected which resulted in strain NX17 (see Table 16 for detailed genotypes).

Example 4: Construction of New NX18

[0466] New strains NX18 was constructed by transforming the intermediate strain IX15 obtained in example 1 with three expression cassettes:

[0467] Expression cassette “fragment D”: 25-Sc_MYO4.pro-Sc_DAK1.orf-Sc_GPM1.ter-2A;

[0468] Expression cassette “fragment E”: 2A-Sc_HHF2.pro-Ec_gldA.orf-Sc_EFM1.ter-2B; and

[0469] Expression cassette “fragment G”: 2B-Sc_HEM13.pro_0001-Zrou_T5.orf-Sc_TEF1.ter_0001-2C.

Fragment D and fragment E were as described above under example 3.

[0470] The third cassette named “fragment G” comprised the *S. cerevisiae* HEM13 promoter (Sc_HEM13.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* terminator (Sc_TEF1.ter_0001). The cassette was decorated with 50 bp connectors 2B and 2C. The cassette was decorated with 50 bp connectors 2B and 2C. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. Connector 2C had a nucleic acid sequence as illustrated in: SEQ ID NO: 69. The nucleic acid sequence of the DNA fragment “fragment G” is illustrated in SEQ ID NO: 65.

[0471] The above three cassettes were integrated in intermediate strain IX15 in the locus INT28 located on a non-coding region on Chromosome IV between YDR345C (HXT3) and YDRT246C (SVF1) of *S. cerevisiae* using CRISPR-Cas9 using the following sequences for homologous integration. INT28_FLANK5 (SEQ ID NO: 72) and INT28_FLANK3 (SEQ ID NO: 73) for homologous integration.

[0472] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT28 locus of the three expression cassettes. Plasmid free colonies were selected which resulted in strain NX18 (see Table 16 for detailed genotypes).

Example 5: Fermentation

[0473] Preculture preparation and conditions: Glycerol stocks (−80° C.) were thawed at room temperature and used to inoculate 0.2 L filter-sterilized mineral medium (as described by Luttkik et al, “*The Saccharomyces cerevisiae ICL2 Gene Encodes a Mitochondrial 2-Methylisocitrate Lyase Involved in Propionyl-Coenzyme A Metabolism*”, (2000), JOURNAL OF BACTERIOLOGY, pages 7007-7013, herein incorporated by reference (Luttkik et al., 2000)) at pH 6.0 (adjusted with 2M H₂SO₄/4N KOH) supplemented with 2% (w/v) glucose, in non-baffled 0.5 L shake-flasks. Precultures were incubated for 16 to 20 hours at 32° C., shaking at 200 RPM. After determination of the yeast biomass (CDW) content of the culture (via OD600 vs CDW calibration), a quantity of preculture corresponding to the required 0.5 g CDW/L inoculum concentration for the propagation was centrifuged (3 min, 5300×g), washed once with one sample volume sterile demineralized water, centrifuged once more, and resuspended in propagation medium.

[0474] Propagation: Propagation media consisted of 20 ml diluted corn mash (70% v/v Corn mash: 30% v/v demineralized water), at pH 5.0 (adjusted with 4N KOH/2M H₂SO₄) in 100 ml non-baffled shake flasks. Urea (1.25 g/L) was added as N-source and a standard antibiotic mix (1 ml 100

μg/L PenG & 1 ml 50 μg/L Neomycin stock per liter of corn mash) was added to prevent outgrowth of bacterial contaminants. The Glucoamylase (Spirizyme, Novozymes) dosage for all the strains was 0.1 g/kg. The amount of preculture material used to inoculate the propagation phase (0.5 g CDW/L) was determined by OD & strain specific OD₆₀₀/CDW conversion factors. The required quantity of preculture was centrifuged (3 min, 4000 rpm), washed once with one culture volume cold (4° C.) sterile demi-water, centrifuged once more, resuspended in 500 μL sterile demi-water and transferred to the propagation. The propagations ran for 6 hrs at 32° C. shaking at 140 rpm.

[0475] Fermentation: Corn mash was used for all test described here. 1 g/L Urea was added as N-source, while the standard antibiotics mix was applied (100 mg/ml PenG stock+50 mg/ml Neomycin stock). pH was adjusted to 5.0 using 2M H₂SO₄/4N KOH. The Glucoamylase (Spirizyme, Novozymes) dosage applied was 0.24 g/kg. Fermentations were performed using 200 ml medium in 500 ml Schott bottles equipped with pressure recording/releasing caps (Ankom Technology, Macedon NY, USA), while shaking at 140 rpm and 32° C. The pressure development was measured in psi units (pound-force per square inch) and the results are illustrated in Table 18 and Table 19. FIG. 1 illustrates the results of Table 18 graphically and FIG. 2 illustrates the results of Table 19 graphically. The pressure listed is the cumulative pressure generated, expressed in psi.

[0476] As illustrated by Table 18 and FIG. 1, over the whole fermentation run, more ethanol and CO₂ was formed by the new strains NX17 and NX18 than by the reference strain RX16, illustrating more conversion of sugars. This is evidenced by the total area below the curve.

[0477] Table 19 and FIG. 2 further illustrate that the strains according to the invention, NX17 and NX18, comprising a promotor as claimed, have a steeper onset in fermentation than reference strain RX16 comprising a standard constitutive promoter, that is, the strains according to the invention are quicker in starting the fermentation.

[0478] pH was not controlled during fermentation. Fermentations were stopped after 66 h.

[0479] Sampling and analysis: All cultivations were sampled at end-of fermentation. Since the fermentation broths contained active GA enzyme, 50 μl of a 10 g/L acarbose stock solution was added to approximately 5 g sample to stop glucoamylase activity. Samples for HPLC analysis were separated from yeast biomass and insoluble components (corn mash) by passing the clear supernatant after centrifugation through a 0.2 μm pore size filter. HPLC analysis was conducted as described in (Sluiter, et al., 2008). The total sugar content (g/L) of the samples at end-of-fermentation (EOF) was determined with HPLC and the results are provided in Table 20.

[0480] As illustrated by Table 20, the total sugar content for the wild-type strain was 13.0 g/L and the total sugar content (g/L) for reference strain RX16 was 14.0 g/L. New strains NX17 and NX18 both had a total sugar content (g/L) at EOF of 12.6 g/L. These results illustrate that the strains according to the invention result in an improved consumption of available sugars.

TABLE 18

ethanol and CO ₂ gas production (in psi) during fermentation			
time (h)	RX16	NX17 (average of duplo)	NX18 (average of duplo)
0	0	0	0
5	5	7	7
10	41	56	55
15	128	160	158
20	216	247	245
25	280	307	305
30	330	358	354
35	376	403	399
40	415	434	433
45	440	447	448
50	452	454	456
55	458	458	460
60	461	460	463
65	463	461	464

TABLE 19

ethanol and CO ₂ gas production (in psi) during fermentation (first 10 hours)			
time (h)	RX16	NX17	NX18
0	0	0	0
1	1	1	1
2	2	2	2
3	2	3	3
4	3	4	4
5	5	7	7
6	8	11	11
7	13	18	17
8	20	27	27
9	29	40	39
10	41	56	55

TABLE 20

Total sugar content (g/L) at end of fermentation (66 hours of fermentation)	
Strain	Total sugar content (g/L)
Wild-type	13
RX16	14
NX17	12.6 *
NX18	12.6 *

* Average of duplicate experiment

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SEQUENCE LISTING

Sequence total quantity: 79

SEQ ID NO: 1 moltype = AA length = 598

FEATURE Location/Qualifiers

REGION 1..598

note = Protein sequence of the glycerol transporter from *S. pombe*

source 1..598

mol_type = protein

organism = *Schizosaccharomyces pombe*

SEQUENCE: 1

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FSSPNLKGLD NTADSDSQPW LGGYLAGRLE DISGQSRNY VDPYYEELNA GRRPNKPVWS 120

LNGPLPHVLG NSVVEKISQK NQEARSRANS RVNSRANSRA NSSVSLAGMD GSPNWKRMK 180

SAVFGSRVKL NDEEAQLPRN KSSVSIAEQ ASRPKVSFSL QSSRQPSIAE EQPQTQRKSS 240

AITVEHAENA EPETPRNNVS FSRKPSIAEQ DSSQDITMPP NEIIAEESLD SGSDTETLYL 300

NYWCKIRHFF REGFAEFLGT LVLVFGVGS NLQATVTNGA GGSFESLSFA WGFGCMLGVY 360

IAGGISGGHV NPAVTISLAI FRKFPWYKVP IYIFFQIWGA FPGGALAYGY HWSITEFEG 420

GKDIRTPATG GCLYTNPKPY VTRNAFFDE FIGTAVLVGC LFAILDITNS PPTQGMTAFI 480

VGLLIAAIGM ALGYQTSFTL NPARDLGPRM FAWWIGYGPH SPFLYHWWWT WGAWGTTIGG 540

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SEQ ID NO: 2 moltype = AA length = 258

FEATURE Location/Qualifiers

REGION 1..258

note = Protein sequence of the glycerol transporter from *Plasmodium falciparum*

source 1..258

mol_type = protein

-continued

organism = *Plasmodium falciparum*

SEQUENCE: 2

MHMLFYKSYV	REFIGEPLGT	FVLMFLGEGA	TANFHTTGLS	GDWYKLC LGW	GLAVFFGILV	60
SAKLSGAHLN	LAVSIGLSSI	NKFDLKKIPV	YFFAQLLGAF	VGTSVYGLY	HGFISNSKIP	120
QFAWETS RNP	SISLTGAFFN	ELILT GILL	VILVVVDENI	CGKFHILKLS	SVVGLIILCI	180
GITFGGNTGF	ALNPSRDLGS	RFLSLIAYGK	DTFTKDNFYF	WVPLVAPCVG	SVVFCQFYDK	240
VICPLVDLAN	NEKDGVDL					258

SEQ ID NO: 3 moltype = AA length = 291

FEATURE Location/Qualifiers

REGION 1..291

 note = Glycerol transporter (aquaglyceroporin AQP9) derived

 from *Daniorerio* (Drer_T3)

source 1..291

 mol_type = protein

 organism = *Danio rerio*

SEQUENCE: 3

MEYLENIRNL	RGRCVLRDI	IREFLAELLG	TFVLILFGCG	SVAQTVLSRE	AKQQLLTIHF	60
GPTLVGMLAV	YMAGGVSGGH	VNPAVSLAMV	VLRLPLKKF	PVYVLAQFLG	AFPGSCAVYC	120
LYYDAFTEFA	NGELAVTGN	VTAGIFASYP	REGLSLLNGF	IDQVIGAGAL	VLCILAVVDK	180
KNIGAPKGME	PLLVLGSILA	IGVSMALNCG	YPINPARDLG	PRLFTAIAAG	GLTVFSAGNG	240
WWWVPVVGPM	VGGVVGAAIY	FLMIEMHHP	NDKNLEDDNS	LKDKYELNTV	N	291

SEQ ID NO: 4 moltype = AA length = 292

FEATURE Location/Qualifiers

REGION 1..292

 note = Protein sequence of the glycerol transporter from

Xenopus tropicalis;

source 1..292

 mol_type = protein

 note = *Xenopus (Silurana) tropicalis*

 organism = *Xenopus tropicalis*

SEQUENCE: 4

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LAFGFAVMLG	ILISGQVSGG	HLNPAVTFAL	CILAREPWVK	FPVYSIAQTL	GAFLGAGI IY	120
GLYYDAIWF	ANDQLYVTGE	NGTAGIFTTF	PSDHLTLMNG	FFDQFIGTAA	LVVCVLAIVD	180
PNNNP IPRGL	EAFTVGVFVL	VIGTSMGFNS	GYAVNPARDF	GPRLFTSLAG	WGTEVFWAGN	240
QWWWVPIVSP	LLGAFAGVLV	YQLMIGCHIE	PVPQSTEQEN	IKLADV KPKD	RI	292

SEQ ID NO: 5 moltype = AA length = 592

FEATURE Location/Qualifiers

REGION 1..592

 note = glycerol transporter (ZYR00E0121) from

Zygosaccharomyces rouxii (Zrou_T5)

source 1..592

 mol_type = protein

 organism = *Zygosaccharomyces rouxii*

SEQUENCE: 5

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FPPTKSNGDN	DRYASVIQGA	VTACYEIGCF	FGSLFVLFFG	DAIGRKPLII	FGAIIVIIGT	120
VISTAPFHHA	WGLGQFV VGR	VITGVGTGFN	TSTIPVWQSE	MTKPNIRGAM	INLDGSVI AF	180
GTMIAFWLDF	GFSPINSSVQ	WRFPVSVQII	FALVLLFGIV	RMPESPRWLM	AKKRPAEARY	240
VLACLNDLPE	NDDAILAEMT	SLHEAVNRSS	NQKSQMKSLF	SMGKQQNFSR	ALIASSTQFF	300
QQFTGCNAAI	YYSTVLFQTT	VQLDRLLAMI	LGGVFATVYT	LSTLPSFYLV	EKVGRRKMF	360
FGALGQGISF	IITFACLVNP	TKQNAKGA AV	GLYLFIIICF	LAILELPWIY	PPEIASMRVR	420
AATNAMSTCT	NWVTNFVAVM	FTPVFIQTSQ	WGCYLFFAVM	NFIYLPVIFV	FYPETAGRSL	480
EEIDIIFAKA	HVDGTLPMV	AHRLPKLSMT	EVEDYSQSLG	LHDDENEKEE	YDEKEAEANA	540
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SEQ ID NO: 6 moltype = DNA length = 876

FEATURE Location/Qualifiers

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 from *Daniorerio* (Drer_T3)

source 1..876

 mol_type = genomic DNA

 organism = *Danio rerio*

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source            1..1779
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                  organism = Zygosaccharomyces rouxii

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ttgcacgacg atgaaaacga aaaggaagaa tacgacgaaa aagaagctga agctaacgct 1620
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FEATURE          Location/Qualifiers
misc_feature      1..12
                  note = Promoter motif
source            1..12
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 9
aaaaattggt ga 12

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SEQ ID NO: 10     moltype = DNA length = 622
FEATURE          Location/Qualifiers
misc_feature      1..622
                  note = Promoter ANB1 from Saccharomyces cerevisiae
source            1..622
                  mol_type = genomic DNA
                  organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 10
ggctcgggtg cagatcggtg aaaacatctt gatcttgctc ccgggaattt tagattcagg 60
taggaaattg attacatcaa tactgttacc ctgaatcata ttcgacgatg tcgtctcaca 120
cggaaatata attcatttct tggttttcca aaaaaatttt catttttttt cacttttttt 180
tttcgtcttc cttttttttt tttttttttt attttttttt ctgtgttcac cttttttttt 240
ttcagttgac atctttctgc attcttttct gtgttttttt tttttttttt cgtttttcca 300
ttgttcgttc gttgctgttt ttttcgccct atgtttctcg agcctaaaaa ttttttctt 360

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-continued

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tcttgcttct ctttcttctg tcaaaagtct ctattccatt gttctctttg gtaaaactcat 420
tggtgtcgga actcagatat attcaggtca atttactgta cttcaattga cttttttctt 480
gaaatttcaa cttgcctttt caacttggtc ttctttttta atcttattct acactttagt 540
tcccttacct tggctctaatt tattgtctag caaaaagaaa acatacacct atttcattca 600
cacactaaaa caatgggaga cc 622

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SEQ ID NO: 11      moltype = DNA length = 825
FEATURE            Location/Qualifiers
misc_feature       1..825
                    note = Promoter DAN1 from Saccharomyces cerevisiae
source             1..825
                    mol_type = genomic DNA
                    organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 11
agaatgcaag ggcaaacagg cctgaggtaa ctttaaaata taatgtacct tgaaaactat 60
gtatacattg cattatttaa gcagtagggg attttgtttg caagaaaata aagctcaaaa 120
tatcttttga gtttgacaat agtatagaaa aaatagctca gcaaccttaa taaaaaaagg 180
gtcattgggc aaagcgtttt aaaaagattt aagtggtagt gtttaaaagt aaagagattg 240
atatagttaa aaattgttga gctcaattca cgctggattc ggcgatccgt tttcttcaat 300
cctcacgtgc tttcttctgt tgagtgcata agttcatatg atgctatctc ccgcttatct 360
tattagtcga aaatggggag aatttcctat tttatctgtc gtttagcaca tacggccagg 420
aaaatacata agatttcgcc gaacgacggg gtcaattcgt ccttttttga cacatcggtt 480
aatttatgag gaaaaaattg tgaaacgtat ctccgtagac gctcctctga aaagtttcat 540
gtttcctgcg cgttcctttg ataggcaata aaacaataca acgctgtgct ttgaaaatgc 600
caagatctat acgagggcctc taacaaaaca tcgttcagga acagagaatg ctaaaaatgc 660
aaaagggtcc ctgggtactc attgaataga aatgattgaa aatactgcgt ataaaatgc 720
acgactaaat gatactatct ttatgtcgac acggtactat ttcttctttt gcagataaaa 780
gtgtagcaga taaaagtgtg gcatactaaa tatatacccc aagta 825

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SEQ ID NO: 12      moltype = DNA length = 1000
FEATURE            Location/Qualifiers
misc_feature       1..1000
                    note = Promoter HEM13 (Sc_HEM13)
source             1..1000
                    mol_type = genomic DNA
                    organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 12
aaaggtagaa ggtattgcac cgattcctag tatcaattct tttcaaaacc acttattttg 60
caaaacatgc gtctaactcg ccatttcagc aacgtaata ataattactt ccatatatat 120
tcttctcccc cagtagcaat ttgacatga caaatttcag cgatagttgc ttcggtatct 180
gaaaatcgta gaagtatgac ttaatgtaga aggttgagaa caaccggatc ttgctggtcat 240
ttttcttttc gaggaaagtg caagtctgcc actttccaga aggcatagcc ttgccctttt 300
gttgatattt ctccccaccg taattgttgc attcgcgata ttttcaacaa tacattttat 360
catcaagccc gcaaatcctc tggagtttgt cctctcgttc actggttgga aaaacaatac 420
gcctaattcg tgattaagat tcttcaaac atttctcgcg gagtttttac tgtgtgttga 480
acggttcaca cgctaaaaaa agtttactat aggcacggta ttttaatttc aattgtttag 540
aaagtgcctt cacaccatta gcccttgagg ttaccgtcat aggcactttc tgctgagctc 600
ctgcgagatt tttgcgtgta aagagtaaaa gaaatcttcc acagcggctc cgcggggcct 660
tctactttta aacgagtcgc aggaacagaa gccaaatttc aaagaacgct acgctttcgc 720
cgtttctggt tctcccacca ataacgctcc agcttgaaca aagcataaga ctgcaaccaa 780
agcgtgacg gacgatccga agataaagct tgccttgccc attgttctcg ttgcgaaagg 840
ctatataagg acacggattt tccttttttt ttccacctat tgtctttctt tgttaagctt 900
ttattctcgg ggtttttttt ttgagcatat cgaaagcttt cttttcgcaa atcaaacata 960
gcaaaccgaa ctcttcgaac acaattaaat acacataaaa 1000

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SEQ ID NO: 13      moltype = AA length = 365
FEATURE            Location/Qualifiers
REGION            1..365
                    note = K. pneumoniae glycerol dehydrogenase (Kpne_gldA)
source            1..365
                    mol_type = protein
                    organism = Klebsiella pneumoniae

```

```

SEQUENCE: 13
MLKVIQSPAK YLQGPDAVL FGQYAKNLAE SFFVIADDFV MKLAGEKVVN GLQSHDIRCH 60
AERFNGECSE AEINRLMAIL QKQGCGRGVV IGGGKTLDTA KAIGYYQKLP VVVIPTIAST 120
DAPTSALSVI YTEAGEFEEY LIYPKNPDMV VMDTAIIAKA PVRLLVSGMG DALSTWFEAK 180
ACYDARATSM AGGQSTEAAL SLARLCYDTL LAEGEKARLA AQAGVVTEAL ERIIEANTYL 240
SGIGFESSGL AAHAHINHF TILIECHHLV HGEKVAFGTL AQLVLQNSPM DEIETVLGFC 300
QRVGLPVTLA QMGVKEGIDA KIAAVAKATC AEGETIHNMP FAVTPESVHA AILTADLLGQ 360
QWLAR 365

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SEQ ID NO: 14      moltype = AA length = 367
FEATURE            Location/Qualifiers
REGION            1..367
                    note = E. aerogenes glycerol dehydrogenase (Eaer_gldA)
source            1..367

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mol_type = protein
organism = Enterococcus aerogenes

SEQUENCE: 14
MDRIIQSPGK YIQGAGAIKR LGEYLKPLAE RWLIIGDKFV LGFAEEQLRT SLGGAGLVAE 60
IAPFGGECSSQ NEINRLRDIA SSAQCHAVLG IGGGKTLDTA KALAHYMHLP VVVVPTIAST 120
DAPCSALSVI YTDDGEFESY LMLPHNPNMV VVDTOIVAAA PARLLAAGIG DALATWFEAR 180
ACSRSGATTM AGGKCTQAAI ALAELCYNTL VEEGEKAMLA AEQHVVPAL ERVIEANTYL 240
SGVGFESGGL AAAHAHINGL TAIPDAHFFY HGEKVAFGTL TQLVLENAPV EEIETAAALC 300
HSVGLPITLA QLDIKGDIPA KMRTVAEAAE AEGETIHNMP GGACADQVYA ALLVADQYGQ 360
RFLQEWEE 367

SEQ ID NO: 15      moltype = AA length = 364
FEATURE           Location/Qualifiers
REGION            1..364
                  note = Y. aldovae glycerol dehydrogenase
source            1..364
                  mol_type = protein
                  organism = Yersinia aldovae

SEQUENCE: 15
MLKVIQSPSK YIQGANALQS IGEFAKLLAN NYFIIADDFV MKLTADTVGT SLQTCELKSH 60
FSRFNGECRS QEIERLTVEL KKYGCNGVIG IGGGKTLDTA KAIAHYQHLP VVVVPTIAST 120
DAPTSALSVI YTEQGEFAEY LIYPKNPDIV LMDTTIIAKA PVRLLVAGMG DALSTYFEAQ 180
ACFDAKAISM AGGASTLAAI TLARLCYETL LAEGYKAKLA VEAGVVTEAV ERIIEANTYL 240
SGIGFESSGL AAAHAHINGL TVLEECHHLY HGEKVAFGTL TQLVLQNSSM EEIETVLSFC 300
QQLGLPITLA EMGVTQDLEC KIRAVAQASC AEGETIHNMP FKVTADSVYA AIIVADRLGQ 360
AFLN 364

SEQ ID NO: 16      moltype = AA length = 367
FEATURE           Location/Qualifiers
REGION            1..367
                  note = E. coli glycerol dehydrogenase (Ec_gldA)
source            1..367
                  mol_type = protein
                  organism = Escherichia coli

SEQUENCE: 16
MDRIIQSPGK YIQGADVNR LGEYLKPLAE RWLVVGDKFV LGFAQSTVEK SFKDAGLVVE 60
IAPFGGECSSQ NEIDRLRGIA ETAQCGAILG IGGGKTLDTA KALAHFMGVP VAIAPTIAST 120
DAPCSALSVI YTDEGEFDRY LLLPNNPNMV IVDTKIVAGA PARLLAAGIG DALATWFEAR 180
ACSRSGATTM AGGKCTQAAI ALAELCYNTL LEEGEKAMLA AEQHVVPAL ERVIEANTYL 240
SGVGFESGGL AAAHAVHNGI TAIPDAHFFY HGEKVAFGTL TQLVLENAPV EEIETVAALS 300
HAVGLPITLA QLDIKEDVPA KMRIVAEAAE AEGETIHNMP GGATPDQVYA ALLVADQYGQ 360
RFLQEWEE 367

SEQ ID NO: 17      moltype = DNA length = 1111
FEATURE           Location/Qualifiers
misc_feature       1..1111
                  note = nucleotide sequeunce encoding E.coli gldA
                  codon-optimised
source            1..1111
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 17
atggacagaa tcatccaatc tccaggtaag tacatccaag gtgctgacgt tatcaacaga 60
ttgggtgaat acttgaagcc attggctgaa agatgggttg ttgttggtga caagtgcgtt 120
ttgggtttcg ctcaatctac cgttgaaaag tctttcaagg acgctgggtt ggttggtgaa 180
atcgtctccat tcggtggtga atgttctcaa aacgaaatcg acagattgag aggtatcgct 240
gaaaccgctc aatgtggtgc tatcttggtg atcgttggtg gtaagacctt ggacaccgct 300
aaggctttgg ctcacttcat ggggtgttcca gttgctatcg ctccaacccat cgcttctacc 360
gacgtctccat gttctgcttt gtctgttata tacaccgacg aaggtgaatt cgacagatac 420
ttgttggtgc caaacaaccc aaacatgggt atcgttgaca ccaagatcgt tgctggtgct 480
ccagctagat tgttggcagc tggtatcggt gacgctttgg ctacctgggt cgaagctaga 540
gcttggttcta gatctggtgc taccaccatg gctggtggtg agtgtaccca agctgctttg 600
gctttggctg aattgtgtta caacaccttg ttggaagaag gtgaaaagcg tatgttggtc 660
gctgaacaac acgttggttac cccagctttg gaaagagtta tcgaagctaa cacctacttg 720
tctggtgttg ttgtcgaaac gctggtgctc acgctgttca caacggtttg 780
accgctatcc cagacgctca ccactactac cacggtgaaa aggttgcttt cggtaccttg 840
acccaattgg ttttggaaaa cgctccagtt gaagaaatcg aaacggttgc tgctttgtct 900
cacgctgttg gtttgccaat caccttggtc caattggaca tcaaggaaga cggtccagct 960
aagatgagaa tcgttgctga agctgcttgt gctgaagggt aaaccatcca caacatgcca 1020
ggtgtgtgta cccagacca agtttacgct gcttggttg ttgtgacca atacggtcaa 1080
agattcctac aagaatggga ataaggcgcg c 1111

SEQ ID NO: 18      moltype = AA length = 584
FEATURE           Location/Qualifiers
REGION            1..584
                  note = S. cerevisiae dihydroxyacetone kinase (S. cerevisiae

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DAK1)
source 1..584
mol_type = protein
organism = *Saccharomyces cerevisiae*

SEQUENCE: 18
MSAKSFEVTD PVNSSSLKGFA LANPSITLVP EEKILFRKTD SDKIALISGG GSGHEPTHAG 60
FIGKGMLSGA VVGEIFASPS TKQILNAIRL VNNASGVLL IVKNYTGDLV HFGLSAERAR 120
ALGINCRVAV IGDDVAVGRE KGGMVGRRAL AGTVLVHKIV GAFAEYSSK YGLDGTAKVA 180
KIINDNLVTI GSSLDHCKVP GRKFESELNE QMELGMGIH NEPGVKVLDP IPSTEDLISK 240
YMLPKLLDPN DKDRAVFKFD EDDEVLLVN NLGGVSNFVI SSITSKTDF LKENYNITPV 300
QTIAGTLMTS PNGNGFSITL LNATKATKAL QSDFEEIKSV LDLLNAFTNA PGWPIADFEK 360
TSAPSVNDDL LHNEVTAKAV GTYDFDKFAE WMKSGAEQVI KSEPHITELD NQVGDGDCGY 420
TLVAGVKGIT ENLDKLSKDS LSQAVAQISD FIEGSMGGTS GGLYSILLSG FSHGLIQVCK 480
SKDEPVTKEI VAKSLGIALD TLKYTKARK GSSTMIDALE PFVKEFTASK DFNKAVKAAE 540
EGAKSTATFE AKFGRASYVG DSSQVEDPGA VGLCEFLKGV QSAL 584

SEQ ID NO: 19 moltype = AA length = 591
FEATURE Location/Qualifiers
REGION 1..591
note = *S. cerevisiae* dihydroxyacetone kinase (*S. cerevisiae* DAK2)
source 1..591
mol_type = protein
organism = *Saccharomyces cerevisiae*

SEQUENCE: 19
MSHKQPKSDG NIVTPYLLGL ARSNPGLTVI KHDRVVFRTA SAPNSGNPPK VSLVSGGSG 60
HEPTHAGFVG EGALDAIAAG AIFASPSTKQ IYSAIKAVES PKGTLIIIVKN YTGDIHFG 120
AAERAKAAGM KVELVAVGDD VSVGKKKGS L VGRRLGATV LVHKIAGAAA SHGLELAEVA 180
EVAQSVVDNS VTIAASLDHC TVPGHKPEAI LGENEYEIGM GIHNESGTYK SSPLPSISEL 240
VSQMLPLLLD EDEDRSVVKF EPKEDVVLV NMGMGSMNLE LGYAAEIVSE QLIDKYQIVP 300
KRTITGAFIT ALNGFGFGIT LMNASKAGGD ILKYFDYPTT ASGWNQMYHS AKDWEVLAKG 360
QVPTAPSLKT LRNEKSGSVK ADYDTFAKIL LAGIAKINEV EPKVTWYDTI AGDGDGCGTTL 420
VSGGEALEEA IKNHTRLLED AALGIEDIAY MVEDSMGGTS GGLYSIYLSA LAQGVDRSGD 480
KELTAETFKK ASNVALDALY KYTRARPGYR TLIDALQPFV EALKAGKGP AAAQAAYDGA 540
EKTRKMDALV GRASYVAKEE LRKLDSEGG L PDPGAVGLAA LLDGFVTAAG Y 591

SEQ ID NO: 20 moltype = AA length = 572
FEATURE Location/Qualifiers
REGION 1..572
note = *K. pneumoniae* dihydroxyacetone kinase (*Kpne_dhaK*)
source 1..572
mol_type = protein
organism = *Klebsiella pneumoniae*

SEQUENCE: 20
MTTKQFQFDS DPLNSALAAE AEASGLAYLP KSKVIYIPLT NDKVTLISGG GAGHEPAQTG 60
FVGPGLLDAA VSGQIFASPS TKQIIAGVNA VKSQRGSIII VMNYTGDLVIH FGMAAEQLRS 120
RYDYHAELVS IGDDISVNKK AGRRLAGTIV LVHKIAGHLA RDGWDVGVLA EALRTTAANL 180
ATVAASLEHC TVPGRKFETE LAADEMEIGM GIHNPEGVKT IKIGKVESLL DELVDKFEP 240
KQDFVPFNKG DEVLLVNSL GGVSSLELHA IANIAQTKEF KVLGVKTVRL IVGNFMAAFN 300
GPGFSLTLN VTTAKKGNF DVLGALDAPV STAAWPSLQQ KDKPANGGVQ EEKETDSDKP 360
AEPTGIKADG KLFKAMIESA VDDLKKEEPQ ITKYDTIAGD GDCGETLLAG GDGILDAIKN 420
KKIDLDDAAG VADISHIVEN SMGGTSGGLY SIFPSGLVVG IKETKAKELS VDVFAKACET 480
ALETLSKYTQ ARVGDRTLMD ALVPFVETLS KTKDFAKAVE AARKGADETS KLPANFGRAS 540
YVNEEGLENI PDPGALGLAV IFEGLLKAWE KK 572

SEQ ID NO: 21 moltype = AA length = 580
FEATURE Location/Qualifiers
REGION 1..580
note = *Y. lipolytica* dihydroxyacetone kinase (*Ylip_DAK1*)
source 1..580
mol_type = protein
organism = *Yarrowia lipolytica*

SEQUENCE: 21
MDKHFINDEP VLVLDGLKSL ADMNKTLTVH EEGKFIYFHD YNKKNVSVIS GGGAGHEPTH 60
SSFVKGKMLT AAVSGSIFAS PSSKQIYTG I KQVESEAGTL VICKNYTGDI LHFGMALEKQ 120
RTAGKAEALI AVADDVSVGR KKSQKVGRRG LSGTVLVHKI AGAAAARGLP LEAVTTIAKA 180
AIDNLVSIGA SLAHVHVPGH EPIAKEDEM K HDAMELGMIH HNEPGCKRIS PIPSIDDLIA 240
QMLKQMLDQS DKDRAYVKIE GDDEVLLMN NLGGLSMLF SAISHKVKEA LAKEYKINPV 300
RIFAGPFTTS LNGLGFGITL LRTTDRVKVE GEEYSLVDLI DQPVEAIGWP LCQPSDLKSK 360
NKIGNVIEE QKQDVKSPVT VDKEKVRQAI VNSMENLIKA EPKITKFDTM AGDGDGCGTTL 420
KRGAEGVLKF VKSDKFSDDP IRIVRDIADV IEDNMDGTSG ALYAIFFHGF AKGMKDTLEK 480
SKDISSKTWA AGLKVALDTL FKYTPARPGD STMCDALVPF VETVFKTNDL NAAVEEARKG 540
ADATADMQAK LGRAVYVGGD VKVPDAGALG VVAIVEGFTK 580

SEQ ID NO: 22 moltype = AA length = 580
FEATURE Location/Qualifiers

-continued

REGION 1..580
 note = *S. pombe* dihydroxyacetone kinase (Spom_DAK1)

source 1..580
 mol_type = protein
 organism = *Schizosaccharomyces pombe*

SEQUENCE: 22

MDKHFINDPE	VLVLVDGLKSL	ADMNKTTLTVH	EEGKFIYPHD	YNKKNVSVIS	GGGAGHEPTH	60
SSFVVGKMLT	AAVSGSIFAS	PSSKQIYTG	KQVESEAGTL	VICKNYTGDI	LHFGMALEKQ	120
RTAGKKABELI	AVADDVSVGR	KKSGKVGRRG	LSGTVLVHKKI	AGAAAARGLP	LEAVTTIAKA	180
AIDNLVLSIGA	SLAHVHVPGH	EPIAKEDEMK	HDEMEELGMI	HNEPGCKRIS	PIPSIDDLIA	240
QMLKQMLDQS	DKDRAYVKIE	GDDEVVLLMN	NLGGLSMLEF	SAISHKVKEA	LAKEYKINPV	300
RIFAGPFTTS	LNGLGFGITL	LRTTDRVKVE	GEEYSLVDLI	DQPVEAIGWP	LCQPSDLKSK	360
NKIGNVSIEE	GQKDVKSPT	VDKEKVRQAI	VNSMENLIKA	EPKITKFDTM	AGDGDGCTTL	420
KRGAEGVLKF	VKSKDFSDDP	IRIVRDIADV	IEDNMDGTSG	ALYAIFFHGF	AKGMKDTLEK	480
SKDISSKTWA	AGLKVVALDTL	FKYTPARPGD	STMCDALVPF	VETFEVKTNDL	NAAVEEARKG	540
ADATADMQAK	LGRAVYVGDD	VKVPDAGALG	VVAIVEGFTEK			580

SEQ ID NO: 23 moltype = DNA length = 1755
 FEATURE Location/Qualifiers
 misc_feature 1..1755
 note = *S. cerevisiae* dihydroxyacetone kinase (*S. cerevisiae* DAK1)

source 1..1755
 mol_type = genomic DNA
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 23

atgtccgcta	aatcgtttga	agtcacagat	ccagtcgaatt	caagtctcaa	agggtttgcc	60
cttgctaacc	cctccattac	gctggtcctc	gaagaaaaaa	ttctcttcag	aaagaccgat	120
tcgcacaaga	tcgcattaat	ttctgggtgt	ggtagtggtg	atgaacctac	acacgccggt	180
ttcattggta	agggatgtgt	gagtggtggt	gtggttggtg	aaatctttgc	atccccctca	240
acaaaacaga	ttttaaattgc	aatccgttta	gtcaatgaaa	atgcgtctgg	cgtttttattg	300
attgtgaaga	actacacag	tgatgttttg	catcttggtc	tgctccgtga	gagagcaaga	360
gccttggtga	ttaactgcg	cgttgctgtc	ataggtgatg	atgttgctag	tgccagagaa	420
aaaggtggta	tggttggtag	aagagcattg	gcaggtaccg	ttttggttca	taagattgta	480
gggtgccttc	cagaagaata	ttctagtaag	tatggcttag	acgggtacag	taaagtggct	540
aaaattatca	acgacaattt	gggtgacctt	ggatcttctt	tagaccattg	taaagttcct	600
ggcagggaat	tcgaaagtga	attaaacgaa	aaacaaatgg	aattgggtat	gggtattcat	660
aacgaacctg	gtgtgaaagt	tttagacctt	attccttcta	ccgaagactt	gatctccaag	720
tatatgttac	caaaactatt	ggatccaaac	gataaggata	gagcttttgt	aaagtttgat	780
gaagatgatg	aagttgtctt	gttagttaac	aatctcggtg	gtgtttctaa	ttttgttatt	840
agttctatca	cttccaaaac	tacggatttc	ttaaaggaaa	attacaacat	aaccccggtt	900
caaaccaattg	ctggcacatt	gatgacctcc	ttcaatggta	atgggttcag	tatcacatta	960
ctaaacgccca	ctaaagctac	aaagggcttg	caatctgatt	ttgaggagat	caaatcagta	1020
ctagacttgt	tgaacgcatt	tacgaacgca	ccgggctggc	caattgcaga	ttttgaaaag	1080
acttctgccc	catctgttaa	cgatgacttg	ttacataatg	aagtaacagc	aaaggccgtc	1140
ggtagcctatg	actttgacaa	gtttgctgag	tggtatgaaga	gtgggtgctga	acaagttatc	1200
aagagcgcaac	cgacacattac	ggaactagac	aatcaagttg	gtgatgggtg	ttgtgggtac	1260
acttttagtg	caggagttaa	aggcatcacc	gaaaaccttg	acaagctgtc	gaaggactca	1320
ttatctcagg	cggttgccca	aatttcagat	ttcattgaag	gctcaatggg	aggtacttct	1380
ggtaggttat	attctattct	tttgtcggtt	ttttcacacg	gattaattca	ggtttgtaaa	1440
tcaagagtag	aaccgcctac	taaggaaaat	gtggctaagt	cactcggaat	tgcatgggat	1500
actttatata	aatatacaca	ggcaagggaag	ggatcatcca	ccatgattga	tgctttagaa	1560
ccattctgta	aagaattttac	tgcatctaa	gatttcaata	aggcggttaa	agctgcagag	1620
gaaggtgcta	aatccactgc	tacattcgag	gccaaatttg	gcagagcttc	gtatgtcggc	1680
gattcatctc	aagttagaaga	tcctggtgca	gtaggcctat	gtgagttttt	gaaggggggt	1740
caaagcgctt	tgtaa					1755

SEQ ID NO: 24 moltype = DNA length = 1776
 FEATURE Location/Qualifiers
 misc_feature 1..1776
 note = *S. cerevisiae* dihydroxyacetone kinase (*S. cerevisiae* DAK2)

source 1..1776
 mol_type = genomic DNA
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 24

atgtctcaca	aacaattcaa	atcagatgga	aacatcggtta	ctccctacct	acttggcctt	60
gctcgaaagca	atcccgccct	tacagtgtat	aagcacgaca	gagtggtttt	ccggactgcg	120
tcagctccta	attcaggga	cctcctctaa	gtttcattgg	ttttctggagg	tgccagtgggt	180
catgagccaa	cgcattgcgg	ttttgttggt	gaaggtgcct	tagatgcgat	tcagcagagt	240
gccatttttg	cttctccttc	aactaaacag	atctattctg	ctattaaagc	tggtgaatct	300
cctaagggtta	ccttgatcat	tgtaaaaaat	tacacgggtg	atattatata	ttttggtctc	360
gctgctgaaa	gagctaaagc	tgctggaatg	aaagtgcga	tggttgctgt	aggagatgat	420
gtctctgtcg	gtaagaagaa	aggttcttta	gtcggggcgt	gaggtctcgg	agccaccgta	480
ttggtgcata	aaattgctgg	ggcagccgct	tctcatggac	tggaagttgg	agaagttgcc	540
gaagttgctc	agtcagtagt	tgacaatagt	gtcacaattg	cggtcatctc	tgatcaactgc	600

-continued

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acggttcctg gccacaaacc tgaagccatt ttgggcgaga atgagtatga aatcggtatg 660
ggtattcata acgagtcttg tactataag tcttctccgc tgccatcgat ttctgagctc 720
gtttcccaga tgettcctct tcttctcgat gaggatgaag accgttctta tgtgaagttt 780
gagcccaaaag aggacgtagt ccttatgggt aacaacatgg gtggtatgtc taatctagaa 840
ttgggttatg ctgcagaggt catttctgaa caattgattg ataagtatca aattgtgccc 900
aagagaacga ttactggagc attcattact gcattgaatg gtctctgggt ttggtatcact 960
cttatgaacg cttcaaaagc tgggtggcgt atccttaagt atttcgatta tcctaccaca 1020
gcgagtggat ggaatcaaat gtaccattct gccaaagatt gggaggtagt tgccaaaggg 1080
caggttccca cgcgccctct tttaaagaca ttgaggaatg aaaaagggtc ggggtgtgaaa 1140
gctgattatg acacttttgc taaaattttg ctgctgggga ttgcaaaaat taacgaggtt 1200
gaaccaaagg ttactttgta cgataccatt gcaggagatg gtgattgcgg aactactctt 1260
gtgtccgggtg gtgaagcatt ggaagaagct attaaaaacc atacgttcgc cctcagggat 1320
gctgctcttg gtatcgaaga tattgcgtat atggttgagg attctatggg tggtagctcc 1380
gggtgctctg actctatcta tcttctgct ctgcacaaag gagttaggga ttctggggac 1440
aaggaaacta ctgcggaaac tttcaaaaag gcatcaaacg ttgcactaga tgctttgtat 1500
aagatataca gagcccgctc tgggttacag actctgatcg atgctctgca acctttgtc 1560
gaagcgctga aagccgggaa ggggtcccaga gccgcgcgcc aagctgctta tgatgggtgc 1620
gaaaagacaa ggaagatgga tgcccttggt gggcgctgct cttacgtggc taaggaggag 1680
ctgagaaaac tcgacagcga aggtggatta ccagatccag gacgagttgg tcttgctgca 1740
ctactcgatg gatttggtac agctgctggg tactag 1776

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SEQ ID NO: 25      moltype = AA length = 459
FEATURE           Location/Qualifiers
REGION           1..459
                  note = Ribulose-1,5-biphosphate carboxylase oxygenase
                  (RuBisCO)
source           1..459
                  mol_type = protein
                  organism = Thiobacillus denitrificans

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SEQUENCE: 25
MDQSARYADL SLKEEDLIK GRHILVAYKM KPKSGYGYLE AAHFPAESS TGTNVEVSTT 60
DDFTKGVDAI VYIDEASED MRIAYPLELF DRNVTDGRFM LVSFLTLAGI NNQMGDIEH 120
AKMIDFYVPE RCIQMFDPGA TDISNLWRIL GRPVVNGGYI AGTIIPKPLG LRPEPFAKAA 180
YQFWLGGDFI KNDEPQGNQV FCPLKKVLPL VYDAMKRAQD DTGQAKLFMS NITADDDHYEM 240
CARADYALEV FGPDAKDLAF LVDGYVGGPG MVTARRQYP GQYLHYHRAG HGAVTSPSAK 300
RGYTAFVLAK MSRLQGASGI HVGTMGYGKM EGEEDDKIIA YMIERDECQG PVYFQKWYGM 360
KPTTPIISGG MNALRLPGFF ENLGHGNVIN TAGGGSYGHI DSPAAGAI SL RQSYECWKQG 420
ADPIEFAKEH KEFARAFESF PKDADKLFPG WREKLGVHS 459

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SEQ ID NO: 26      moltype = DNA length = 1380
FEATURE           Location/Qualifiers
misc_feature      1..1380
                  note = synthetic; based on cbbM gene from Thiobacillus
                  denitrificans, codon optimized for S. cerevisiae
source           1..1380
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 26
atggatcaat ctgcaagata tgcgtacttg tctttaaagg aagaagattt gattaaagg 60
ggtagacata ttttgggtgc ttacaaaatg aaaccaaatt ctggttatgg ttatttggaa 120
gctgctgctc attttgctgc tgaatcttct acaggtagaa atgttgaagt ttctactaca 180
gatgatttta caaaagggtg tgatgcttta gtttactaca tcgatgaagc ttcagaagat 240
atgagaattg cttatccatt ggaattattc gacagaaatg ttactgacgg aagattcatg 300
ttagtttctt ttttgacttt ggctatttgg aacaatcaag gaatgggaga tatagaacat 360
gcaaaaatga tagattttta cgttcagaaa agatgtattc aaatggttga tgggtccagct 420
acagatattt ctaatttggg gagaattttg ggtagaccag tagttaatgg tgggttatatt 480
gctgtgacta ttattaagcc aaaattgggt ttaagaccag aaccatttgc taaagctgct 540
tatcaatttt ggttgggttg agattttatc aagaatgacg aaccacaagg taatcaagtt 600
ttttgtccat gtaaaaaagt tttgccattg gtttacgatg ctatgaaaag agcacaagat 660
gatactggtc aagcaaaaat gttttctatg aatattactg cagacgatca ttatgaaatg 720
tgtgcaagag ctgattatgc tttggaagtt ttcggtccag atgcagataa attggctttt 780
ttggttagatg gttacgttgg aggtccagga atggttacta ctgctagaag gcaatatcct 840
ggtcaatatt tgcattatca tagagcaggt cacggtgctg ttacttctcc atctgctaaa 900
agaggttata ctgcttttgg tttggctaaa atgtctagat tgcaaggcgc ttcaggtatt 960
catgttggtg ctatgggtta tggaaaaaag gcgacgataa gattatttgc 1020
tatatgatag aaagggacga atgtcaaggt ccagtttatt ttcaaaaatg gtacggtatg 1080
aaaccaacta ctccaattat ctccggagga atgaatgctt tgagattgcc tgggtttttc 1140
gaaaatttgg gtcattgtaa cgttattaat actgcaggtg gtggttctta cggtcattat 1200
gattctcctg ctgctggtgc tatttctttg agacaactct acgaatgttg gaaacaagg 1260
gcagatccaa ttgaatttgc taaggacat aaggaaattg caagagcttt tgaatctttt 1320
ccaaaagatg ctgataagtt atttccagga tggagagaaa aattgggagt tcattcttaa 1380

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SEQ ID NO: 27      moltype = AA length = 352
FEATURE           Location/Qualifiers
REGION           1..352
                  note = S. oleracea phosphoribulokinase (prk)

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source          1..352
                mol_type = protein
                organism = Spinacia oleracea

SEQUENCE: 27
MSQQQTIVIG LAADSGCGKS TFMRLTSVF GGAEPPKGG NPDSENTLISD TTVICLDDF 60
HSLDRNGRKV EKVTALDPKA NDFDLMEYQV KALKEGKAVD KPIYNHVSGL LDPPELIQPP 120
KILVIEGLHP MYDARVRELL DFSIYLDISN EVKFAWKIQR DMKERGHSLE SIKASIESRK 180
PDFDAYIDPQ KQHADVVIEW LPTELIPDDD EGKVLVRVMI QKEGVKFFNP VYLFDEGSTI 240
SWIPCGRKLT CSYPGIKFSY GPDTFYGNV TVVEMDGMFD RLDELIYVES HLSNLSTKPY 300
GEVTQQMLKH QNFPGSNNGT GFFQTIIGLK IRDLFEQLVA SRSTATATAA KA 352

SEQ ID NO: 28      moltype = DNA length = 1059
FEATURE            Location/Qualifiers
misc_feature        1..1059
                    note = synthetic; based on prk gene from S. oleracea -
                    codon optimizedfor S. cerevisiae

source             1..1059
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 28
atgtcacacaac aacaacaat tgtgattggt ttagcagcag attcagggtg tggtaaagagt 60
acattcatga ggagggttaac aagtgttttc ggtggcgcgcg cgcagccacc aaagggtggt 120
aaccagatt caaacacatt gattagtgac actactactg ttatctgttt ggatgatttt 180
cattcccttg atagaaatgg caggaaagtg gaaaaagtta ctgctttaga cccaaaagct 240
aatgattttg atcttatgta tgaacaagtt aaggctttga aagaaggtaa agctgttgat 300
aaacctattt ataatacatgt ttctggtttg ttggaccctc ctgagcttat tcaacctcct 360
aagatcttgg tcattgaagg gttacacccc atgtatgacg cactgtgtgag ggaattgcta 420
gacttcagca tctacttgga cattagcaat gaagttaaat ttgcctggaa aattcagaga 480
gacatgaaag aacagaggaca cagttctttaa agcatcaaaag ccagtattga atccagaaag 540
ccagattttg atgctttacat tgaccacaaa aagcagcatg ctgatgtagt gattgaagta 600
ttgccaaactg aactcattcc tgatgatgat gaaggcaaaag tgttgagagt aaggatgatt 660
cagaaagaag gagtcaagtt tttcaaccca gtttacttgt ttgatgaag atctaccatt 720
tcatggattc catgtggtag aaaattaaca tgttcttacc ctggtatcaa attttcctat 780
ggcccagaca ccttctatgg caacgaggtg acagtagtag agatggatgg gatgtttgac 840
agattagacg aactaatcta cgtcgaaagc catttgagca atctatcaac caagttttat 900
ggtgaagtca ctcaacaaat gttgaagcac caaaatttcc caggaaagcaa caatggaact 960
ggtttcttcc aaaccataat tggattgaag atcagagact tgttcgagca gctcgttgct 1020
agcaggtcta cagcaactgc aacagctgct aaagcctag 1059

SEQ ID NO: 29      moltype = AA length = 548
FEATURE            Location/Qualifiers
REGION             1..548
                    note = Translated protein sequence, based on GroEL from E.
                    coli
source             1..548
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 29
MAAKDVKFGN DARVKMLRGV NVLADAVKVT LGPKGRNVVL DKSFGAPTIT KGVSVVAREI 60
ELEDKFENMG AQMVKEVASK ANDAAGDGTT TATVLAQAI TEGLKAVAAG MNPMDLKRG I 120
DKAVTAAVEE LKALSVPCSD SKAIAQVGTI SANSDETVGK LIAEAMDKVG KEGVITVEDG 180
TGLQDELDDV EGMQFDRGYL SPYFINKPET GAVELESFPI LLADKKISNI REMLPVLEAV 240
AKAGKPLLI AEDVEGEALA TLVNTMRGI VKVAAVKAPG FGDRRKAMLQ DIATLTGGTV 300
ISEEIGMELE KATLEDLQQA KRVVINKDIT TIIDGVGEEA AIQGRVAQIR QQIEEATSDY 360
DREKLQERVA KLAGGVAIVK VGAATEVEMK EKARVEDAL HATRAAVEEG VVAGGGVALI 420
RVASKLADLR QONEDQNVGI KVALRAMEAP LRQIVLNCGE EPSVVANTVK GGDGNYGYNA 480
ATEEYGNMID MGILDPTKVT RSLAQYAASV AGLMITTECM VTDLPKNDAA DLGAAGGMGG 540
MGGMGGMM 548

SEQ ID NO: 30      moltype = DNA length = 1647
FEATURE            Location/Qualifiers
misc_feature        1..1647
                    note = Synthetic, based on GroEL from E. coli - codon
                    optimized
source             1..1647
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 30
atggctgcta aggacgttaa gttcggtaac gacgctagag ttaagatggt gagaggtggt 60
aacgttttgg ctgacgctgt taaggttact ttgggtccaa agggtagaaa cgttgttttg 120
gacaagtctt tcggtgtctc aactatcact aaggacggtg tttctgttgc tagagaaatc 180
gaattggaag acaagttcga aaacatgggt gctcaaatgg ttaaggaagt tgettctaag 240
gctaacgacg ctgctgttga cgttactact actgtactg ttttggctca agctatcatc 300
actgaaggtt tgaaggctgt tgctgctggt atgaacccaa tggacttgaa gagaggtatc 360
gacaaggctg ttaactgctg tgttgaaaga ttgaaggctt tgtctgttcc atgttctgac 420
tctaaggcta tcgctcaagt tggtactatc tctgctaact ctgacgaaac tgttggttaa 480

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ttgatcgctg aagctatgga caaggttggt aaggaaggtg ttatcactgt tgaagacggt 540
actgggtttg aagacgaatt ggacgttggt gaaggtatgc aattcgacag aggttacttg 600
tctccatact tcatcaacaa gccagaaact ggtgctgttg aattggaatc tccattcattc 660
ttgttggtcg acaagaagat ctctaactc agagaaatgt tgccagtttt ggaagctggt 720
gctaaggctg gtaagccatt gttgatcatc gctgaagacg ttgaaggtga agctttggct 780
actttgggtg ttaacactat gagagggtatc gttgaaggtg ctgctgttaa ggctccaggt 840
ttcggtgaca gaagaaaggc tatgttgcaa gacatcgcta ctttgactgg tggtactggt 900
atctctgaag aaatcggtat ggaattggaa aaggctactt tggaagactt gggccaagct 960
aagagagttg ttatcaacaa ggacactact actatcatcg acggtgttgg tgaagaagct 1020
gctatccaag gtagagttgc tcaaatcaga caacaaatcg aagaagctac ttctgactac 1080
gacagagaaa agttgcaaga aagagttgct aagttggctg gtggtgttgc tgttatcaag 1140
gttggtgctg ctactgaagt tgaatgaag gaaaagaagg cttagagtga agacgctttg 1200
cacgctacta gagctgctgt tgaagaaggt gttgttgctg gtggtggtgt tgctttgatc 1260
agagttgctt ctaagttggc tgacttgaga ggtcaaaacg aagacaaaaa cggttggtatc 1320
aaggttgctt tgagagctat ggaagctcca ttgagacaaa tcgttttgaa ctgtggtgaa 1380
gaaccatctg ttgttgctaa cactgttaag ggtggtgacg gtaactacgg ttacaacgct 1440
gctactgaag aatacgttaa catgatcgac atgggtatct tggaccaaac taaggttact 1500
agatctgctt tgcaatacgc tgcttctggt gctggtttga tgatcactac tgaatgatg 1560
gttactgact tggcaagaa gacgctgctg gacttgggtg ctgctggtgg tatgggtggt 1620
atgggttggt tgggtggtat gatgtaa 1647

SEQ ID NO: 31      moltype = AA length = 97
FEATURE           Location/Qualifiers
REGION           1..97
source           note = Translated protein sequence, based on GroES of E.coli
                 1..97
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 31
MNIRPLHDRV IVKRKEVETK SAGGIVLTGS AAAKSTRGEV LAVGNRGILE NGEVKPLDVK 60
VGDIVFPNDG YGVKSEKIDN EEVLIMSED ILAIVEA 97

SEQ ID NO: 32      moltype = DNA length = 294
FEATURE           Location/Qualifiers
misc_feature      1..294
source           note = Synthetic, based on GroES from E. coli - codon
                 1..294
                 optimized
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 32
atgaacatca gaccattgca cgacagagtt atcgtttaaga gaaaggaagt tgaactaag 60
tctgctgggt gtaatcgttt gactggttct gctgctgcta agtctactag aggtgaagtt 120
ttggctggtg gtaacggttag aatcttgtaa aacggtgaag ttaagccatt ggacgttaag 180
gttggtgaca tcgttatctt caacgacggt tacggtgtta agtctgaaaa gatcgacaac 240
gaagaagttt tgatcatgtc tgaatctgac atcttggcta tcgttgaagc ttaa 294

SEQ ID NO: 33      moltype = AA length = 825
FEATURE           Location/Qualifiers
REGION           1..825
source           note = B. animalis xylulose-5P/fructose-6P
                 1..825
                 phosphoketolase(Bani_XFP.orf) amino acid sequence
                 mol_type = protein
                 organism = Bifidobacterium animalis

SEQUENCE: 33
MTNPVIGTPW QKLDPRVSEE AIEGMDKYWR VANYMSIGQI YLRSNPLMKE PFTRDDVKHR 60
LVGHWGTTTPG LNFLLAHINR LIADHQQNTV FIMGPGHGGP AGTAQSYIDG TYTEYYPNIT 120
KDEAGLQKFF RQFSYPGGIP SHFAPETPGS IHEGGELGYA LSHAYGAIMD NPSLFVPCII 180
GDGEAETGPL ATGWQSNKLV NPRTDGIVLP ILHLNGYKIA NPTILARISD EELHDFFRGM 240
GYHPYEFVAG FDNEDHLSIH RRFaelFETI FDEICDIKAA AQDDMTTRPF YPMLIFRTPK 300
GWTCPKFIDG KKTEGWSRAH QVPLASARDT EAHFEVLKGW MESYKPEELF NADGSIKEDV 360
TAFMPKGELR IGANPNANGG RIREDLKLPE LDQYEITGVK EYGHGWGQVE APRSLGAYCR 420
DIIKNPNDSF RVFGPDETAS RRLNATYEVY KKQWDNGYLS ALVDENMAVT QQVVEQLSEH 480
QCEGFLEAYL LTGRHGIWSS YESFVHVIDS MLNQHAKWLE ATVREIPWRK PISSVNLLVS 540
SHVWRQDHNG FSHQDPGVTS VLLNKTFFND HVTNIYFATD ANMLLAIAEK CFKSTNKINA 600
IFAGKQPAAT WITLDEVRAE LEAGAAEWKW ASNAKSNDEV QVVLAAGADV PTQEIIMASD 660
ALNKMGIKFK VVNVVDLIK LQSSKENDEAM SDEDPADLFT ADKPVLPAYH SYAQDVRGLI 720
YDRPNHNDFT VVGYPEQGST TTPFDMVRVN DMDRYALQAK ALELIDADKY ADKINELNEF 780
RKTAQFQFVD NGYDIPEFTD WVYPDVKVDE TSMLSATAAT AGDNE 825

SEQ ID NO: 34      moltype = AA length = 825
FEATURE           Location/Qualifiers
REGION           1..825
source           note = B. adolescentis xylulose-5P/fructose-6P
                 1..825
                 phosphoketolase(Bado_XFP.orf) amino acid sequence

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source 1..825
mol_type = protein
organism = Bifidobacterium adolescentis

SEQUENCE: 34

MTSPVIGTPW	KKLNAVSE	AIEGVDKYWR	AANYLSIGQI	YLRSNPLMKE	PFTREDVKHR	60
LVGHWGTTTG	LNFLIGHNR	LIADHQQNTV	IIMGPGHGGP	AGTAQSYLDG	TYTEYFPNIT	120
KDEAGLQKFF	RQFSYPGGIP	SHYAPETPGS	IHEGGELGYA	LSHAYGAVMN	NPSLFVPAIV	180
GDGEAETGPL	ATGWQSNKLI	NPRTDGIVLP	ILHLNGYKIA	NPTILSRISD	EELHEFFHGM	240
GYEPYEFVAG	FDNEDHLSIH	RRFAELFETV	FDEICDIKAA	AQTDDMTRPF	YPMIIFRTPK	300
GWTCPKFIDG	KKTEGWSRSH	QVPLASARDT	EAHFEVLKNW	LESYKPEELF	DENGAVKPEV	360
TAFMPTGELR	IGENPNANGG	RIREELKLPK	LEDYEVKEVA	EYGHGWGQLE	ATRRRLGVYTR	420
DIKNNPDSF	RIFGPDETAS	NRLQAAYDVT	NKQWDAGYLS	AQVDEHMAVT	GQVTEQLSEH	480
QMEGFLEGYL	LTGRHGIWSS	YESFVHVIDS	MLNQHAKWLE	ATVREIPWRK	PISSMNLVLS	540
SHVWRQDHNG	FSHQDPGVTS	VLLNKCFFND	HVIGIYFPVD	SNMLLAVAEK	CYKSTNKINA	600
IITAGQPAAT	WLTLDDEARAE	LEKGAAEWKW	ASNVSNDDEA	QIVLAAATGDV	PTQEIMAAAD	660
KLDAMGIKFF	VVNVVDLVLK	QSAKENNEAL	SDEEPAELFT	EDKPVLFAYH	SYARDVRGLI	720
YDRPNHDNPN	VHGVEEQGST	TPPYDMVRVN	NIDRYELQAE	ALRMIDADKY	ADKINELEAF	780
RQEAQFQFVD	NGYDHPDYTD	WVYSGVNTNK	QGAISATAAT	AGDNE		825

SEQ ID NO: 35 moltype = AA length = 825
FEATURE Location/Qualifiers
REGION 1..825
note = B. lactis xylulose-5P/fructose-6P phosphoketolase
(Blac_XFP.orf) amino acid sequence

source 1..825
mol_type = protein
organism = Bifidobacterium lactis

SEQUENCE: 35

MTNPVIGTPW	QKLDPRVSEE	AIEGMDKYWR	VANYMSIGQI	YLRSNPLMKE	PFTRDDVKHR	60
LVGHWGTTTG	LNFLLAHNR	LIADHQQNTV	FIMGPGHGGP	AGTAQSYIDG	TYTEYFPNIT	120
KDEAGLQKFF	RQFSYPGGIP	SHFAPETPGS	IHEGGELGYA	LSHAYGAIMD	NPSLFVPCII	180
GDGEAETGPL	ATGWQSNKLV	NPRTDGIVLP	ILHLNGYKIA	NPTILARISD	EELHDFFRGM	240
GYHPYEFVAG	FDNEDHLSIH	RRFAELFETI	FDEICDIKAA	AQTDDMTRPF	YPMIIFRTPK	300
GWTCPKFIDG	KKTEGWSRAH	QVPLASARDT	EAHFEVLKGW	MESYKPEELF	NADGSIKEDV	360
TAFMPKGELR	IGANPNANGG	RIREDLKLPE	LDQYEITGVK	EYGHGWGQVE	APRSLGAYCR	420
DIKNNPDSF	RVFGPDETAS	NRLNATYEV	KKQWDNGYLS	ALVDENMAVT	GQVVEQLSEH	480
QCEGFLEAYL	LTGRHGIWSS	YESFVHVIDS	MLNQHAKWLE	ATVREIPWRK	PISSVNLVLS	540
SHVWRQDHNG	FSHQDPGVTS	VLLNKTFFND	HVTNIYFATD	ANMLLAIAEK	CFKSTNKINA	600
IFAGQPAAT	WITLDEARAE	LEAGAAEWK	ASNKSNDDEV	QVVLAAAGDV	PTQEIMAAAD	660
ALNKMGIKFF	VVNVVDLVLK	QSSKENDEAM	SDEDFADLFT	ADKPVLFAYH	SYAQDVRGLI	720
YDRPNHDNFT	VVGKEQGST	TPPYDMVRVN	DMDRYALQAK	ALELIDADKY	ADKINELNEF	780
RKTAQFQFVD	NGYDIPEFTD	WVYPDVKVDE	TSMLSATAAT	AGDNE		825

SEQ ID NO: 36 moltype = AA length = 813
FEATURE Location/Qualifiers
REGION 1..813
note = L. mesenteroides xylulose-5P/fructose-6P
phosphoketolase(Lmes_XFP.orf) amino acid sequence

source 1..813
mol_type = protein
organism = Leuconostoc mesenteroides

SEQUENCE: 36

MADFDSKEYL	ELVDKWRAT	NYLSAGMIFL	KSNPLFSVTN	TPIKAEDVKV	KSIGHWGTIS	60
GQTFLYAHAN	RLINKYGLNM	FYVGGPGHGG	QVMVTNAYLD	GAYTEDYPEI	TQDIEGMSHL	120
FKRFSFPGGI	GSHMTAQTPG	SLHEGGELGY	SLSHAFGAVL	DNPQVAFAV	VGDGEAETGP	180
SMASWHSIKF	LNAKNDGAVL	PVLDLNGFKI	SNPTIFSRMS	DEEITKFFEG	LGYSRPFIEI	240
DDIHDIATYH	QLAANILQQA	IEDIQAIQND	ARENGKYQDG	EIPAWPVIIA	RLPKGWGGPT	300
HDASNNPIEN	SFRAHQVPLP	LEQHDLATLP	EFEDWMNSYK	PEELFNADGS	LKDELKAIAP	360
KGDKRMSANP	ITNGGADRSD	LKLPNWREFA	NDINDNTRGK	EFADSKRNMD	MATLSNYLGA	420
VSQLNPTRFR	FFGPDETMSN	RLWGLFNVTP	RQWMEIEKEP	QDQLLSPTGR	IIDSQQLSEHQ	480
AEGLWLEGYTL	TGRVGIFASY	ESFLRVVDTM	VTQHFQWLRH	ASEQAWRNDY	PSLNLIATST	540
AFQQDHNGYT	HQDPGMLTHL	AEKKSNFIRE	YLPADGNSLL	AVQERAPSER	HKVNLLIASK	600
QPRQWFTVE	EAEVLANEGL	KIIDWASTAP	SGDVDTIFAS	AGTEPTIETL	AALWLINQAF	660
PVDKFRYVNV	VELLRLQKKS	EIPNMNDEREL	SABEENKYFQ	ADTPVIFGFH	AYENLIESFF	720
FERKFTGDVY	VHGTYREDGDI	TTYDMMRVYS	HLDRFHQAKE	AAEILSANGK	IDQAAADTFI	780
AKMDDTLAKH	FQVTRNEGRD	IEEFTDWTWS	PLK			813

SEQ ID NO: 37 moltype = AA length = 323
FEATURE Location/Qualifiers
REGION 1..323
note = B. subtilis phosphotransacetylase (Bs_PTA.orf) amino
acidsequence

source 1..323
mol_type = protein
organism = Bacillus subtilis

SEQUENCE: 37

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MADLFSTVQE	KVAGKDVKIV	FPEGLDERIL	EAVSKLAGNK	VLNPIVIGNE	NEIQAKAKEL	60
NLTGGLVKIY	DPHTYEGMED	LVQAFVERRK	GKATEEQARK	ALLDENYFGT	MLVYKGLADG	120
LVSGAAHSTA	DTVRPALQII	KTKEGVKKTS	GVFIMARGE	QYVFADCAIN	IAPDSQDLAE	180
IAIESANTAK	MFDIEPRVAM	LSFSTKGSAK	SDETEKVADA	VKIAKEKAPE	LTLDGFEQFD	240
AAFPVPSVAEK	KAPDSEIKGD	ANVFVFPSPLE	AGNIGYKIAQ	RLGNFEAVGP	ILQGLNMPVN	300
DLSRGCNAED	VYNLALITAA	QAL				323

SEQ ID NO: 38 moltype = AA length = 325
 FEATURE Location/Qualifiers
 REGION 1..325
 note = L. plantarum phosphotransacetylase (Lpla_PTA.orf)
 amino acidsequence
 source 1..325
 mol_type = protein
 organism = Lactobacillus plantarum

SEQUENCE: 38

MDLFESLSQK	ITGQDQITVF	PEGTEPRIVG	AAARLAADGL	VKPIVLGATD	KVQAVAKDLN	60
ADLTGVQVLD	PATYPADKQ	AMLDSLVERR	KGKNTPEQAA	KMLEDENYFG	TMLVYMGKAD	120
GMVSGAVHPT	GDTVRPALQI	IKTKPGSHRI	SGAFIMQKGE	ERYVFADCAI	NIDPDADTLA	180
EIATQSAATA	KVFPDIDPKVA	MLSFSSTKGS	KGDMVTKVQE	ATAKAQAAAP	ELAIDGEMQF	240
DAAFVEKVG	LQKAPGSKVAG	HANVFVFPPEL	QSGNIGYKIA	QRFHGFHFAVG	PVLQGLNKPV	300
SDLSRGCSEE	DVYKVAIITA	AQGLA				325

SEQ ID NO: 39 moltype = AA length = 556
 FEATURE Location/Qualifiers
 REGION 1..556
 note = B. adoloscensis phosphotransacetylase (Bado_PTA.orf)
 amino acidsequence
 source 1..556
 mol_type = protein
 organism = Bifidobacterium adoloscensis

SEQUENCE: 39

MSFTSVTIIS	PEAANGNRNV	ALGVTKTLLA	AGKTGVFRPA	VCRKDTFTDV	LIEASNAGLS	60
REQSVGVCPK	RARNDKEGSR	ADIVAAYTQA	VETARPDAMV	IVGTDRSAVN	DPAMFSEFNAD	120
VAADLQSPVL	LAVCTIERTP	EQVKSTVEAS	TKVIEDAGSK	VVGVFITGCD	DTQPNPLKAC	180
FVDYVPVWVT	LPAYDFNDDD	AISKADEAFA	TNVDVELTV	ALESPPFDAPT	TPYAFQYGLL	240
GKAKADKTTI	VLPEGNEDRI	IKAADYLLER	DIVDLIIVGD	ENAILARGQE	LGLKSLGKAK	300
FQAKDDETVL	EPMVAKLCEL	RAKKGMTTEE	ARKQLADDSY	FGTMLVVMGM	ADGLVSGSVN	360
STANTVRPAL	QVIKTKPGTS	LVSAGFLMCF	KDHAADFAD	AINLNPNAEQ	LAEIAIQSAE	420
TAKAFGLEPK	VGMLSYSTLG	SGKGPVDVLV	EEATTIVKDK	APDLAVVSGI	QFDAAWSPTV	480
AATKAKGDPV	AGHVNVPVFP	DLCAGNIAYK	AVQRSSGAAA	VGPVLQGLNR	PVNDLSRGAT	540
VQDIINTIAL	TAIEAQ					556

SEQ ID NO: 40 moltype = AA length = 333
 FEATURE Location/Qualifiers
 REGION 1..333
 note = M. thermophila phosphotransacetylase (Mthe_PTA.orf)
 amino acidsequence
 source 1..333
 mol_type = protein
 organism = Methosarcina thermophila

SEQUENCE: 40

MVTFLEKISE	RAKKLNKTIA	LPETEDIRTL	QAAAKILERG	IADIVLVGNE	ADIKALAGDL	60
DLKAKIVDP	KTYEKKDEYI	NAPYELRKHK	GITLENAAEI	MSDYVYFAVM	MAKLGEVDGV	120
VSGAAHSSSD	TLRPAVQIVK	TAKGAALASA	FFIISVPDCE	YGSFGTFLFA	DSGMVEMPSV	180
EDVANIAVIS	AKTFELLVQD	VPKVAMLSYS	TKGSAKSKLT	EATIASTKLA	QELAPDIAID	240
GELQVDAAIV	PKVAASKAPG	SPVAGKANVF	IFPDLNCGNI	AYKIAQRLAK	AEAYGPITQG	300
LAKPINDLSR	GCSDEDIVGA	VAITCVQAAA	QDK			333

SEQ ID NO: 41 moltype = AA length = 409
 FEATURE Location/Qualifiers
 REGION 1..409
 note = B. adoloscensis acetate kinase (Bado_ACK) amino acid
 sequence
 source 1..409
 mol_type = protein
 organism = Bifidobacterium adoloscensis

SEQUENCE: 41

MAKTVLVINS	GSSSIKYQLV	DLETGEGIAS	GLVEKIGEPV	DGHYKHEYNG	EKHELEEPIH	60
DHEQGLKRVL	GFFDEFQPKL	ADAGIVAVGH	RVVQGSIFP	KPALVNDKTI	GQVKDLAVLA	120
PLHNGPEAKG	AEVMSRLLPD	VPQIFVFDSS	FFFQLPKASS	TYALNKEVAQ	QYHIRRYGAH	180
GTSHEFISSV	VPSVIGKPAE	GLKQIVLHIG	NGASASAEIS	GKPVETSMGL	TPLEGLVMGG	240
RTGDIDPAVV	PHLRNAHMS	VDELDTLFLN	RSMMGLTGF	GDLREVHRLV	EEGNEDAKLA	300
LDIYVHRIVG	YIGNYTAQMG	GVDVITFTAG	VGENDDVVRK	MVCDKLAPFG	VKLDEEKNAT	360
RSKEPRIIST	PDSAVTICVI	PTNEELAIAI	KSAATAEEGK	DSYGNVFSK		409

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SEQ ID NO: 42 moltype = AA length = 420
 FEATURE Location/Qualifiers
 REGION 1..420
 note = A. nidulans ACK (Ani_ACK) amino acid sequence
 source 1..420
 mol_type = protein
 organism = *Aspergillus nidulans*

SEQUENCE: 42
 MPRKSILSVN AGSSSVKITF YSYTKTPSVI ATAQVSGITA PPAATFKYSVG SKQKKEELKE 60
 KISSGPDFAK LLLHRCFTDS DLKDVASADD LAYICHRVVH GGDFESPVVI NEETYHQLED 120
 LEDLAPLHNF AALEIVRLCK KELPNVQSIT FFDSSFHKS L PPYVKTYPID QETARRNKLK 180
 KYGFHGISYS FILRSVAEYL NKPVEKTS LI ALHIGSGASV CAIKDGKSID TSMGLTPLAG 240
 LPGATRSGDI DPSLVFHYTN EAGKLSFAST KEMHISTAE E ILNKKSGWKV LTGTTDFSQI 300
 AVEDPPSEQH KLAFLDILVDR IVGYIGNYYV KLDGQVEGIV FAGGIGEKSA LLRKAVIEQT 360
 RCLGFAVDPE KNQHPGDDE TVVDITASGR SDVKRVFICQ TDEQFEMAYN CTKTQGLDKQ 420

SEQ ID NO: 43 moltype = AA length = 891
 FEATURE Location/Qualifiers
 REGION 1..891
 note = E. coli bifunctional NAD+ dependent
 acetylatingacetaldehyd/alcohol dehydrogenase (E. coli
 adhE) amino acidsequence
 source 1..891
 mol_type = protein
 organism = *Escherichia coli*

SEQUENCE: 43
 MAVTNVAELN ALVERVKKAQ REYASFTQEQ VDKIFRAAAL AAADARIPLA KMAVAESGMG 60
 IVEDKVIKMH FASEYIYNAY KDEKTCGVLS EDDTPGTITI AEPIGIICGI VPTNPTSTA 120
 IPKSLISLKT RNAIIFSPHP RAKDATNKAA DIVLQAAIAA GAPKDLIGWI DQPSVELSNA 180
 LMHHDPDINLI LATGGPGMVK AAYSSGKPAI GVGAGNTPV IDETADIKRA VASVLMSKTF 240
 DNGVICASEQ SVVVVDSVYD AVRERFATHG GYLLQGKELK AVQDVILKNG ALNAAIVGQP 300
 AYKIAELAGF SVPENTKILI GEVTVVDESE PFAHEKLSPT LAMYRAKDFE DAVEKAELKV 360
 AMGGIGHTSC LYTDQDNQPA RVSYFGQKMK TARILINTPA SQGGIGDLYN FKLAPSLTLG 420
 CGSWGNGNIS ENVGPKHLIN KKTVAKRAEN MLWHKLPKSI YFRRGSLPIA LDEVITDGHK 480
 RALIVTDRFL FNNGYADQIT SVLKAAGVET EVFVEVEADP TLSIVRKGAE LANSFKPDVI 540
 IALGGGSPMD AAKIMWVME HPETHFEELA LRFMDIRKRI YKPKMGVKA KMIATVTTSG 600
 TGSEVTPFAV VTDDATGQKY PLADYALTPD MAIVDANLVM DMPKSLCAFG GLDAVTHAME 660
 AYVSVLASEF SDGQALQALK LLKEYLPASY HEGSKNPVAR ERVHSAATIA GIAFANAFLG 720
 VCHSMAHKLK SQPHIPHLA NALLICNVIR YNANDNPTKQ TAFSQYDRPQ ARRRYAEIAD 780
 HLGLSAPGDR TAAKIEKLLA WLKTLKAEELG IPKSIREAGV QEADFLANVD KLSDAFDDQ 840
 CTGANPRYPL ISELKQILLD TYYGRDYVEG ETAAKKEAAP AKAEEKKAKKS A 891

SEQ ID NO: 44 moltype = AA length = 467
 FEATURE Location/Qualifiers
 REGION 1..467
 note = E. coli ethanolamine utilizing protein (Ec_eutE)
 amino acidsequence
 source 1..467
 mol_type = protein
 organism = *Escherichia coli*

SEQUENCE: 44
 MNQDDIEQVV KAVLLKMQSS DTPSAAVHEM GVFASLDDAV AAKVAQQGL KSVAMRQLAI 60
 AAIREAGEKH ARDLAELAVS ETGMGRVEDK FAKNVAQARG TPGVECLSPQ VLTGDNGLTL 120
 IENAPWGVVA SVTPSTNPAA TVINNAISLI AAGNSVIFAP HPAAKKVSQR AITLLNQAIV 180
 AAGGPENLLV TVANPDIIETA QRLFKFPGIG LLVVTGGEAV VEAARKHTNK RLIAAGAGNP 240
 PVVVDETADL ARAAQSIKVG ASFDNNIICA DEKVLIVVDS VADELMLRME GQHAVKLTA E 300
 QAQQQLQPVLL KNIDERGKGT VSRDWWGRDA GKIAAAIGLK VPQETRLLFV ETTAEHPFAV 360
 TELMMPVLPV VRVANVADAI ALAVKLEGGC HHTAAMHSRN IENMNQMANA IDTSIFVKNG 420
 PCIAGLGLGG EGWTTMTITT PTGEGVTSAR TFRVLRRCVL VDAFRIV 467

SEQ ID NO: 45 moltype = AA length = 455
 FEATURE Location/Qualifiers
 REGION 1..455
 note = L. plantarum acetaldehyde dehydrogenase (Lpla_acdH)
 source 1..455
 mol_type = protein
 organism = *Lactobacillus plantarum*

SEQUENCE: 45
 MLKEMEETTV SRSIDRLVLN ASLAANRLEV MDQSQVDQAV AAMARAAHAA RGMLAAMAVE 60
 ETGRGNRYRK VAKNDFAAKN VYNYIKDDKT VGIINDDPVS GVMKVAEPVG IIAGVTPVTN 120
 PTSTVIFNAM LALKTRNP II FGFHPFAQKS CVETGRIIRD AAIASGAPKD WIQWIKTPSL 180
 EATNTLMNHP GVATIIATGG AGMVKTAYST GKPALGVGPG NVPCFIEQTA DIQQAVSDV 240
 TSKSFDNGMI CASESNLIVA DQIYDQVKRE LSHNGVYFVG TENFKALEAT VMNLDKQAVD 300
 PKVAGQTPWQ IAQWAGFDVP SDTKVLAVEL PSIGGDQVLS REKLSPLVAV VHAKDTEAGF 360
 NLMKRSALG GLGHTAALHT TDEAVMNKFA LEMTACRALI NVPSSQGAIG YKYDNVAPSL 420
 TLGCGTGWGHN SISHNLEDWD LLNIKTVAKR LTKIR 455

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SEQ ID NO: 46 moltype = AA length = 469
 FEATURE Location/Qualifiers
 REGION 1..469
 note = L. innocua acetaldehyde dehydrogenase (Linn_acdH)
 source 1..469
 mol_type = protein
 organism = Listeria innocua

SEQUENCE: 46
 MESLELEQLV KKVLEKLAEE QKEVPTKTTT QGAKSGVFDT VDEAVQAQAVI AQNCYKEKSL 60
 EERRNVVKAI REALYPEIET IATRAVAETG MGNVTDKILK NTLAIEKTPG VEDLYTEVAT 120
 GDNGMTLYEL SPYGVIGAVA PSTNPTETLI CNSIGMLAAG NAVFYSPHPG AKNISLWLE 180
 KLNTIVRDSG GIDNLIVTVA KPSIQAAQEM MNHPKVPLLV ITGGPGVVLQ AMQSGKKVIG 240
 AGAGNPPSIV DETANIEKAA ADIVDGASFD HNILCIAEKS VVAVDSIADF LLFQMEKNGA 300
 LHVTNPDSIQ KLEKVAVTDK GVTNKKLVGK SATEILKEAG IACDFTPLRI IVETEKSHPF 360
 ATVELLMPIV PVVRVPDFDE ALEVAIELEQ GLHHTATMHS QNISRLNKAA RDMQTSIFVK 420
 NGPSFAGLGF RGEGSTTFTI ATPTGEGTTT ARHFARRRRR VLTGDFSIR 469

SEQ ID NO: 47 moltype = AA length = 869
 FEATURE Location/Qualifiers
 REGION 1..869
 note = S. aureus acetaldehyde/alcohol dehydrogenase
 (Saur_adhE)
 source 1..869
 mol_type = protein
 organism = Staphylococcus aureus

SEQUENCE: 47
 MLTIPEKENR GSKEQEVAIM IDALADKGGK ALEALSKKSQ EEIDHIVHQM SLAAVDQHMV 60
 LAKLAHEETG RGIYEDKAIK NLYASEYIWN SIKDNKTVGI IGEDKEKGLT YVAEPIGVIC 120
 GVTPTTNPTS TTIFKAMIAI KTGNIPIFAF HPSAQESSKR AAEVVLEAAM KAGAPKDIIQ 180
 WIEVPSIEAT QOLMNHKGIA LVLATGGSGM VKSAYSTGKP ALGVGPGNVP SYIEKTAHIK 240
 RAVNDIIGSK TFDNGMICAS EQVVVIDKEI YKDVTFNEFK HQAYFVKKDE LQRLNAIMN 300
 EQKTGIKPII VGKSAVEIAE LAGIPVPENT KLIIAEISGV GSDYPLSREK LSPVLALVKA 360
 QSTKQAFQIC EDTLHFGGLG HTAVIHTED TLQKDFGLRM KACRVLVNTP SAVGGIGDMY 420
 NELIPSLTLG CGSYGRNSIS HNVSATDLLN IKTIARRNN TQIFKVPAQI YFEENAIMSL 480
 TTMDKIEKVM IVCDPGMVEF GYTKTVENVL QRTEQPQIK IFSEVEPNPS TNTVYKGLEM 540
 MVDFQPDTHI ALGGGSAMDA AKAMWMFFEHE PETSFFGAKQ KFLDIGKRTY KIGMPENATF 600
 ICIPTTSGTG SEVTPFAVIT DSETNVKYPL ADFALTPDVA IIDPQFVMSV PKSVTADTGM 660
 DVLTHAMESY VSVMASDYTR GLSLQAIIKT FEYLKSSVEK GDKVSREKMH NASTLAGMAF 720
 ANAFLGIAHS IAHKIGGEYG IPHGRANAIL LPHIIRYNAK DPQKHALFPK YEFFRADTDY 780
 ADIAKFLGLK GNTTEALVES LAKAVYELGQ SVGIEMNLKS QGVSEBELNE SIDRMAELAF 840
 EDQCTTANPK EALISEIKDI IQTSYDYKQ 869

SEQ ID NO: 48 moltype = AA length = 314
 FEATURE Location/Qualifiers
 REGION 1..314
 note = acetylating acetaldehyde dehydrogenase (Ec_mhpF)
 source 1..314
 mol_type = protein
 organism = Escherichia coli

SEQUENCE: 48
 MSKRKVAIIG SGNIGTDLMI KILRHGQHLE MAVMVGIDPQ SDGLARARRM GVATTHEGVI 60
 GLNMPEFAD IDIVDFATSA GAHVKNDAAL REAKPDIRLI DLTAAIGPY CVPWNLEAV 120
 DQLNVNMVTC GGQATIPMVA AVSRVARVHY AEIIASIASK SAGPGTRANI DEFTETTSRA 180
 IEWGGAAGK AIIVLNPAEP PLMMRDVVV LSDEASQDDI EASINEMAE VQAYVPGYRL 240
 KQRVQFEVIP QDKPVNLPV GQFSGLKTA VLEVEGAHY LPAYAGNLDI MTSSALATAE 300
 KMAQSLARKA GEAA 314

SEQ ID NO: 49 moltype = AA length = 713
 FEATURE Location/Qualifiers
 REGION 1..713
 note = Acetyl-coenzyme A synthetase 1 (Sc_ACS1)
 source 1..713
 mol_type = protein
 organism = Saccharomyces cerevisiae

SEQUENCE: 49
 MSPSAVQSSK LEEQSSEIDK LKAKMSQSAA TAQQKKEHEY EHLTSVKIVP QRPISDRLQP 60
 AIATHYSPLH DGLQDYQRLH KESIEDPAKF FGSKATQFLN WSKPFDKVFI PDPKTGRPSF 120
 QNNAWFLNGQ LNACYNVDR HALKTPNKKA IIFEGDEPGQ GYSITYKELL EEVCQVAQVL 180
 TYSMGVRKGD TVAVYMPMVP EAIITLLAIS RIGAIHSVVF AGFSSNSLRD RINDGDSKV 240
 ITTDESNRGG KVIETKRIVD DALRETPGVR HVLVYRKTNM PSVAFHAPRD LDWATEKKKY 300
 KTYYPCTPVD SEDPLFLLYT SGSTGAPKGV QHSTAGYLLG ALLTMRYTFD THQEDVFFTA 360
 GDIGWITGHT YVVYGPLLYG CATLVFEGTP AYPNYSRYWD IIDEHKVTQF YVAPTALRL 420
 KRAGDSYIEN HSLKSLRCLG SVGEPIAAEV WEWYSEKIGK NEIPIVDITYW QTESGSHLVT 480
 PLAGGVTPMK PGSASFPPFG IDAVLDPNP GEELNTSHAE GVLAVKAAWP SFARTIWKNH 540
 DRYLDTYLNP YPGYYFTGDG AAKDKDGYIW ILGRVDDVVN VSGHRLSTAE IEAAIIEDPI 600

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VAECAVVGFN	DDLTGQAVAA	FVVLKKNSSW	STATDDELQD	IKKHLVFTVR	KDIGPFAAPK	660
LILVDDLPK	TRSGKIMRRI	LRKILAGESD	QLGDVSTLSN	PGIVRHLIDS	VKL	713

SEQ ID NO: 50 moltype = AA length = 683
 FEATURE Location/Qualifiers
 REGION 1..683
 note = Acetyl-coenzyme A synthetase 2 (Sc_ACS2)
 source 1..683
 mol_type = protein
 organism = Saccharomyces cerevisiae

SEQUENCE: 50
 MTIKEHKVVY EAHNVKALKA PQHFYNSQPG KGYVTDMQHY QEMYQQSINE PEKFFDKMAK 60
 EYLHWDAPYT KVQSGSLNNG DVAWFLNGKL NASYNVCDRH AFANPDKPAL IYEADDESND 120
 KIITFGELLR KVSQIAGVLK SWGVKKGDTV AIYLPPIPEA VIAMLAVERI GAIHSVVFAG 180
 FSAGSLKDRV VDANSKVIT CDEGKRGGKT INTKKIVDEG LNGVDLVSRI LVFQRTGTGEG 240
 IPMKAGRQYW WHEEAAKQRT YLPPVSCDAE DPLFLLYTSG STGSPKGVVH TTGGYLLGAA 300
 LTRYVFDIH PEDVLFAGD VGWITGHTYA LYGPLTLGTA SIIFESTPAY PDYGRYWRRI 360
 QRRKATHFYV APTALRLIKR VGEAEIAKYD TSSLRVLGVS GEPISPDLWE WYHEKVGKNK 420
 CVICDTMWQT ESGSHLIAPL AGAVPTKPGS ATVPFPGINA CIIDPVTGVE LEGNDVEGVL 480
 AVKSPWPSMA RSVVNHHDYR MDTYLPKYPG HYFTGDGAGR DHDGYYWIRG RVDDVVNVSG 540
 HRLSTSEIEA SISHENHNSV AAVVGIPDEL TGQTVVAVVS LKDGYLQNNR TEGDAEHITP 600
 DNLRRELILQ VRGEIGPFAS PKTIILVRDL PRTRSGKIMR RVLKRVASNE AEQLGDLTTL 660
 ANPEVVPAIL SAVENQFSSQ KKK 683

SEQ ID NO: 51 moltype = AA length = 348
 FEATURE Location/Qualifiers
 REGION 1..348
 note = Alcohol dehydrogenase 1 (Sc_ADH1)
 source 1..348
 mol_type = protein
 organism = Saccharomyces cerevisiae

SEQUENCE: 51
 MSIPETQKGV IFYESHGKLE YKDIPVPKPK ANELLINVKY SGVCHTDLHA WHGDWPLPVK 60
 LPLVGGHEGA GVVVGMGENV KGNKIGDYAG IKWLNGSCMA CEYCELGNES NCPHADLSGY 120
 THDGSFQQYA TADAVQAHI PQGTDLAQVA PILCAGITVY KALKSANLMA GHWWAISGAA 180
 GGLGSLAVQY AKAMGYRVLG IDGGEGKEEL FRSIGGEVFI DFTKEKDIVG AVLKATDGGG 240
 HGVINVSVEE AAIEASTRYC RANGTTVLVG MPAGAKCCSD VFNQVVKSSIS IVGSYVGNRA 300
 DTREALDFFA RGLVKSPIKV VGLSTLPEIY EKMEKGQIVG RYVVDTSK 348

SEQ ID NO: 52 moltype = AA length = 348
 FEATURE Location/Qualifiers
 REGION 1..348
 note = Alcohol dehydrogenase 2 (Sc_ADH2)
 source 1..348
 mol_type = protein
 organism = Saccharomyces cerevisiae

SEQUENCE: 52
 MSIPETQKAI IFYESNGKLE HKDIPVPKPK PNELLINVKY SGVCHTDLHA WHGDWPLPTK 60
 LPLVGGHEGA GVVVGMGENV KGNKIGDYAG IKWLNGSCMA CEYCELGNES NCPHADLSGY 120
 THDGSFQEYA TADAVQAHI PQGTDLAQVA PILCAGITVY KALKSANLRA GHWWAISGAA 180
 GGLGSLAVQY AKAMGYRVLG IDGGEGKEEL FTSLGGEVFI DFTKEKDIVS AVVKATNGGA 240
 HGVINVSVEE AAIEASTRYC RANGTTVLVG LPAGAKCCSD VFNHVVKSIS IVGSYVGNRA 300
 DTREALDFFA RGLVKSPIKV VGLSSLPEIY EKMEKGQIAG RYVVDTSK 348

SEQ ID NO: 53 moltype = AA length = 375
 FEATURE Location/Qualifiers
 REGION 1..375
 note = Alcohol dehydrogenase 3 (Sc_ADH3)
 source 1..375
 mol_type = protein
 organism = Saccharomyces cerevisiae

SEQUENCE: 53
 MLRTSTLFTR RVQPSLFSRN ILRLQSTAAI PKTQKGVIFY ENKGLHYKD IPVPEPKPNE 60
 ILINVKYSGV CHTDLHAWHG DWPLPVKLPL VGGHEGAGVV VKLGSNVKGW KVGDLAGIKW 120
 LNGSCMTCEF CESGHESNCP DADLSGYTHD GSFQQFATAD AIQAAKIQQG TDLAEVAPIL 180
 CAGVTYVKAL KEADLKAGDW VAISGAAGGL GSLAVQYATA MGYRVLGIDA GEEKEKLFKK 240
 LGGEVFIDFT KTKNMVSDIQ EATKGGPHGV INVSVSEAAI SLSTEYVRPC GTVVLVGLPA 300
 NAYVKSEVFS HVVKSINIKG SYVGNRADTR EALDFFSRGL IKSPIKIVGL SELPKVYDLM 360
 EKGKILGRIV VDTSK 375

SEQ ID NO: 54 moltype = AA length = 382
 FEATURE Location/Qualifiers
 REGION 1..382
 note = Alcohol dehydrogenase 4 (Sc_ADH4)
 source 1..382
 mol_type = protein

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organism = *Saccharomyces cerevisiae*

SEQUENCE: 54

MSSVTGFYIP	PISFFGEGAL	EETADYIKNK	DYKKALIVTD	PGIAAIGLSG	RVQKMLEERD	60
LNVAIYDKTQ	PNPNIANVTA	GLKVLKEQNS	EIVVSIGGGS	AHDNAKAIAL	LATNGGEIGD	120
YEGVNQSKKA	ALPLFAINTT	AGTASEMTRF	TIISNEEKKI	KMAIIDNNVT	PAVAVNDPST	180
MFGLPPALTA	ATGLDALTHC	IEAYVSTASN	PITDACALKG	IDLINESLVA	AYKDGKDKKA	240
RTDMCYAEYL	AGMAFNNASL	GYVHALAHL	GGFYHLPHGV	CNAVLLPHVQ	EANMQCPKAK	300
KRLGEIALHF	GASQEDPEET	IKALHVLNRT	MNIPRNLKEL	GVKTEDFEIL	AEHAMHDACH	360
LTNPVQFTKE	QVVAIIKAY	EY				382

SEQ ID NO: 55 moltype = AA length = 351

FEATURE Location/Qualifiers

REGION 1..351

 note = Alcohol dehydrogenase 5 (Sc_ADH5)

source 1..351

 mol_type = protein

 organism = *Saccharomyces cerevisiae*

SEQUENCE: 55

MPSQVIPEKQ	KAIVFYETDG	KLEYKDVTVP	EPKPNEILVH	VKYSQVCHSD	LHAWHGDWPF	60
QLKFPLIGGH	EGAGVVVKLG	SNVKGWKVGD	FAGIKWLNGT	CMSCEYCEVG	NESQCPYLDG	120
TGFTHDGTFQ	EYATADAVQA	AHIPPNVNLA	EVAPILCAGI	TVYKALKRAN	VIPGQWVTIS	180
GACGGLGSLA	IQYALAMGYR	VIGIDGGNAK	RKLFEQLGGE	IFIDFTEKED	IVGAIKATN	240
GGSHGVINVS	VSEAAIEAST	RYCRPNGTVV	LVGMPAHAYC	NSDVFNQVVK	SISIVGSCVG	300
NRADTREALD	FFARGLIKSP	IHLAGLSDVP	EIFAKMEKGE	IVGRYVVETS	K	351

SEQ ID NO: 56 moltype = AA length = 561

FEATURE Location/Qualifiers

REGION 1..561

 note = *P. strigosozonata* glucoamylase

source 1..561

 mol_type = protein

 organism = *Punctularia strigosozonata*

SEQUENCE: 56

SSKSVSAYIS	SESPIAHSKL	LDNIGPDGAK	APGAPPGVVV	ASPSTDNPNY	YYSWIRDSSL	60
VFKTLIDYV	NGKNTSKSLR	SLIDDFVTAS	SVFQQTPNPS	GNVSTGGLGE	PKFYVNETAF	120
LDSWGRPQRD	GPALRSTALI	TYANYLLDND	NTTWVKDTLW	PIIELDVNYV	SDFWNYTTFD	180
LWEEVASSSF	FTTAVQHRAL	RQASKLAKTL	DKTDNIDSWN	TQADNVLCFL	QSYWNGSAII	240
ANTGGGRSGI	DANTVLASIH	TFDSSAGCDA	TTFQPCSDKA	LSNLKVYVDA	FRSIYEINSG	300
IDPNAAVATG	RYPEDVFYDG	NPWYLATAAV	AEQLYDALYV	WNTTGSLEIT	DISLPFFQQF	360
DSDVKTGTYS	DDDTFDSLIS	SIQSFADGFL	EIHAKYTPDD	GALSEEFSTK	DGSQTSAADL	420
TWSYAAALTA	FDARSRAAV	KWGAKGLQVP	DGTCKTNEGG	DDGLGVPVTF	LVKDAETVEG	480
QSVYITGSIA	TLKSWSPDDA	LLMSPSDYPT	WTLTVNLSAS	ESVQYKYIKK	DTAGTVIWES	540
DPNNSLLVPS	GGSVTTDDTW	R				561

SEQ ID NO: 57 moltype = AA length = 578

FEATURE Location/Qualifiers

REGION 1..578

 note = *P. strigosozonata* glucoamylase with native signal

 sequence FGA09

source 1..578

 mol_type = protein

 organism = *Punctularia strigosozonata*

SEQUENCE: 57

MLSSLIVSGL	LASGVCASSK	SVSAYISSES	PIAHSKLLDN	IGPDGAKAPG	AFPGVVVASP	60
STDNPNYYY	WIRDSSLVFK	TLIDDVVNGK	NTSKSLRSLI	DDFVTASSVF	QQTPNPSGNV	120
STGGLGEPKF	YVNETAFLDS	WGRPQRDGP	LRSTALITYA	NYLLDNDNTT	WVKDTLWPII	180
ELDVNYVSDF	WNYTTFDLWE	EVASSSFFTT	AVQHRALRQA	SKLAKTLDKT	DNIDSWNTQA	240
DNVLCFLQSY	WNGSAIIANT	GGGRSGIDAN	TVLASIHTFD	SSAGCDATTF	QPCSDKALSN	300
LKVYVDAPRS	IYEINSIDP	NAAVATGRYP	EDVFDGNPW	YLATAAVAEQ	LYDALYVWNT	360
TGSLEITDIS	LPFFQQFDSD	VKTGTYSDDD	TFDSLISIQ	SFADGFLEIH	AKYTPDDGAL	420
SEEFSTKDG	QTSAADLTWS	YAAALTAFDA	RSRDAAVKNG	AKGLQVPDGT	CKTNEGGDDG	480
LGVPVTFVLK	DAETVEGQSV	YITGSIATLK	SWSPDDALLM	SPSDYPTWTL	TVNLSASESV	540
QYKYIKKDTA	GTWIESDPN	NSLLVPSGGS	VTTDDTWR			578

SEQ ID NO: 58 moltype = AA length = 580

FEATURE Location/Qualifiers

REGION 1..580

 note = *P. strigosozonata* glucoamylase with Sc alpha mating

 factor signalsequence

source 1..580

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 58

MRFPSIFTAV	LFAASSALAS	SKSVSAYISS	ESPIAHSKLL	DNIGPDGAKA	PGAPPGVVVA	60
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QADNVLCPLQ	SYWNGSAIIA	NTGGGRSGID	ANTVLASIHT	FDSSAGCDAT	TFQPCSDKAL	300
SNLKVVYDAF	RSIYEINSIGI	DPNAAVATGR	YPEDVIFYDGN	PWYLATAAVA	EQLYDALYVW	360
NTTGSLEITD	ISLPFFQQFD	SDVKTGTYS	DDTFDSLIS	IQSPADGFLE	IHAKYTPDDG	420
ALSEEFSKTD	GSQTSAADLT	WSYAAALTAF	DARSRDAAVK	WGAKGLQVPD	GTCKTNEGGD	480
DGLGVPVTF	VKDAETVEGQ	SVYITGSIAT	LKSWSPDDAL	LMSPSDYPTW	TLTVNLSASE	540
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SEQ ID NO: 59 moltype = DNA length = 2716
 FEATURE Location/Qualifiers
 misc_feature 1..2716
 note = Artificial Sequence, fragment A,
 25-EFT2p.DAK1.ENO1t-2A(pDSM005479)
 source 1..2716
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 59

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SEQ ID NO: 60 moltype = DNA length = 2062
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 note = Artificial Sequence 2A-HHF2p.gldA.CTC1t-2B
 source 1..2062
 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 61      moltype = DNA length = 2741
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                   note = Artificial Sequence, fragment C,
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source            1..2741
                   mol_type = other DNA
                   organism = synthetic construct

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SEQ ID NO: 62      moltype = DNA length = 2717
FEATURE           Location/Qualifiers
misc_feature       1..2717
                   note = Artificial Sequence, fragment
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source            1..2717
                   mol_type = other DNA
                   organism = synthetic construct

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SEQ ID NO: 63      moltype = DNA length = 1837
FEATURE           Location/Qualifiers
misc_feature       1..1837
                   note = Artificial Sequence, fragment
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source            1..1837
                   mol_type = other DNA
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gctaagatga gaattgttgc tgaagctgct tgtgctgaag gtgaaacatc tcacaacatg 1620
ccaggtgggt ccaccccaga ccaagtctac gctgctttgt tgggtgctga ccaatcgggt 1680
caaaagattct tgcaagaatg ggagtaaat tgaactgtag cctaagtata aaattctacg 1740
tatgtatata tttacatgca attttttctt ttccaattc atgcctctcc tcggatcggc 1800
cagttgggag caaggtggta cttctgtcgt cgtattt 1837

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SEQ ID NO: 64

FEATURE

misc_feature

moltype = DNA length = 2789

Location/Qualifiers

1..2789

note = Artificial Sequence, fragment

F,2B-Sc_ANB1.pro_0001-Zrou_T5.orf-Sc_TEF1.ter_0001-2C
(lvDSM012515)

source

1..2789

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 64

```

tctcgggac ggccagttgg gagcaagggt gtacttctgt cgtcgtattt gtgcagatcg 60
gtgaaaacat cttgatcttg ctcccgggaa ttttagattc aggtaggaaa ttgattacat 120
caatactgtt accctgaatc atattcgacg atgtcgtctc acacggaaat ataattcatt 180
tcttgggttt ccaaaaaaat tttcattttt tttcactttt ttgtttcgtc ctctttttt 240
ttttttttt tttattttt ttctgtgtt cactttttt ttttccagt gacatcttc 300
tgcattcttt tctgtgtttt tttttttt ttctgtttt ccattgttcg ttcgttgct 360
gttttttcgc cctattgttc tcgagcctaa aaatttttct ctttctgct ttcctttct 420
cgttcaaatg ttctatttcc atgtttctct ttggtaaact cattgtgttc ggaactcaga 480
tatattcagg tcaatttact gtaacttcaat tgactttttt cttgaaattt caacttgct 540
tttcaacttg tttctttt ttaattcttat tctacacttt agttccctta cttgttctct 600
aattattgtc tagcaaaaaa aaaacataca cctatttcat tcacacacta aaacaatggg 660
taagagaact caaggtttca tggactacgt tttctccaga acttctactg ctggtttgaa 720
aggtgccaga ttgagataca tgcgtgctgc cgttgcgtgt atcgggttcg ctttattcgg 780
ttacgaccaa ggtttgatgt ctggtttgat taccgggtgac caattcaaca aggaattccc 840
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tgcttgttac gaaattgggt gtttcttcgg tttcttgcgt gcttattctt ttggtgatgc 960
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ctccactgct ccatttccacc acgcttgggg ttgggtcaa tttgttgcg gttagattat 1080
cactgggtgc ggtactgggt tcaacacttc caccatccca gtctggcaat ctgaaatgac 1140
caagccaaac atcagagggt ctatgattaa cttggatggt tccgtcattg ctttcggtag 1200
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tatcttgaaa tttccatgga tctaccacc agaaattgct tccatgagag tcagagctgc 1920
taccacgct atgtccacct gtaccaactg ggttaccac tttgcggtg tcatgttcac 1980

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cccgatcttt atccaaactt ctcaatgggg ttgttacttg ttcttcgctg tcatgaactt 2040
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aatcgatatt attttcgcta aggcctcatgt tgacgggtact ctaccatgga tgggttgcga 2160
cagattacca aagttgtcca tgactgaagt cgaagattac tctcaatctt tgggtttgca 2220
cgacgatgaa aacgaaaagg aagaatacga cgaaaaagaa gctgaagcta acgctgcctt 2280
gttccaagtt gaaacctctt ccaagtctcc atcctccaac agaaaggatg acgacgctcc 2340
aattgaacac aacgaagttc aagaatctaa cgacaactcc tctaactctt ctaacgttga 2400
agctccaatt ccagttccac acaacgacct ataaaaggaga ttgataagac ttttctagtt 2460
gcatactctt tatatttaaa tcttatctat tagttaattt tttgtaattt atccttatat 2520
atagtcgtgt tattctaaaa tatcatttca gtatctaaaa attcccctct ttttccagtt 2580
atatcttaac aggcgacagt ccaaatgttg atttatccca gtccgattca tcagggttgt 2640
gaagcatttt gtcaatggtc gaaatcacat cagtaatagt gcctcttact tgctcatag 2700
aatttcttct tcttaacgtc accggttggt ctttctctca tttggaacag caggcgagggt 2760
gtattggtgc gagaagaggt gtttccgag 2789

SEQ ID NO: 65      moltype = DNA length = 3188
FEATURE           Location/Qualifiers
misc_feature       1..3188
                   note = Artificial Sequence, fragmentG, (lvDSM012518)
source             1..3188
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 65
tctcgggato ggccaggttg gagcaagggtg gtacttctgt cgctgatttt gtgcaaaggt 60
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atgctgtctaa cgtgccattt cagcaacgta aataataatt acttccatat atattcttct 180
ccccagtag caattttgac atgacaaatt tcacgcatag ttgcttcggc attcgaaaat 240
cgtagaagta tgtcttaatg tagaagggtg agaacaaccg gatcttgccg tcatttttct 300
tttcgaggaa agtgcaagtc tgccactttc cagaaggcat agccttgccc ttttgttgat 360
atttctcccc accgtaattg ttgcattcgc gatcttttca acaatacatt ttatcatcaa 420
gcccgcaaat cctctggagt ttgtcctctc gttcactgtt gggaaaaaca atacgcctaa 480
ttcgtgatta agattcttca aaccatttcc tgcggagttt ttactgtgtg ttgaacggtt 540
cacagcgtaa aaaaaagtta ctataggcac ggtattttta tttcaattgt ttagaagtg 600
ccttcacacc attagccctc gggattaccg tcataggcac tttctgctga gctcctgcga 660
gatttttgcg ctgaaagagt aaaagaaatc tttcacagcg gctccgcggt ccttctact 720
tttaaacgag tcgcaggaac agaagccaaa tttcaaaaga cgctacgctt tcgcccgtttc 780
tgggtctccc accaataacg ctccagcttg aacaaagcat aagactgcga ccaaagcgct 840
gacggacgat ccgaagataa agcttgcttt gcccatgtt ctcgtttcga aaggctatat 900
aaggcacggt atttctctt ttttttccac ctattgtctt tctttgttaa gcttttattc 960
tcggggtttt ttttttgagc atatcgaaag ctttcttttc gcaaatcaaa catagcaaac 1020
cgaaactctc gaacacaaat aaatacacat aaaaatgggt aagagaactc aagggttcat 1080
ggactacgtt ttctccagaa ctctactgct tgggttgaaa ggtgccagat tgagatacac 1140
tgctgctgoc ttgtctgtta tcgggttctc tttattcggg tacgaccaag gtttgatgtc 1200
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caacgacaga tacgcctcgc tcatccaaag tgctgtcact gcttggttac aaattgggtg 1320
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agaaaacgac gatgccatct tggccgaaat gacttctttg caggaagctg tcaacagatc 1860
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cagagctttg attgcttctc ctacccaatt cttccaacaa ttcaccggtt gtaatgctgc 1980
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gactgaagtc gaagattact ctcaatcttt ggggttgacg gacgatgaaa acgaaaagga 2640
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caagtctcca tctccaaca gaaaggatga cgacgctcca attgaacaca acgaagtcca 2760
agaatctaac gacaactcct ctaactcttc taacgttgaa gctccaattc cagttcacca 2820
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caaatgttga tttatcccag tccgattcat cagggttggtg aagcattttg tcaatggctg 3060
aaatcacatc agtaaatgtg cctcttactt gcctcataga atttcttctc cttaacgtca 3120
ccgtttggtc ttttctcat ttggaacagc aggcgaggtg tattggtgag agaagaggtg 3180

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tttccgag 3188

SEQ ID NO: 66 moltype = DNA length = 50
 FEATURE Location/Qualifiers
 misc_feature 1..50
 note = Connector 25
 source 1..50
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 66
 atccgtacac gtttgtgcgc aaccggaagt attgcaacag gtcgaagtcc 50

SEQ ID NO: 67 moltype = DNA length = 50
 FEATURE Location/Qualifiers
 misc_feature 1..50
 note = Connector 2A
 source 1..50
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 67
 acccgatccc gaagaaagtc caactccagt tcgattccct gctttcgagt 50

SEQ ID NO: 68 moltype = DNA length = 50
 FEATURE Location/Qualifiers
 misc_feature 1..50
 note = Connector 2B
 source 1..50
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 68
 tcctcggatc ggccagttgg gagcaagggt gtacttctgt cgtcgtattt 50

SEQ ID NO: 69 moltype = DNA length = 50
 FEATURE Location/Qualifiers
 misc_feature 1..50
 note = Connector 2C
 source 1..50
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 69
 atttgaaca gcaggcgagg tgtattggtg cgagaagagg tgtttccgag 50

SEQ ID NO: 70 moltype = DNA length = 461
 FEATURE Location/Qualifiers
 misc_feature 1..461
 note = Sc_INT7.03_FLANK5
 source 1..461
 mol_type = unassigned DNA
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 70
 atagcgccac atgcatgaa ccggagcttt tatcttcatt gatatatagt aattacaaca 60
 tcttattgaa atgcacagca cttatccaga gttatttgat ataaagatta tgactcaaat 120
 tcttctcgac aaacaagaaa ctttctcacc tacaatgtc ttcaacaatt ctaataattg 180
 acacataagt atattgtata tataacatgg atataaatgg ccctatatgg aaagctatgt 240
 caaaaatgga atccttcctt cctcagcgct ttttatcaag ggccaattga tctgatacac 300
 ctagttagtg actataagtg tagaaagcat actatactat tcgacacttc ctttcaatcc 360
 tggaattaac agtcactttt aaaaaagaca tctaccgtga aggtgccgta gagtatcgcg 420
 ttaccatata gccaaaaact gatatacgcc gcggaaccca g 461

SEQ ID NO: 71 moltype = DNA length = 554
 FEATURE Location/Qualifiers
 misc_feature 1..554
 note = Sc_INT7.03_FLANK3
 source 1..554
 mol_type = unassigned DNA
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 71
 gcacctcatg cacagcaata acacaacaca atgggttagta gcaacctgaa ttcggtcatt 60
 gatgcatgca tgtgccgtga agcgggacaa ccagaaaagt cgtctataaa tgccggcacg 120
 tgcgatcatc gtggcggggt tttaagagtg catatcacaa attgtcgcat taccgcggaa 180
 ccgcccagata ttcattactt gcgcacaaaag cgtttgaaat aatgacgaaa aagaaggaag 240
 aaaaaaaaaa aaaaataaccg cttctaggcg gggtatctac tgatccgagc ttccactagg 300
 atagcaccca aacacctgca tatttggacg acctttactt acaccaccaa aaaccacttt 360
 cgctctctcc gccctcgata acgtccacta attgagcgat tacctgagcg gtcctctttt 420
 gtttgacgca tgagacttgc atactgcaaa tcgtaagtag caacgtctca aggtcaaaac 480
 tgtatggaaa cctgtgcacc tcacttaatt cttagtagcc taccctgcaa gtcaagaggt 540

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ctccgtgatt ccta 554

SEQ ID NO: 72 moltype = DNA length = 100
 FEATURE Location/Qualifiers
 misc_feature 1..100
 note = INT28_FLANK5
 source 1..100
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 72
 ttattttcttt ttacagaaa aatgaaagca atagaattga atttcaacaa gtcacaaaca 60
 aactttgttag actttttatt acaaaaattt attaagtgcac 100

SEQ ID NO: 73 moltype = DNA length = 100
 FEATURE Location/Qualifiers
 misc_feature 1..100
 note = INT28_FLANK3
 source 1..100
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 73
 cattttcctc atcggggacg ccagcttttg tgcggggatga aggcaatatc caaaccagca 60
 tcaggaaaag cctgaagaaa tgggtgcct aatgctctta 100

SEQ ID NO: 74 moltype = AA length = 910
 FEATURE Location/Qualifiers
 REGION 1..910
 note = Nitrate reductase from Zea mays (Zm_NR1)
 source 1..910
 mol_type = protein
 organism = Zea mays

SEQUENCE: 74
 MAAVEPRQFG RLEPGSSPVR VATNGAKAYP PPASHLPERR DSPVRGCSFP PLVSPPRRLD 60
 DASDDEDEEQ EDWRELYGSH LQLEVEPAVQ DARDEGTADA WIERNPCLVR LTGKHPLNCE 120
 PPLARLMHHG FITPAPLHYV RNHGAVPRGD WATWTVETVG LVRRPARLTM EELARDFPAV 180
 EIPVTLACAG NRRKEQNMVQ QTVGFNWGAA GVSTSVWRGA RLRDVLRRCG IVPRKGGALN 240
 VCFEGAEDLP GGGGSKYGTS VTREWALDPS RDIMLAYMQN GEPLLPDHGF PVRVPIPGCI 300
 GGRMVKWLKR IIVTPAESDN YYHFKDNRVL PSHVDAELAN AEAWWYKPEY IINELNINSV 360
 IITPGHDEIL PINSITTQRG YTMKGAYASG GKKVTRVEV TLDGGETWLV CHLDHPEKPN 420
 KYGKYWCWCF WSVEVEVDL LGAKEIAVRA WDQSLNTQPE KLIWNLGMM NNCWFKVKVN 480
 VCRPHKGEIG LVFEHPTQPG NQPGGWMARQ KHLETAEEAA PGLKRSTSTP FMNTTDVGKQ 540
 FTMSVVRKHA SQESAWIAVH GHVYDCTKFL KDHPGGADSI LINAGTDCTE EPDAIHSDKA 600
 KALLDTYRIG ELITGTGYS SDNSVHGGSV LSHLAPIREA VRAPALSNPR DKIHCRVLVGK 660
 KELSRRDVRLF RFSLPSPDQV LGLPIGKHIF VCASIEGKLC MRAYTPSMV DEIGHFDLLV 720
 KVFYKNEHPK FPNGGLMTQY LDSLPVGSYI DVKGPLGHVE YTGSGSPVIN GKQRHASRLA 780
 MICGSGGITP MYQIIQAVLR DQPEDHTEMH LVYANRTEDD ILLRDELDWR AAEYPRDLKV 840
 WYVIDQVKRP EEGWKYSVGF VTEAVLREHV PEGGDDTLAL ACGPPPMIQF AISPNNLEKMK 900
 YDMANSFVVF 910

SEQ ID NO: 75 moltype = AA length = 926
 FEATURE Location/Qualifiers
 REGION 1..926
 note = Nitrate reductase from Spinacia oleracea (Spo_NIA)
 source 1..926
 mol_type = protein
 organism = Spinacia oleracea

SEQUENCE: 75
 MAASVDRQYH PAPMSGVVRT PFSNNHRSDS PVRNGYTFSN PPSSNGVVKP GEKIKLVDNN 60
 SNSNNGSNMN NNRYSDSSEE DDDENEMNVW NEMIKKGNS LEPSSVDSRD EGTADQWIER 120
 NPSMIRLTGK HPFNSEPLPT RLMHHGFLTP VPLHYVRNHG PVPNAKWEDW TVEVTGLVKR 180
 PIRFTMDQLV NDFQSRFPV TLVCAGNRRK EQNMTKQSIG FNWGSAAVST SVVRGPVLRD 240
 VLKRCGVMS LKALNVCFE GAEDLPGGGG SKYGTSVKRE FAMDPARDII LAYMQNGEKL 300
 SPDHGYPVRM IIPFIGGRM VKNLKRIIVT TTESDNYHY KDNRVLP SHV DAELANSEAW 360
 WYKQEIINE LNVNSVITSP CHEEILPINA WTTQRPYTMR GYAYSGGGRK VTRVEVTMDG 420
 GDTWDICELD HQERGSKYGK FWCWCFWSLE VEVLDDLGA EIGVRAWDES LNTQPEKLIW 480
 NVMGMMNNCW FRVKTNVCKP HKGEIGIVFE HPTQPGNKGSG GWMARERHLE ISDSGPTLKR 540
 TASTPFMNTT SKMYSMEVVK KHNTADSAMI VVHGNVYNAT RFLKDHPPGS DSILINAGTD 600
 CTEEFDAIHS DAKAKRLLED F RIGELISTGY TSDSSSPGNS VHGGSVYSGL AGLAPITEAV 660
 PLRNVALNPR VKIPCKLIEK VSLSHDVRRP RFLPSEDQV LGLPVGKHIF LKANVDDKLC 720
 MRAYTPSSTI DVVGYFDLVV KVFYKDVHPR FPNGGVMSQH LDSLSLGSIV DVKGPLGHIE 780
 YLKGNGFTVH GKPKFAKLA MISGGTGITP IYQVMQAILK DPEDKTEMHV VYANRTEEDI 840
 LLREELDKWA DEFDRVKVW YVVEKAEAGW KYDTGFISEK ILRDHVPVAV DDVLALTCGP 900
 PPMIQFAVQP NLDKMGFDIK EQLLIF 926

SEQ ID NO: 76 moltype = AA length = 108
 FEATURE Location/Qualifiers

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REGION 1..108
note = nirD Nitrite reductase (small subunit) from *E. coli*
(*Ec_nirD*)

source 1..108
mol_type = protein
organism = *Escherichia coli*

SEQUENCE: 76
MSQWKDICKI DDILPETGVC ALLGDEQVAI FRPYHSDQVF AISNIDPFFE SSVLSRGLIA 60
EHQGLWVAS PLKKQRFRLS DGLCMEDEQF SVKHYEARVK DGVVQLRG 108

SEQ ID NO: 77 moltype = AA length = 847
FEATURE Location/Qualifiers
REGION 1..847
note = nirB Nitrite reductase (large subunit) from *E. Coli*
(*Ec_nirB*)

source 1..847
mol_type = protein
organism = *Escherichia coli*

SEQUENCE: 77
MSKVRLAIIG NGMVGHRFIE DLLDKSDAAN FDITVFCEEP RIAYDRVHLS SYFSHHTAEE 60
LSLVREGFYE KHGKVLVGE RAITINRQEK VIHSSAGRTV FYDKLIMATG SYPWIPPIKG 120
STQDCFCVYR TIEDLNAIES CARRSKRGAV VGGGLLGLEA AGALKNLGIE THVIEFAPML 180
MAEQLDQMGG EQLRRKIESM GVRVHTSKNT LEIVQEGVEA RKTMRFADGS ELEVDFFVFS 240
TGIRPRDKLA TQCGLDVAPR GGVINDSCQ TSDPDIIAIG ECASWNNRVF GLVAPGYKMA 300
QVAVDHILGS ENAFEGADLS AKLKLGLVDV GGIGDAHGR TPGARSYVYLD ESKEIYKRLI 360
VSEDNKTLLG AVLVGDTSDY GNLLQLVLNA IELPENPDSL ILPAHSGSGK PSIGVDKLPD 420
SAQICSCFDV TKGDLIAAIN KGCHTVAALK AETKAGTGCG GCIPLVTVQL NAEALAKQGIE 480
VNNNLCEHFA YSRQELFHLI RVEGIKTFEE LLAKHGKGYG CEVCKPTVGS LLASCWNEYI 540
LKPEHTPLQD SNDNFLANIQ KDGTVSVIPR SPGGEITPEG LMAVGRIARE FNLYTKITGS 600
QRLAMFGAQK DDLPEIWRQL IEAGFETGHA YAKALMAKT CVGSTWCERYG VGDSVGLGVE 660
LENRYKGIRT PHKMKGVSQ CTRECSAQG KDVGIATEK GWNLYVCNGG GMKPRHADLL 720
AADIDRETLI KYLDRFMMFY IRTADKLTRT APWLENLEGG IDYLKAVIID DKLGLNAHLE 780
EEMARLREAV LCEWTETVNT PSAQTRFKHF INSDKRDPNV QMVPEREQHR PATPYERIPV 840
TLVEDNA 847

SEQ ID NO: 78 moltype = AA length = 1104
FEATURE Location/Qualifiers
REGION 1..1104
note = niiA Nitrite reductase from *Emericella nidulans*
(*Aspergillusnidulans*) (*En_niiA*)

source 1..1104
mol_type = protein
organism = *Emericella nidulans*

SEQUENCE: 78
MPLLDGPRNG ETVTASAHNG IPIIDGVDPS TLRGDIDQDP NRRQKIVVVG LGMVAVAFIE 60
KLVKLDSEER KYDIVVIGEE PHIAYNRVGL SSYFEHRKIE DLYLNPKEWY GSFKDRSFDY 120
YLNTRVTDVF PQHKTVKTST GDIVSYDILV LATGSDAVLP TSTPGHDAKG IFVYRTISDL 180
ERLMEFAANH KGQGTGVTVGG GLLGLEAAGA MTDLEDGFSV KLIDRNKWL ARQLDGDAGS 240
LVTKKIRDLG LEVLHEKRVK KIHTDDNNV TGILFEDGQE LDCCCICFAI GIRPRDELGG 300
STGIQCAKRG GFVIDESLRT SVNDIYAIGE CASWENQTFG IAPGIEMAD VLSFNLNTPD 360
KEPKRFRNRP LSTKLKLLGV DVASFGDFFA DRDGPKFLPG QRPSAESIGA ADPNREEEPQ 420
VKALTYRDPF GGVIKKYLFV MDGKYLGGM MIGDTKDYVK LNQMVKSQKP LEVPPSEFIL 480
GAQSGGEENA DDLDDSTQIC SCHNVTKGDV VESVKSCTK TIADVKSCTK AGTGC GGCMF 540
LVQSI FNKTM LDMGQEVSN LCVHI PYSRA DLYNVIAIRQ LRTFDDVMKS AGKCPDSLGC 600
ETCKPAIASI LSSLFNPHLM DKEYHELQET NDRFLANIQR NGTFSVVPV PGGEITADKL 660
IAIGQVAKKY NLYCKITGGQ RIDMFGARKQ DLLDIWTEL V DAGMESGHAY AKSLRTVKSC 720
VGTTCWCRFG GDSVGMAIRL EQRYKSIRAP HKFKGAVSGC VRECAEAQNK DFGLIATEKG 780
FNIFVGGNGG AKPRHSELLA KDVPPPEVIP ILDRYVIFYI RTADKLQRTA RWLES LPGGI 840
EYLDKVV LND KLGIAAEMER QMQELVDSYF CEWTETVRNP KRRKYFQQFA NTDET VENVE 900
IVKEREQVRP TYWPKDGANE DFKGHQWSSL SWQPVKADY FSDGPPAISS ANIKRGDTQL 960
AIFKVKGYKY ATQCMCPHKR TFLVSDGLIG DDDNGKYWVS CPYHKRNFEL NGEQAGRCQN 1020
DEAMNIATFP VEEREDGWYI MKLPPVEELD SVLGTEKWKV KKGEAVDPFE AYDKKYS GMK 1080
GKRAGAKGIE GSKPTRSPSN TIDW 1104

SEQ ID NO: 79 moltype = AA length = 508
FEATURE Location/Qualifiers
REGION 1..508
note = YNT1 Nitrate/Nitrite transporter from *Pichia angusta*
(*Hansenulapolymorpha*)

source 1..508
mol_type = protein
organism = *Ogataea angusta*

SEQUENCE: 79
MQLSTLWEPV IVNPNRLKAT SIPIFNLWNV YGRNFFFGWF GFFVCFLSWF AFPPLLHGML 60
KIDLRLTAVD ISNNNICGLT GTLLGRFILG PLNDKYGPRI TLTGVLVAGA IPTAFVPLVT 120
NVAGLHAIRF FISFLGSSFI CCSQFCVAVF DNNIIGTANA ISAGWGNAGG GVAFFVMPAI 180

-continued

SNALENRGYS	LHHSWSYSFV	IGPFLILMIT	AILIFVFGSD	CPRGRWSLRG	DILGINMDNM	240
LVKSVSVTRH	FSKEGELTSV	FVEPVNAIDK	VVEPNQDQE	ILEVADIING	DEIEDPSLN	300
DVVKICLSPR	TMLVGLCYMC	SFGTELAVES	IISNLFQGM	TNWSTSKAGA	WGSMLGLLN	360
VARPAGGIIS	DFLYQRFKTT	KAKKFWMIPT	GLMQGIFLIW	IGLVPELSIA	GLIVSVSFLC	420
LWFEMNGAN	YACVPVNRH	HSGIVSGVTG	AMGNLGGILF	SLVFRYTISN	GVNNYKAPW	480
IIGIVCTAVN	LVCVLIPIRE	ERPRKAEN				508

1. A recombinant yeast cell that functionally expresses:
 - a nucleic acid sequence encoding a protein having glycerol dehydrogenase activity;
 - a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity; and
 - a nucleic acid sequence encoding a protein having glycerol transporter activity, wherein the expression of the nucleic acid sequence encoding the protein having glycerol transporter activity is under control of a promoter (the "GT promoter"), which GT promoter has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more.
2. The recombinant yeast cell according to claim 1, wherein the GT promoter is the promoter of a gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C, SML1, TIR2, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4.
3. The recombinant yeast strain according to claim 1, wherein the GT promoter is a synthetic oligonucleotide.
4. The recombinant yeast cell according to claim 1, wherein the protein having glycerol transporter activity is a protein having glycerol-proton symporter activity.
5. The recombinant yeast cell according to claim 1, wherein the protein having glycerol dehydrogenase activity is a protein having NAD⁺ dependent glycerol dehydrogenase activity.
6. The recombinant yeast cell according to claim 1, wherein the nucleic acid sequence encoding the protein having glycerol dehydrogenase activity is a heterologous nucleic acid sequence.
7. The recombinant yeast cell according to claim 1, wherein the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is a native nucleic acid sequence.
8. The recombinant yeast cell according to claim 1, wherein the expression of the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is under control of a promoter.
9. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell comprises one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink.
10. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses:
 - a nucleic acid sequence encoding a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or
 - a nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity; and/or
 - a nucleic acid sequence encoding one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.
11. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses:
 - a nucleic acid sequence encoding a protein comprising phosphoketolase activity (EC 4.1.2.9 or EC 4.1.2.22, PKL); and/or
 - a nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8); and/or
 - a nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).
12. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10).
13. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses a nucleic acid sequence encoding an enzyme having NADH-dependent nitrate reductase activity and/or a nucleic acid sequence encoding an enzyme having NADH-dependent nitrite reductase activity.
14. The recombinant yeast cell according to claim 13, wherein the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding an enzyme having nitrate and/or nitrite transporter activity.
15. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further comprises a deletion or disruption of a nucleic acid sequence encoding a protein having glycerol-3-phosphate dehydrogenase (GPD) activity and/or a nucleic acid sequence encoding a protein having glycerol phosphate phosphatase (GPP) activity.
16. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding a protein having glucosylase activity (EC 3.2.1.20 or 3.2.1.3).
17. A process for the production of ethanol, comprising converting a carbon source, preferably a carbohydrate, using a recombinant yeast cell according to claim 1.
18. The recombinant yeast cell according to claim 1, wherein the protein having glycerol transporter activity is an STL1 protein.
19. The recombinant yeast cell according to claim 1, wherein the protein having glycerol transporter activity is an STL1 protein derived from *Zygosaccharomyces rouxii*.

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